Three dimensional Cauchy-Riemann geometry, Burns-Epstein invariant and Cartan flow

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To my beloved grandmother

I have always cherished the time we spent together

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Introduction

Historically, the Cauchy-Riemann (abbreviated as "CR" for short) structure arises in the study of complex analysis when mathematicians attempted to extend the Riemann mapping theorem in $\mathbb{C}$ to general dimensions. In general, there is no such strong result in $\mathbb{C}^{n+1}$ for $n \geq 1$. Dating back to more than a hundred years ago, as the oldest result, it was first showed by Henri Poincaré that two real hypersurfaces in $\mathbb{C}^2$ are generally not biholomorphically equivalent. One of the most studied type of the complex domains in $\mathbb{C}^{n+1}$ is given by the plurisubharmonic functions. These domains are called (strictly) pseudo-convex, and their boundary inherit the so-called strictly pseudo-convex CR structures. So these boundaries are real hypersurfaces with some special geometric structures known as strictly pseudo-convex CR structures. They are the most common examples in the CR geometry and are vastly studied. The basic ingredient that determine a strictly pseudo-convex CR structure is the Levi form, which is a Hermitian form on complexified tangent bundle of the hypersurface. It appears naturally in the context of complex analysis, for instance from the differential of some plurisubharmonic function. The reason the name "strictly pseudo-convex" given is literally because the Levi form in this scenario is a non-degenerate positive definite complex bilinear form. A given Levi form on the real hypersurface determines a maximally non-integrable hyperplane distribution, which is namely a contact structure, and also a integrable complex structure to this subbundle. Therefore, a strictly pseudo-convex CR structure is described intrinsically by a pair consisting of a contact structure and a integrable complex structure. It is natural to make the definition of strictly pseudo-convex CR structure on an arbitrary odd dimensional orientable real manifold. Further, extending the definitions to the cases with various codimensions, the notions of CR geometry are developed in full generality. Due to its origin in complex analysis, much of the early literature on CR geometry is decidedly analytic in nature.

One of the fundamental problems in CR geometry is The local/global CR equivalence. By "CR equivalence", it means the CR structures are related by local diffeomorphisms. In the case of hypersurfaces, two strictly pseudo-convex domains in $\mathbb{C}^{n+1}$ are biholomorphically equivalent if and only if their boundaries are globally CR equivalent. The geometrically flavored study on such problem was initiated by Elie Cartan in 1930's who solved the local equivalence problem for 3-dimensional strictly pseudo-convex hypersurfaces [19]. In 1960's and 1970's, Noboru Tanaka [24], Shiin-Shen Chern and Jürgen Moser [21] as well as Sidney Webster [20] generalized Cartan's result to higher dimensional cases. The key technique involved is Cartan's moving frame method. In this fashion, people can study the strictly pseudo-convex CR manifolds in the way of Riemannian geometry and Cartan geometry with the help of local geometric invariants, g.e. connection and curvature. Especially from the Cartan geometry point of view, the CR structure is understood as a G-structure, the principal bundle theory impact massively. The CR equivalence problem then turns into G-structure equivalence problem. The Klein geometry or the geometry of homogeneous spaces are the model geometry for Cartan geometry. Such spirit is brought to the study of CR geometry, and the spherical CR structure is of great interest since the spheres (as homogeneous spaces) are always one of the most simple and very first learned examples in all kind of geometries.

Shiin-Shen Chern and James Harris Simons wrote a paper on characteristic class and geometric invariant in 1974, where they introduce the transgression forms to encode geometric information [15]. They applied their construction to orientable Riemannian 3-manifolds to get a transgression 3-form and integating over the manifold obtain a global conformal invariant. This global invariant is nowadays called Chern-Simons invariant and the transgression 3-form is called Chern-Simons form. In particular, by the fact that all orientable 3-manifolds are parallelizable, the Chern-Simons form is globally defined. D. M. Burns and Charles L. Epstein made an analogous construction on compact

strictly pseudo-convex CR 3-manifolds whose holomorphic tangent bundles are trivial, and defined a global CR invariant [30]. Similar to Chern-Simons conformal invariant who has critical point as specific conformal structure called conformal flat structure, Burns and Epstein in their work show that their global invariant has critical point as the spherical CR structure. Afterwards, Jih-Hsin Cheng and John M. Lee [32] have generalized this global CR invariant to any closed strictly pseudo-convex CR 3-manifold relaxing the condition of trivial holomorphic tangent bundle imposed by Burn and Epstein in their construction. What they have obtained is a relative version of Burns-Epstein invariant.

The main purpose of this thesis is to study the global CR invariant on closed pseudo-convex CR 3-manifolds constructed by Burns-Epstein as well as Cheng-Lee, which we refer all as Burns-Epstein invariant. The CR invariance and variational properties are discussed and analyzed. We will see explicitly what the critical point of Burns-Epstein invariant is and how it relates to spherical CR structure. At the end, we review the Cartan flow introduced and studied by Cheng and Lee which is the negative gradient flow of the Burns-Epstein invariant of CR structures. In terms of the analytical setting developed by Gerald B. Folland and Elias M. Stein, we described the existence result for the Cartan flow, and take a glimpse into its geometric consequence. Below we make an outline of the structure of this thesis.

In section 0, we give a preliminary on the principal bundle theory, introducing the basic concepts including principal connection, curvature as well as the extension and restriction of principal bundles. This further leads to the definition of Cartan connection and curvature, which are elementary quantities in Cartan geometry. For this reason, we also give a brief description on G-structure whereas the moving frame method of Cartan is introduced.

In section 1, we give some background of symplectic geometry, contact geometry and complex geometry. We will see how strictly pseudo-convex hypersurfaces arise and lie in the intersection of all these topics. By learning how symplectic structures and contact structures have no local geometric invariant, and we will realize when complex structures are brought in endowing some Riemannian metrics from the Levi forms, the geometry becomes local and has local invariant.

In section 2, we make definitions of basic notions in CR geometry, such as what is a CR structure and pseudo-Hermitian structure, what is CR isomorphisms and how it is related to biholomorphisms in complex analysis. The embedding and filling of CR manifolds are also discussed shortly, since these are also fundamental problems concerned by the CR geometry community aside from CR equivalence problem. Based on the fundamental work by Chern and Moser, we discuss the normal form of strictly pseudo-convex hypersurfaces, which is the hyperquadric. Of course, we particularly focus on three-dimensional case.

In section 3, Webster's curvature theory on strictly pseudo-convex CR manifolds is introduced. It characterizes the pseudo-Hermitian structure on such CR manifolds. In particular, we present how to use Cartan's moving frame method to extract connection and curvature.

In section 4, following Webster and Chern-Moser, we interpret strictly pseudo-Hermitian structures and pseudo-convex CR structures on 3-manifolds as G-structures. By Cartan's moving frame method, connection forms and curvature forms are derived. The study of hyperquadrics will provide the basic recipes on constructing Cartan connection and curvature out of these local data for the CR structure. We show that the Cartan connection is indeed CR invariant. Since the concrete connection and curvature quantities are from pseudo-Hermitian structure by Webster's approach, we relates the pseudo-Hermitian structure and CR structure from G-structure point of view, and represent the Cartan curvature for CR structure in terms of Webster's pseudo-Hermitian curvature and torsion.

In section 5, we make general description on the spherical CR structures. By the work of Cartan and Chern-Moser, we will see the necessary and sufficient condition for a strictly pseudo-convex CR structure to be spherical is the vanishing of certain curvature quantities. In particular for 3-manifolds, it is the so-called Cartan tensor.

In section 6, we give the construction of Burns-Epstein invariant via transgression form (introduced in appendix A in detail) in its original definition and show that it is indeed global CR invariant. Then we turn to discuss the generalization by Cheng-Lee. First we will show how it is a natural generalization of the original Burns-Epstein invariant, and further analyze its variational properties (hence the variation properties of the original Burns-Epstein invariant) by viewing the invariant as a functional for space of CR structures on 3-manifolds. The first variation is proved to be exactly the Cartan tensor, which implies that the critical point of Burns-Epstein invariant is actually spherical CR structure. The second variation is particularly computed for $\mathbb{S}^3$, and it turns out to be strictly positive at the standard CR structure.

In section 7, the Cartan flow is introduced with comparison to Ricci flow. We collect the results by Cheng-Lee on short-time existence and certain weak uniqueness of Cartan flow. The proof of short-time existence is based on Folland-Stein space theory (brief introduction in appendix B), and we make a sketch of it. At the end, a CR geometry consequence, namely how fillability is preserved under Cartan flow is illustrated. We make some outlook for the long-time existence of Cartan flow.

Last but not least, through out the entire thesis, we employ the so-called Einstein summation convention on abstract indices without declaration.

I claim no originality in this thesis, all the materials presented are known. I put effort to organize and expose them in my own way which seems comprehensive at least to me.

0 Principal bundles and G-structures

0.1 Connections and curvatures on principal bundles

Definition 0.1. *Let G be any Lie group. A principal G-bundle is a triple $(\mathscr{P}, p, M)$ where $p : \mathscr{P} \to M$ is a smooth submersion between smooth manifolds and there is right action of G on $\mathscr{P}$ such that:*

(i) $\mathscr{P}_x = p^{-1}(x)$, for any $x \in M$ are G-orbits.

(ii) (local trivialization) Every $x \in M$ has a neighborhood U and a diffeomorphism $\varphi : p^{-1}(U) \to U \times G$ such that the diagram commutes

$$\begin{array}{ccc} p^{-1}(U) & \xrightarrow[\cong]{\varphi} & U \times G \\ & \searrow^{p} & \downarrow{\scriptstyle pr_1} \\ & & U \end{array}$$

and φ is equivariant, i.e. $\varphi(u \cdot g) = \varphi(u)g \ \forall\, u \in p^{-1}(U), g \in G$ where G acts on $U \times G$ by $(x, g') \cdot g = (x, g'g)$.

Example. The frame bundle of a rank-n complex or real vector bundle $\mathscr{V} \to M$ is $\mathrm{Fr}(\mathscr{V}) = \bigsqcup_{x \in M} \mathrm{Fr}(\mathscr{V}_x) \cong \bigsqcup_{x \in M} \mathrm{End}(\mathscr{V}_x)$, which is a $\mathrm{GL}(n, \mathbb{C})$- or $\mathrm{GL}(n, \mathbb{R})$-bundle.

For any G-bundle $(\mathscr{P}, p, M)$ pick an open cover $\{U_i\}_{i \in I}$ of M and local trivializations $\varphi_i : \mathscr{P}|_{U_i} \xrightarrow{\cong} U_i \times G$. For $U_i \cap U_j \neq \varnothing$ we consider the isomorphism $\varphi_j \circ \varphi_i^{-1} : (U_i \cap U_j) \times G \to (U_i \cap U_j) \times G$, it can be shown to have the form $\varphi_j \circ \varphi_i^{-1}(x, g) = (x, g_{ji}(x)g)$ where $g \in G$, $x \in U_i \cap U_j$ and $g_{ji} : U_i \cap U_j \to G$ is smooth map. $\{g_{ji}\}_{j,i \in I}$ are called the transition functions for E w.r.t to local trivialization neighborhood $(U_i, \varphi_i)_{i \in I}$. For $i, j, k \in I$ such that $U_i \cap U_j \cap U_k \neq \varnothing$ the following cocycle condition holds:

- $g_{kj}(x)g_{ji}(x) = g_{ki}(x) \ \forall\, x \in U_i \cap U_j \cap U_k$
- $g_{ii}(x) = e_G \ \forall\, x \in U_i$

The transition functions define an equivalence relation: $(x, g)_i \sim (y, h)_j$ if and only if $x = y$ and $h = g_{ji}(x)g$. The total space $\mathscr{P} = (\bigsqcup_{i \in I} U_i \times G)/\sim$. This provides an alternative equivalent definition of principal G-bundle as a smooth manifold $\mathscr{P}$ with a smooth free G-action on it such that the projection to the orbit space $M = \mathscr{P}/G$ is a smooth fiber bundle with local trivialization satisfying equivariant condition as in definition 0.1.

Let G be a Lie group, $\mathfrak{g} = T_eG$ is its Lie algebra. For a G-bundle over M, there is a short exact sequence:

$$0 \to \mathfrak{g} \xrightarrow{\nu_u} T_u\mathscr{P} \xrightarrow{p_*} T_{p(u)}M \to 0$$

where $\nu_u := dR_g = (R_g)_* : \mathfrak{g} \hookrightarrow T_u\mathscr{P}$ the differential of right action map $R_g : g \mapsto u \cdot g$ and $\mathrm{im}(\nu_u) \subseteq T_u\mathscr{P}$ is called the vertical space $V_u\mathscr{P} = \ker(p_*|_u)$. We then want to choose a horizontal space $H_u\mathscr{P}$ so that $p_*(H_u\mathscr{P}) = T_{p(u)}M$, this is equivalent to a choice of linear map $\omega_u : T_u\mathscr{P} \to \mathfrak{g}$ such that $\omega_u \circ \nu_x = \mathrm{id}_{\mathfrak{g}}$ (causing the short exact sequence of vector spaces split) and hence $H_u\mathscr{P} = \ker(\omega_u)$. This defines a horizontal subbundle of $T\mathscr{P}$ which is pointwise complementary to the vertical subbundle $V\mathscr{P}$. We see that $p_*|_{H_u\mathscr{P}}$ is an isomorphism onto $T_{p(u)}M$, which implies that for every vector field on the base M, there is a unique horizontal lift to one in $H\mathscr{P}$ (the horizontal lifting property). A

connection on $\mathscr{P}$ is a choice of smooth horizontal subbundle $H\mathscr{P} \subset T\mathscr{P}$ which is invariant under the right G-action on $\mathscr{P}$. We call the 1-form with $\mathfrak{g}$-values $\omega \in \Omega^1(\mathscr{P};\mathfrak{g})$ the defining form for $H\mathscr{P}$. The map $\nu_u = (R_g)_*$ maps a $X \in \mathfrak{g} \cong T_eG$ to a vector denoted as $X^*(u) \in T_u\mathscr{P}$ which forms a vector field X^* on $\mathscr{P}$. Clearly $\omega(X^*) = X$, and this vector field is called a *fundamental vector field* corresponding to the Lie algebra element $X \in \mathfrak{g}$. In fact, if a 1-form $\omega \in \Omega^1(\mathscr{P};\mathfrak{g})$ satisfies $\omega(X^*)$ for all fundamental vector fields X^* and $(R_g)^*\omega = \mathrm{Ad}(g^{-1}) \circ \omega$ for all $g \in G$, then $\ker(\omega)$ actually defines a connection on $\mathscr{P}$. This motivates the following definition.

Definition 0.2. *A connection 1-form on G-bundle is a 1-form $\omega \in \Omega^1(\mathscr{P};\mathfrak{g})$ such that*

(i) $\omega_u \circ \nu_u = id_{\mathfrak{g}}$.

(ii) (Ad-equivariant) $R_g^*\omega = Ad(g^{-1}) \circ \omega \ \forall\, g \in G$ (this condition is equivalent to $(R_g)_* H_u\mathscr{P} = H_{R_g(u)}\mathscr{P} \ \forall\, u \in \mathscr{P}, g \in G$).

A connection 1-form determines uniquely up to isomorphisms a subbundle $\ker(\omega) \subset T\mathscr{P}$, and this is called a connection on $\mathscr{P}$.

The above definition of connection was first introduced by Charles Ehresmann (it actually makes sense for any smooth fiber bundles), and hence one can more specifically calls it Ehresmann connection.

Proposition 0.3. *Every principal G-bundle has a connection.*

Proof. Let $\{U_i\}_{i\in I}$ be an open cover of M such that $\mathscr{P}$ is trivialized over each open set U_i. By paracompactness of M, there exists partition of unity subordinate to the cover, i.e. a family of smooth functions $\rho_i : M \to \mathbb{R}$ with $\rho_i \geq 0$ and $\mathrm{supp}(\rho_i) \subset U_i$, $\sum_{i\in I} \rho_i \equiv 1$. Then $p^*\rho_i = \rho_i \circ p : \mathscr{P} \to \mathbb{R}$, $\sum_{i\in I} p^*\rho_i \equiv 1$. Since $p|_{p^{-1}(U_i)} : p^{-1}(U_i) \to U_i$ is isomorphic to a trivial bundle $U_i \times G$, there is a trivial connection on it. Let $\omega_i \in \Omega^1(p^{-1}(U_i);\mathfrak{g})$ be a connection 1-form on this product bundle, and define $\omega := \sum_{i\in I} p^*\rho_i\omega_i$. One easily checks that this is indeed a connection 1-form on $\mathscr{P}$. Namely, we construct global connection 1-form by gluing local trivial connection 1-forms together via partition of unity. □

For each $i \in I$, we have local section $s_i : U_i \to \mathscr{P}$, $p \mapsto \varphi_i^{-1}(x, e)$ via local trivialization. Set $\omega_i = s_i^*\omega \in \Omega^1(U_i;\mathfrak{g})$, $\kappa_{ij} = g_{ij}^*\kappa_G$ where $(\kappa_G)_g = (L_{g^{-1}})_*$ is the unique G-invariant 1-form in $\Omega^1(G;\mathfrak{g})$ such that $\kappa_G(X) = X$ for any $X \in \mathfrak{g}$ (here X are also considered as left-invariant vector field on TG since $\mathfrak{g} \cong T_eG \cong C^\infty(G, TG)^G$), when G is matrix Lie group, e.g. $GL(n,\mathbb{R})$, then it is of the form $(\kappa_G)_g = g^{-1}dg \ \forall\, g \in G$. On $U_i \cap U_j \neq \varnothing$, we have $\omega_j = \mathrm{Ad}(g_{ij}^{-1}) \circ \omega_j + \theta_{ij}$ and ω_i completely determines $\omega|_{p^{-1}(U_i)}$. So conversely a family of local 1-forms on M satisfying such transition relation in fact determines uniquely a global connection 1-form. These can be summarized as following.

Proposition 0.4. *Let $(\mathscr{P}, p, M)$ be a principal G-bundle, and $\{U_i\}_{i\in I}$ be a open cover of M with local trivializations $\{\varphi_i\}_{i\in I}$ and transition functions $\{g_{ij}\}_{i,j\in I}$. Then there is a one-to-one correspondence between connection 1-forms on $\mathscr{P}$ and a family of local 1-forms $\{A_i \in \Omega^1(U_i;\mathfrak{g})\}_{i\in I}$ satisfying $A_j = Ad(g_{ij}^{-1}) \circ A_i + g_{ij}^*\kappa_G$ on $U_i \cap U_j$ for all i, j.*

Now we want to define the curvature form on $\mathscr{P}$: Let $\mathrm{pr_h} : T\mathscr{P} \to H\mathscr{P}$ be the horizontal projection, the curvature 2-form is defined as $\Omega := d\omega \circ \mathrm{pr_h}$. Since $\omega \in \Omega^1(\mathscr{P};\mathfrak{g})$ we have $\omega \wedge \omega \in \Omega^2(\mathscr{P};\mathfrak{g}\otimes\mathfrak{g})$ and define $[\omega,\omega] \in \Omega^2(\mathscr{P};\mathfrak{g})$ to be the image of $\omega\wedge\omega$ under Lie bracket $[\cdot,\cdot] : \mathfrak{g}\otimes\mathfrak{g} \to \mathfrak{g}$. We have following structure equation.

Proposition 0.5. *The curvature 2-form $\Omega \in \Omega^2(\mathscr{P};\mathfrak{g})$ for connection 1-form ω is defined by the structural equation $\Omega = d\omega + \frac{1}{2}[\omega,\omega]$.*

Proposition 0.6. *The curvature 2-form satisfies the Bianchi identity* $d\Omega = [\Omega, \omega]$.

This can be shown by differentiating the structural equation, the key step is using the Jacobi identity.

Note that ω is Ad-equivariant, hence Ω is also Ad-equivariant. As a consequence, the curvature $\Omega \equiv 0$ if and only if $H\mathscr{P}$ is closed under the Lie bracket of horizontal vector fields. A connection with vanishing curvature is said to be flat. When a principal bundle $\mathscr{P}$ admits a flat connection, further by Frobenius theorem, it means $H\mathscr{P}$ is integrable. If in addition M is simply connected, then the principal bundle $\mathscr{P}$ is trivial due to the horizontal lifting property, namely there exists a nowhere vanishing global section to $\mathscr{P}$ trivializing it. The following proposition and corollary reveal the relations between flat connections, Maurer-Cartan forms and transition functions.

Proposition 0.7. *A connection 1-form* ω *on principal* G*-bundle* $p : \mathscr{P} \to M$ *is flat if and only if around every point in* M *there is a trivializing neighborhood* U *with* $\varphi : p^{-1}(U) \xrightarrow{\cong} U \times G$ *such that* $\omega|_{\mathscr{P}|_U} = \varphi^* pr_2^* \kappa_G$ *(*$pr_2 : U \times G \to G$ *is the projection to the second factor).*

Corollary 0.8. *Let* $p : \mathscr{P} \to M$ *be a principal* G*-bundle. The following are equivalent:*

(1) $\mathscr{P}$ *admits a flat connection.*

(2) There is an trivializing open cover $\{U_i\}_{i\in I}$ *with trivialization maps* $\{\varphi_i\}_{i\in I}$ *such that all transition functions* $g_{ij} : U_i \cap U_j \to G$ *are constant.*

Definition 0.9. *A* k*-form* $\omega \in \Omega^k(\mathscr{P}; V)$ *(*V *is a vector space) is called horizontal if* $\omega(X_1, \dots, X_k) = 0$ *for* $X_1, \dots, X_k \in T_u\mathscr{P}$ *among which at least one is vertical.*

Let $\rho : G \to \mathrm{GL}(V)$ be a representation of G on V. Then $\omega \in \Omega^k(\mathscr{P}; V)$ is called ρ-equivariant if and only if $R_g^*\omega = \rho(g^{-1}) \circ \omega \ \forall\, g \in G$. A horizontal ρ-equivariant form with ρ trivial is then called basic (for non-trivial representation it is called semi-basic).

Proposition 0.10. *Let* $\mathscr{P}$ *be a* G*-bundle over* M, $\{U_i\}_{i\in I}$ *be a open cover of* M *with trivializations* $\{\varphi_i : \mathscr{P}|_{U_i} \to U_i \times G\}_{i\in I}$ *and transition functions* $\{g_{ij} : U_i \cap U_j \to G\}_{i,j\in I}$. *Then is a 1-1 correspondence between horizontal* ρ*-equivariant* k*-forms on* $\mathscr{P}$ *and* k*-forms in* $\bigcup_{i\in I} \Omega^k(U_i; V)$ *satisfying* $\omega_j = \rho(g_{ji}) \circ \omega_i$ *on* $U_i \cap U_j$ *(Here* ω_i *is the pullback of* $\omega \in \Omega^k(\mathscr{P}; V)$ *by the local section* $s_i : U_i \to \mathscr{P}|_{U_i}$ *sending* $x \in U_\alpha$ *to* $\varphi_i^{-1}(x, e)$*).*

Corollary 0.11. $p^* : \Omega^k(M; V) \to \Omega^k(\mathscr{P}; V)$ *gives an isomorphism onto the basic forms on* $\mathscr{P}$.

Let $\mathscr{V} = \mathscr{P} \times_\rho V$ be the associated vector bundle, the space of bundle-valued differential forms is defined as $\Omega^k(M; \mathscr{V}) := \mathrm{Alt}^k_{C^\infty(M)}(C^\infty(M, TM), C^\infty(M, \mathscr{V}))$ where $C^\infty(M, TM)$ together with Lie bracket of vector fields is regarded as Lie algebra of smooth vector fields on M. It has local description as $\Omega^k(U_i; \mathscr{V}|_{U_i}) \cong \Omega^k(U_i; V)$ where U_i is the trivializing open neighborhoods of $\mathscr{P}$. We write $\Omega^k_{\mathrm{hor}}(\mathscr{P}; V)^G$ for the space of G-equivariant horizontal k-forms on $\mathscr{P}$.

Corollary 0.12. *There is a* $C^\infty(M)$*-linear bijection* $\Omega^k(M; \mathscr{V}) \cong \Omega^k_{hor}(\mathscr{P}; V)^G$ *for all* k.

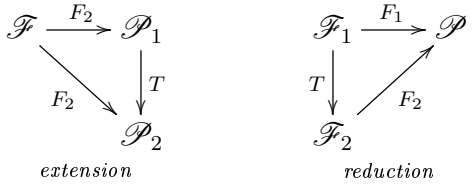
For special case, consider the adjoint representation $\mathrm{Ad} : G \to \mathrm{GL}(\mathfrak{g})$, then the above result yields a one-to-one correspondence between horizontal Ad-equivariant forms on $\mathscr{P}$ to forms on M with bundle value $\mathrm{Ad}\mathscr{P} := \mathscr{P} \times_{\mathrm{Ad}} \mathfrak{g}$. In fact, curvature form Ω is horizontal Ad-equivariant 2-form (despite the connection ω is not horizontal form). The set of all connection 1-forms on $\mathscr{P}$ is an affine space denoted as $\mathcal{A}$ whose vector space of translations is $\Omega^1(M; \mathrm{Ad}\mathscr{P})$, i.e. $T_\omega\mathcal{A} = \Omega^1(M; \mathrm{Ad}\mathscr{P})$ for any $\omega \in \mathcal{A}$.

We often need to compare bundles with different structure groups, for example, regarding one as a subbundle of the other via the inclusion of structure groups. Let G and $H \subset G$ be two Lie groups. Suppose there is a non-surjective Lie group homomorphism between them $\phi : H \to G$. Then for a G-bundle $(\mathscr{P}, p_{\mathscr{P}}, M)$ and a H-bundle $(\mathscr{F}, p_{\mathscr{F}}, M)$, we have the following notions of extension and reduction.

Definition 0.13. *A based-preserving, fiberwise injective and ϕ-equivariant smooth map $F : \mathscr{F} \to \mathscr{P}$ is called an extension of $\mathscr{F}$ to G or a reduction of $\mathscr{P}$ to H, i.e. $F \circ p_{\mathscr{P}} = p_{\mathscr{F}}$, $F|_{\mathscr{F}_x}$ maps $\mathscr{F}_x$ into $\mathscr{P}_x$ for all $x \in M$, and $F(v \cdot h) = F(v) \cdot \phi(h)$ for all $v \in \mathscr{F}$, $h \in H$.*

Remark. Of course, the definition above also makes sense when ϕ is surjective with non-trivial kernel. However, in such case, it is called a lifting of $\mathscr{P}$ to H. All the results below also hold for lifting, but we do not actually need this notion here.

Definition 0.14. *Two extensions of $F_1 : \mathscr{F} \to \mathscr{P}_1$ and $F_2 : \mathscr{F} \to \mathscr{P}_2$ are said to be equivalent if there is an bundle isomorphism $T : \mathscr{P}_1 \to \mathscr{P}_2$ such that $T \circ F_1 = F_2$. Similarly, two reductions $F_1 : \mathscr{F}_1 \to \mathscr{P}$ and $F_2 : \mathscr{F}_2 \to \mathscr{P}$ are said to be equivalent if there is bundle isomorphism $T : \mathscr{F}_1 \to \mathscr{F}_2$ such that $F_1 = F_2 \circ T$. In terms of commutative diagrams:*



Proposition 0.15. *Let $\phi : H \to G$ be a non-surjective Lie group homomorphism and $(\mathscr{F}, p_{\mathscr{F}}, M)$ be a prnicipal H-bundle. Then there exists an unique up to equivalence extension of $\mathscr{F}$ to G.*

For the proof of this, one should notice that H can acts on G from the left via ϕ as $h \cdot g = \phi(h)g$ for all $h \in H$. Hence we can construct a principal G-bundle $\mathscr{F} \times_G G$ with right G-action given by $(v, g) \cdot g' = (v \cdot g, gg') = ((x, h \cdot g), gg')$ for any $g, g' \in G$, $v \in \mathscr{F}$. Furthermore there is a map $F : \mathscr{F} \to F \times_G G$ given by $F(v) = [(v, e_G)]$ making it an extension of $\mathscr{F}$ to G. If $F' : \mathscr{F} \to \mathscr{P}$ is any other extension, then there is a isomorphism $T : \mathscr{F} \times_G G \to \mathscr{P}$ given by $T([(v, g)]) = F'(v) \cdot g$ for all $v \in \mathscr{F}$, $g \in G$. Clearly $F' = T \circ F$. By definition, F is equivalent to F'.

Remark. Although the extension exists and unique up to equivalence, reduction does not always exit, and even if it does exist, it is usually not unique.

In local description, given a Lie group homomorphism $\phi : H \to G$, a principal G-bundle admits a reduction to H if and only if there is a trivializing open cover $\{U_i\}_{i \in I}$ of M such that the transition functions is of the form $\phi \circ h_{ij}$ on $U_i \cap U_j \neq \varnothing$ where h_{ij} satisfy the cocycle condition. The most important case is when $H \subset G$ is a closed (embedded) Lie subgroup. Then in this case, ϕ is an inclusion of Lie group.

Example. Over a n-dimensional Riemannian manifold, the orthogonal frame bundle is a $\mathrm{O}(n)$-bundle, the inclusion $\mathrm{O}(n) \subset \mathrm{GL}(n, \mathbb{R})$ gives an extension of the orthogonal frame bundle to the general frame bundle. The Riemannian metric therefore defines a reduction of the principal $\mathrm{GL}(n, \mathbb{R})$-bundle to a $\mathrm{O}(n)$-bundle.

It is well-known that G/H ($H \subset G$ Lie subgroup) is a smooth manifold (it is called homogeneous space). The right H-actions on G then induces a fiberwise right H-action on the total space $\mathscr{P}$, we called the quotient space $\mathscr{P}/H$ the reduction of G-bundle $\mathscr{P}$ to H. One can show that $\mathscr{P}/H \cong$

$\mathscr{P}\times_G(G/H)$ as a fiber bundle with fiber diffeomorphic to G/H and base is M. Furthermore following from the fact $G \to G/H$ as a H-bundle over G/H, $\mathscr{P} \to \mathscr{P}/H$ is a principal H-bundle over $\mathscr{P}/H$. Using these we can prove following theorem.

Theorem 0.16. *There is a 1-1 correspondence between the set of sections of the fiber bundle $\mathscr{P}/H \to M$ with fiber G/H and the set of equivalence classes of reductions of $\mathscr{P}$ to H. In particular, $\mathscr{P}$ admits a reduction to H if and only if $\mathscr{P}/H$ has a section.*

Proof. Given a reduction $F : \mathscr{F} \to \mathscr{P}$ where $\mathscr{F}$ is a principal H-bundle, it induces a map on quotient spaces $s_F : \mathscr{F}/H \to \mathscr{P}/H$. Notice $M \cong \mathscr{F}/H$, and then one can check easily that s_F is indeed a smooth section using local trivialization of $\mathscr{P}/H \to M$. Conversely, given a section $s : M \to \mathscr{P}/M$, we get a pullback bundle $s^*\mathscr{P} \to M$ along the section

$$\begin{array}{ccc} s^*\mathscr{P} & \longrightarrow & \mathscr{P} \\ \downarrow{\scriptstyle \bar{s}} & & \downarrow \\ M & \xrightarrow{s} & \mathscr{P}/H \end{array}$$

and it is also a H-bundle (recall $s : \mathscr{P} \to \mathscr{P}/H$ is a H-bundle). It follows straightforwardly that $\bar{s} : s^*\mathscr{P} \to \mathscr{P}$ is a reduction of G-bundle $\mathscr{P} \to M$ to H. □

Remark. Putting $H = \{e_H\}$ (trivial group), the theorem 0.16 gives a one-to-one correspondence between (global) trivializations and (global) sections of $\mathscr{P} \to M$.

Since every connected Lie group G is diffeomorphic to $K \times \mathbb{R}^N$ for some $N \in \mathbb{N}$ and $K \subset G$ a maximal compact Lie subgroup. As a consequence of above theorem:

Corollary 0.17. *Every principal G-bundle with G a connected Lie group has a reduction to K where $K \subset G$ is a maximal compact Lie subgroup.*

After the discussion of extension and reductions of principal bundle with respect to inclusion of Lie group, we would also like to somehow extend the value of connection forms on prescribed bundles. For this purpose, we next introduce the Cartan connection and the corresponding curvature. Here we mainly follow the description in [26] (one can also refer to more original literatures cited therein) and [9] chapter 5 especially §3.

Definition 0.18. *Let G be a Lie group, and H be a closed Lie subgroup with $\dim G/H = n$. Denote $\mathfrak{g}$ and $\mathfrak{h}$ be their Lie algebras receptively. For a principal H-bundle $\mathscr{P}$ over a smooth n-manifold M. A G/H-type Cartan connection 1-form on $\mathscr{P}$ is a $\mathfrak{g}$-valued 1-form ω on $\mathscr{P}$ satisfying following conditions:*

i) $\omega(X^*) = X$ for any $X \in \mathfrak{h}$.

ii) $R_h^*\omega = Ad(h^{-1}) \circ \omega$ for all $h \in H$.

iii) For any $\widetilde{X} \in T_u\mathscr{P}$, $u \in \mathscr{P}$, $\omega(\widetilde{X}) = 0$ if and only if $\widetilde{X} = 0$.

We want to compare Cartan connection with Ehresmann connection. For instance, Ehresmann connection 1-form on $\mathscr{P}$ gives isomorphism from $V_u\mathscr{P}$ (spanned by fundamental vector fields at $u \in \mathscr{P}$) to $\mathfrak{h}$, while the Cartan connection 1-form instead gives an isomorphism from $T_u\mathscr{P}$ to $\mathfrak{g}$ due to condition *iii*). So although $\omega(X^*) = X$ only for element in $\mathfrak{h}$ in condition *i*), we can actually define a vector field $\widehat{X}$ for all $X \in \mathfrak{g}$ such that $\omega(\widehat{X}) = X$ via this isomorphism (the inverse map of ω). Such vector fields have the properties:

- $\widehat{X} = 0$ implies $X = 0 \in \mathfrak{g}$.
- $\widehat{X} = X^*$ for any $X \in \mathfrak{h}$.
- $(R_h)_* \widehat{X} = \widehat{(\mathrm{Ad}(h^{-1})X)}$ for all $h \in H$ and $X \in \mathfrak{g}$.

In this sense, we can think of the Cartan connection as an extension of the Ehresmann connection.

Fixing a complementary subspace $\mathfrak{m}$ of $\mathfrak{h}$ in $\mathfrak{g}$, i.e. $\mathfrak{g} = \mathfrak{h} + \mathfrak{m}$ (note that $\mathfrak{m}$ is not necessary a Lie subalgebra), we can decompose ω with respect to these two subspaces as $\omega = \mathrm{pr}_{\mathfrak{h}} \circ \omega + \mathrm{pr}_{\mathfrak{m}} \circ \omega = \omega_{\mathfrak{h}} + \omega_{\mathfrak{m}}$. It follows that for $\widetilde{X} \in T_u \mathscr{P}$, $\omega_{\mathfrak{m}}(\widetilde{X}) = 0$ if and only if $\widetilde{X}$ is vertical. To make a local description, one can pick a local section $s : U \to \mathscr{P}$ over an open set $U \subset M$, set $\pi = s^* \omega$ and $\pi = \pi_{\mathfrak{h}} + \pi_{\mathfrak{m}}$. Then one can show directly for any $X \in TM|_U$, $\pi_{\mathfrak{m}}(X) = 0$ implies $X = 0$.

For a trivializing open cover $\{U_i\}_{i \in I} \subset M$ with trivializations $\psi_i : p^{-1}(U_i) \to U_i \times H$ and corresponding transition functions $h_{ij} : U_i \cap U_j \to H$ given by $h_{ij} = \psi_i \circ \psi_j^{-1}$, define local sections $s_i : U_i \to \mathscr{P}$ as $s_i(x) = \psi_i^{-1}(x, e_H)$ and put $\pi_i = s_i^* \omega$. Analogously, we have $\pi_{i,\mathfrak{m}}(X) = 0$ implies $X = 0 \in TM|_{U_i}$, and this is equivalent to that $q_* \circ \pi_i : T_x M \to \mathfrak{g}/\mathfrak{h}$ is an isomorphism where $q_* : \mathfrak{g} \to \mathfrak{g}/\mathfrak{h}$ is the quotient map. Under identification $\mathfrak{m} \cong \mathfrak{g}/\mathfrak{h}$, this map can be written as $\mathrm{pr}_{\mathfrak{m}} \circ \pi_i$. The following proposition tell us the local description completely determines the Cartan connection.

Proposition 0.19. *If a family $\{\pi_i \in \Omega^1(U_i; \mathfrak{g})\}_{i \in I}$ of local $\mathfrak{g}$-valued 1-forms on M satisfies $\pi_j = Ad(h_{ij}^{-1}) \circ \pi_i + \kappa_{ij}$ on $U_i \cap U_j \neq \varnothing$ ($\kappa_{ij} := h_{ij}^* \kappa_H$) and $q_* \circ \pi_i : T_x M \to \mathfrak{g}/\mathfrak{h}$ is an isomorphism for all $x \in M$, then $\omega := \psi_i^*(Ad(h^{-1}) \circ \pi_i + \kappa_H)$ defines a Cartan connection 1-form on $\mathscr{P}$ pointwise at $u = \psi_i^{-1}(x, h) \in \mathscr{P}$ for all $x \in M$ and $h \in H$.*

The curvature 2-form corresponds to the Cartan connection 1-form $\omega \in \Omega^1(\mathscr{P}; \mathfrak{g})$ is naturally defined by the structure equation $\Omega = d\omega + \frac{1}{2}[\omega, \omega] \in \Omega^2(\mathscr{P}; \mathfrak{g})$. We also have Bianchi identity just as for Ehresmann connection forms. Cartan curvature 2-form Ω annihilates $\widehat{X}$ for any $X \in \mathfrak{h}$. The Cartan curvature also yields decomposition with respect to subspaces $\mathfrak{h}$ and $\mathfrak{m} \cong \mathfrak{g}/\mathfrak{h}$, and the component $\mathrm{pr}_{\mathfrak{m}} \circ \Omega$ is called the *torsion 2-form* of the Cartan connection (c.f. [9] Page 184 below Definition 3.1).

Example. On the H-bundle $q : G \to G/H$, the Maurer-Cartan form κ_G is a Cartan connection 1-form, and clearly it has vanishing Cartan curvature. If furthermore, G/H is reductive (i.e. there exists complementary vector space $\mathfrak{m}$ such that $\mathrm{Ad}(H)\mathfrak{m} \subset \mathfrak{m}$), then the Ehresmann connection 1-form on this H-bundle is $\mathrm{pr}_{\mathfrak{h}} \circ \kappa_G \in \Omega^1(G; \mathfrak{h})$ and the corresponding curvature 2-form is given by $-\frac{1}{2}\mathrm{pr}_{\mathfrak{h}} \circ [\mathrm{pr}_{\mathfrak{m}} \circ \kappa_G, \mathrm{pr}_{\mathfrak{m}} \circ \kappa_G] \in \Omega^2(G; \mathfrak{h})$.

A G/H-type Cartan connection on a H-bundle $\mathscr{P}$ over M can be regarded as Ehresmann connection for a reduction of a G-bundle over M to H. Let $\widetilde{\mathscr{P}} := \mathscr{P} \times_H G$, then $\tilde{p} : \widetilde{\mathscr{P}} \to M$ is a principal G-bundle over M. $\mathscr{P}$ regarded as a subbundle of $\widetilde{\mathscr{P}}$ through the inclusion $\mathscr{P} \to \mathscr{P} \times G$. A Cartan connection form ω on $\mathscr{P}$ can be uniquely extended to an Ehresmann connection form $\tilde{\omega}$ on $\widetilde{\mathscr{P}}$ in the following way: The inclusion $\iota : P \hookrightarrow \widetilde{\mathscr{P}}$ is a reduction, we can define local trivializations of $\widetilde{\mathscr{P}}$ from those of $\mathscr{P}$ as $\tilde{\psi}_i : \tilde{p}^{-1}(U_i) \to U_i \times G$, $\tilde{\psi}_i(\tilde{u}) = \psi_i(u) \cdot g$ if $\tilde{u} = u \cdot g$ for $\tilde{u} \in \widetilde{\mathscr{P}}, u \in \mathscr{P}$. The transition functions of two bundles coincide, i.e. $h_{ij} = \psi_i \circ \psi_j^{-1} = \tilde{\psi}_i \circ \tilde{\psi}_j^{-1}$. $\tilde{\kappa}_{ij} = h_{ij}^* \kappa_G$, since h_{ij} takes value in H, so this is actually $h_{ij}^* \kappa_H = \kappa_{ij}$. Thus $\{\pi_i\}_{i \in I}$ also defines a global $\mathfrak{g}$-valued 1-form $\tilde{\omega}$ on $\widetilde{\mathscr{P}}$ given pointwise by $\tilde{\omega} = \tilde{\psi}_i^*(\mathrm{Ad}(g^{-1}) \circ \pi_i + \kappa_G)$ at $\tilde{u} = \tilde{\psi}_i^{-1}(x, g)$ for all $g \in G$, which is an usual connection 1-form on $\widetilde{\mathscr{P}}$ in Ehresmann sense. We have $\iota^* \tilde{\omega} = \omega$ and $\iota^* \tilde{\Omega} = \Omega$. Locally $\Omega = 0$ implies $\tilde{\Omega}$. Hence we have following local flatness result.

Proposition 0.20. *If Cartan curvature $\Omega = 0$ near $u \in \mathscr{P}$, then there exists a neighborhood U of $p(u) \in M$ and an open set $V \subset G/H$, and an H-bundle isomorphism $\varphi : p^{-1}(U) \to q^{-1}(V)$ such*

that $\omega = \varphi^*\kappa_G$ on $p^{-1}(U)$. Moreover if M is simply connected, then there exists a global H-invariant map $\varphi : P \to G$ such that $\omega = \varphi^*\kappa_G$.

In conclusion, any H-bundle with global flat Cartan connection of G/H-type is modeled on the H-bundle $G \to G/H$. In fancy words, one says the Cartan geometry (characterized by local geometric data like Cartan connections and curvatures) is the geometry of spaces that modeled on the Klein geometry (which describes the geometry of homogeneous space G/H, initiated by Klein's famous Erlangen program). Rather complete descriptions of Cartan geometry as well as Klein geometry in detail can be found in [9].

0.2 G-structures and connections

Historically, the concept of connection was initiated from moving frame method developed extensively by Élie Cartan. In terms of modern notation of principal bundles, a moving frame or a local frame field of a vector bundle (e.g. tangent bundle or its subbundles) is a local smooth section of the corresponding frame bundle. It is well known that a vector bundle admitting a global frame field is trivial, and equivalently the corresponding frame bundle is also trivial since it has a global section as a principal bundle. Cartan's moving frame method is very useful and generic approach to extract local data such as connection and curvature from manifold with certain geometric structure. We shall introduce the concept of G-structures to describe 'geometric structures' formally and study them with moving frame method. Standard references are [6],[7]. This machinery will be the fundamental tool involved massively in the studying of Cauchy-Riemann structures later.

Consider the vector space $V = \mathbb{R}^n$ (respectively $V_{\mathbb{C}} := V \otimes \mathbb{C} = \mathbb{C}^n$), and $\mathrm{GL}(V) = \mathrm{GL}(n, \mathbb{R})$ (respectively $\mathrm{GL}(n, \mathbb{C})$). Recall that a tangent bundle over any real n-dimensional smooth manifold has generic fiber isomorphic to $V = \mathbb{R}^n$, the corresponding frame bundle forms a principal $\mathrm{GL}(n, \mathbb{R})$-bundle. Dually, the collection of all frames of the cotangent bundle forms also a $\mathrm{GL}(n, \mathbb{R})$-bundle, which is called the coframe bundle. A local section of the coframe bundle is a moving coframe or local coframe field on M. Frame bundle and coframe bundle are non-canonically isomorphic, which is followed from isomorphism between tangent bundle and cotangent bundle. Therefore, we denote in this sense the $\mathrm{GL}(n, \mathbb{R})$-bundle abstractly as $\mathscr{F}_{\mathrm{GL}}$.

Definition 0.21. *A G-structure on a smooth manifold M is a reduction of $\mathscr{F}_{GL}$ to a subbundle $\mathscr{F}_G \subset \mathscr{F}_{GL}$, where $G \subset GL(V)$ is a matrix Lie group.*

Equivalently, a G-structure can be viewed as a smooth section of the quotient bundle $\mathscr{F}_{\mathrm{GL}}/G \to M$, since all sections in $\mathscr{F}_G$ (frames/coframes) induces the same element in $\mathscr{F}_{\mathrm{GL}}/G$. Given a section $\bar{s} : M \to \mathscr{F}_{\mathrm{GL}}/G$, $\mathscr{F}_G$ can be recovered as the set of all sections in $\mathscr{F}_{\mathrm{GL}}$ representing $\bar{s}$. This means that $\mathscr{F}_{\mathrm{GL}}/G$ is the space of G-structure on M.

Given a diffeomorphism between two manifolds $\phi : M \to M'$, we obtain a bundle isomorphism $(\mathscr{F}_{\mathrm{GL}}, M) \cong (\mathscr{F}'_{\mathrm{GL}}, M')$ and induced the identification of spaces of G-structures $(\mathscr{F}_{\mathrm{GL}}/G, M) \cong (\mathscr{F}'_{\mathrm{GL}}/G, M')$.

Definition 0.22. *Two G-structures $(\mathscr{F}_G, M)$ and $(\mathscr{F}'_G, M')$ are equivalent if there exists a diffeomorphism $\phi : M \to M'$ such that the induced identification of $\mathscr{F}_{GL}$ with $\mathscr{F}'_{GL}$ makes $\mathscr{F}_G$ identified with $\mathscr{F}'_G$.*

The most natural differential invariant to characterize a G-structure is the principal connection form on the corresponding G-bundle $\mathscr{F}_G$. In practice, we often prefer to use differential forms on M

for computations, hence we would like to push down the principal connection forms to M. We know already that G-structure is determined by a G-action on V the generic fiber of TM, namely the group transformation of the frames in TM. This is equivalent to a representation $\rho : G \to \mathrm{GL}(V)$. Then we have a special case of associated vector bundle to the principal bundle $\mathscr{F}_G$, i.e. $\mathscr{V} = \mathscr{F}_G \times_\rho V = TM$ the tangent bundle itself. In general, the principal connections on principal bundles and the affine connections or covariant derivatives on associated vector bundles are closely related, the parallel transport on the principal bundle defines uniquely a parallel transport on the associated vector bundle from which the affine connections is defined along any curve on M and it is G-equivariant. These aspects are discussed in detail in [4] Chapter 3. Applying the general theory to the special case on tangent bundle and G-structure here: A principal connection 1-form ω on $\mathscr{F}_G$ induces a G-equivariant affine connection $\nabla : C^\infty(M, TM) \to C^\infty(M, TM \otimes T^*M)$. To discuss local expression, we pick a local section (local frame field) $s : U \subseteq M \to \mathscr{F}_G$ whose components $s_i \in C^\infty(U, TM)$. Then

$$\nabla s_j = \omega_j^i \otimes s_i \qquad \forall\, i, j = 1, \dots, \dim M$$

where $\omega_j^i = (s^*\omega)_j^i$ (see for instance Proposition 8.4.5 in [6]). On the other way around, an affine connection ∇ on TM is said to be compatible with a G-structure $\mathscr{F}_G \subset \mathrm{Fr}(TM)$ if along any curve $c : [0,1] \to M$ the parallel transport with respect to ∇ yields isomorphisms $T_{c(0)}M \cong T_{c(1)}M$ and $\mathscr{F}_{G,c(0)} \cong \mathscr{F}_{G,c(1)}$. Then the connection matrix $(\omega_j^i) \in \Omega^1(U; \mathfrak{gl}(V))$ for ∇, which is defined with respect to a local frame field (s_i) on U in TM by $\nabla s_i = \omega_j^i$, actually takes value in $\mathfrak{g} \subset \mathfrak{gl}(V)$, and it induces a principal connection $\hat{\omega} \in \Omega^1(\mathrm{Fr}(TM); \mathfrak{gl}(V))$ on $\mathrm{Fr}(TM)$ such that $\omega = \hat{\omega}|_{\mathscr{F}_G} \in \Omega^1(\mathscr{F}_G; \mathfrak{g})$ is a principal connection on $\mathscr{F}_G$ locally represented by (ω_j^i) on U. So we see the one-to-one correspondence between G-structure compatible affine connections ∇ on TM and principal connections ω on $\mathscr{F}_G$. Dually from the viewpoint of local coframe fields, we relate the local coframe dual to the local frame with the principal connection via the following so-called Cartan's first structure equation:

Proposition 0.23. *Let $(\theta^i) : U \subseteq M \to \mathscr{F}_G$ be a local coframe field, $\omega \in \Omega^1(\mathscr{F}_G; \mathfrak{g})$ is a connection 1-form if and only if it satisfies the structures equation*

$$d\theta^i = -\omega_j^i \wedge \theta^j + T_{jk}^i \theta^j \wedge \theta^k$$

where $(\omega_j^i) \in \Omega^1(U; \mathfrak{g})$ and with $T_{jk}^i = -T_{kj}^i$.

If we pick $v_i \in V$ and $v^j \in V^*$ a basis and a dual basis, $\theta = \theta^i v_i$ defines a *solder form* on G-bundle $\mathscr{F}_G$ which is a G-equivariant horizontal 1-form. Then we have $\omega_j^i v_i \otimes v^j \in \Omega^1(U; \mathfrak{g})$ representing connection $\omega \in \Omega^1(\mathscr{F}_G; \mathfrak{g})$ locally, where we have used $\mathfrak{g} \subset \mathfrak{gl}(V) \subset \mathrm{End}(V) \cong V \otimes V^*$. Similarly, $(T_{jk}^i) = T_{jk}^i v_i \otimes v^j \wedge v^k \in C^\infty(U, V \otimes \Lambda^2 V^*)$ also determines a global quantity $T^\omega \in C^\infty(\mathscr{F}_G, V \otimes \Lambda^2 V^*)$ called the *apparent torsion* or simply *torsion* of the connection ω. Then the structure equation can also be written globally ([6] Proposition 8.3.3):

$$d\theta = -\omega \wedge \theta + \frac{1}{2} T(\theta \wedge \theta).$$

Back to affine connection ∇ on TM, we have also notion of torsion for any ∇ defined as

$$T^\nabla(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y] \qquad \forall X, Y \in C^\infty(M, TM).$$

When ∇ is compatible to G-structure, then we can actually identified T^∇ with T^ω for corresponding ω by expressing T^∇ in terms of local frame and matching the components T_{ij}^k with that of T^ω.

The curvature 2-form is given by the Cartan second structure equation discussed earlier:

$$\Omega_j^i = d\omega_j^i + \omega_k^i \wedge \omega_j^k.$$

We now introduce a very important notion for G-structure, namely its integrability or flatness. We follow the definition in [8] §3.

Definition 0.24. *A G-structure is said to be locally flat or integrable if for any $x \in M$ there exists local coordinate $(x^1, \ldots, x^n)$ on a neighborhood of x such that $(dx^1, \ldots, dx^n)$ is a local section of $\mathscr{F}_G \subset Fr(T^*M)$ respectively $(\frac{\partial}{\partial x_1}, \ldots, \frac{\partial}{\partial x_n})$ is a local section of $\mathscr{F}_G \subset Fr(TM)$. This kind of coordinate chart is called adapted coordinate chart while the local section is called adapted coframe respectively adapted frame.*

If there exits a global adapted frame field exists, which means there is a global section $M \to \mathscr{F}_G$, consists of global vector fields on TM that are pairwise commuting under Lie bracket. This globally flat scenario by Frobenius theorem is evidently describing an integrable distribution on M. An integrable G-structure locally looks like a certain integrable distribution in TM. Therefore, we see the motivation underlying the term "integrability".

Proposition 0.25 ([7] Proposition 1.2)**.** *If a G-structure on M is integrable, then it admits a torsion-free connection.*

Proof. Let U be a adapted chart neighborhood. On each such U we can always construct a locally flat principal connection ω_U inducing flat affine connection ∇_U such that $\frac{\partial}{\partial x_1}, \ldots, \frac{\partial}{\partial x_n}$ are parallel with respect to the affine connection on U. We cover M with adapted charts $\{U_i\}_i$ and glue these local connections together via partition of unity $\{\rho_i\}_i$ subordinating to the open cover, i.e. $\nabla = \sum_i \rho_i \nabla_i$, and by checking easily that locally $T^{\nabla}(X, Y) = 0 \ \forall X, Y \in TM$ with respect to any $\rho_i \nabla_i + \rho_j \nabla_j$ in $U_i \cap U_j \neq \varnothing$, we obtain the desired connection. □

Remark. The existence of torsion free connection is the necessary condition for the G-structure to be integrable. The converse of this statement is in general not true, namely this condition is not a sufficient condition. By the way we construct this connection, not only is its torsion vanishing but also its curvature vanishes. Comparing to integrability of affine connection on M: an affine connection ∇ on M is said to be integrable if around each point in M one can find a coordinate chart $(x_1, \ldots, x_n)$ such that $\frac{\partial}{\partial x_i}$ are all flat, i.e. $\nabla_X \frac{\partial}{\partial x_i} = 0$ for any vector field X on M. An affine connection is integrable if and only if its curvature and torsion vanish. For G-structure, the compatible affine connection is integrable implies the principal connection is flat and torsion free.

We have following weaker flatness for G-structures for finer characterization:

Definition 0.26. *A G-structure is said to be 1-flat if the torsions for all connections are vanishing, and it is said to be 2-flat if it admits connection with both torsion and curvature vanishing.*

In fact, all the obstructions to flatness for a G-structure lie in the so-called Spencer cohomology groups $H^{k,2}(\mathfrak{g})$ of the Lie algebra $\mathfrak{g}$. To check whether a G-structure is flat or not, we merely need to calculate the invariants taking value in those $H^{k,2}(\mathfrak{g}) \neq 0$. This leads to a hierarchy of k-torsion and k-flatness, c.f. [8]. For instance, the torsion T^{ω} for a connection ω on $\mathscr{F}_G$ introduced above is a representative of the so-called 1-torsion $[T^{\omega}] \in C^{\infty}(\mathscr{F}_G, H^{0,2}(\mathfrak{g}))$ of the G-structure, it is independent of ω and the G-structure is 1-flat if its 1-torsion vanishes, c.f. [6] §8.3. A precise characterization of integrability of G-structure in terms Spencer cohomology is found by Victor Guillemin, c.f. [8] Theorem 4.1 and Theorem 5.1.

Examples.

(1) A $\mathrm{GL}(n, \mathbb{R})$-structure on M is nothing but a frame bundle $\mathrm{Fr}(TM)$ which is clearly integrable.

(2) A $\mathrm{GL}^+(n,\mathbb{R})$-structure on a manifold can exists if and only if M is orientable. This can be viewed as an equivalent definition orientation on M. In such case, we know at any point in M, we can find a coordinate chart $(x_1,\ldots,x_n)$ around that point such that the orientation is induced by $\frac{\partial}{\partial x_1},\ldots,\frac{\partial}{\partial x_n}$ hence imply integrability. And the space of orientations is $\mathrm{GL}(n,\mathbb{R})/\mathrm{GL}^+(n,\mathbb{R})=\mathbb{Z}_2$.

(3) The $\mathrm{SL}(n,\mathbb{R})$-structure is the volume forms on M. This can be seen by the $\mathrm{GL}(n,\mathbb{R})$-action on $\Lambda^n V$: $A\nu=\det(A)\nu$ for $A\in\mathrm{GL}(n,\mathbb{R})$ and $\nu\in\Lambda^n V$ $(V=\mathbb{R}^n)$. This action is transitive on $\Lambda^n V\backslash\{0\}$ with isotropy subgroup $\mathrm{SL}(n,\mathbb{R})$. Therefore $\Lambda^n V\backslash\{0\}\cong\mathrm{GL}(n,\mathbb{R})/\mathrm{SL}(n,\mathbb{R})$ is the space of volume forms. This G-structure is integrable since at any point in M there is always certain coordinate $(x_1,\ldots,x_n)$ such that the volume form can be expressed locally as $dx_1\wedge\cdots\wedge dx_n$.

(4) One of the most studied G-structure is $\mathrm{O}(n)$-structure, which is also known as Riemannian structure since the group $\mathrm{O}(n)$ is formed by orthogonal frame with respect to the given Riemannian metric g on M. Using moving frame method, one can derive the Levi-Civita connection (the unique torsion free connection compatible with the Riemannian metric) and Riemannian curvature, c.f. [1] Chapter 7. Riemannian curvatures are the local differential invariant to tell whether the Riemannian metrics are locally isometric or not. An $\mathrm{O}(n)$-structure is integrable if and only if the Riemannian curvature vanishes, i.e. the Levi-Civita connection is flat. In this case, one also say the Riemannian metric is flat or integrable. Then locally one can always find coordinate $(x_1,\ldots,x_n)$ such that $g=\sum_{i=1}^n dx_i\otimes dx_i$.

In the language of G-structure, we expect to detect local equivalence of some geometric structures via these local differential invariants derived by moving frame method. For the our main interest in this thesis, the concerning geometric structures will be the CR structures.

1 Symplectic, contact and complex manifolds

In this section, we give some very narrow and preliminary descriptions on three types of manifolds, i.e. symplectic, contact and complex manifolds. Basic definitions, properties and common examples will be presented, and their relations will be discussed. We will see how some very first Cauchy-Riemann manifolds, before we make formal definitions, arise naturally in the intersection of these categories. The contents refer mainly to [10],[11],[12].

1.1 Symplectic manifolds

Definition 1.1. *Let M be a real $2n$-dimensional smooth manifold. A symplectic structure on M is a closed non-degenerate 2-form $\omega \in \Omega^2(M)$ (i.e. at each point $x \in M$, $\forall\, u \neq 0 \in T_xM$, $\exists\, v \neq 0 \in T_xM$: $\omega_x(u,v) \neq 0$). This is equivalent to: $d\omega = 0$ and $\omega^n \not\equiv 0$ is a volume form on M. The pair (M,ω) is called a symplectic manifold.*

Given a smooth manifold, if there exists a symplectic structure, its non-degeneracy enforces the dimension of the manifold to be even. Since ω^n is a volume form, a symplectic manifold (M,ω) is oriented by its symplectic structure. If M is further compact, one can show using Stokes' theorem that $[\omega^n] \neq 0 \in H^{2n}_{\mathrm{dR}}(M) \cong H^{2n}(M;\mathbb{R})$.

Examples.

(1) The Euclidean space $\mathbb{R}^{2n}$ with the standard symplectic form which in terms of coordinates $(x_1,\dots,x_n,y_1,\dots,y_n) \in \mathbb{R}^{2n}$ reads $\omega_{\mathrm{std}} = \sum_i^n dx_i \wedge dy_i$ is the prototype of symplectic manifolds. It is also called the standard symplectic $\mathbb{R}$-vector space.

(2) On the cotangent bundle T^*M of a given smooth real n-manifold Q, there is a tautological or Liouville 1-form defined pointwise: On each point $x = (q,p) \in T^*Q$, $\lambda_{\mathrm{std}}|_x := (d\,\mathrm{pr}_x)^* p$ where $p \in T^*_qQ$, projection map $\mathrm{pr} : T^*Q \to Q$, pushforward $d\,\mathrm{pr}_x : T_x(T^*Q) \to T_qQ$ and pullback $\mathrm{pr}_x^* : T^*_qQ \to T^*_x(T^*Q)$. For any $v \in T_x(T^*Q)$ we have $\lambda_{\mathrm{std}}(v) = p(d\,\mathrm{pr}_x(v))$ while for any 1-form $\alpha : Q \to T^*Q$ we have $\alpha^*\lambda_{\mathrm{std}} = \alpha$. On a local coordinate chart $(q_1,\dots,q_n,p_1,\dots,p_n)$, it can be written as $\lambda_{\mathrm{std}} = \sum_{i=1}^n p_i dq_i$. This leads to the standard/canonical symplectic 2-form $\omega_{\mathrm{std}} = d\lambda_{\mathrm{std}}$ and locally in coordinate chart as $\omega_{\mathrm{std}} = \sum_{i=1}^n dp_i \wedge dq_i$.

(3) Take $\mathbb{S}^2$ as the unit sphere around the origin in $\mathbb{R}^3$. Regard a point x on the 2-sphere as a unit normal vector orthogonal to the tangent plane $T_x\mathbb{S}^2$ with respect to the Euclidean inner product $\langle\cdot,\rangle$ in $\mathbb{R}^3$. The standard symplectic 2-form on 2-sphere is defined pointwise via Euclidean inner product and exterior product as: $\omega_x(u,v) := \langle x, u \times v\rangle\ \forall\, u,v \in T_x\mathbb{S}^2 \subset \mathbb{R}^3$. The form is closed since it is a top form on $\mathbb{S}^2$, and it is clearly non-degenerate.

Remark. In above examples, (1), (2) are non-compact symplectic manifolds with exact symplectic 2-forms, hence they are the so-called exact symplectic manifold. One can check conversely by using Stokes' theorem again that any exact symplectic manifold can not be closed manifold.

(3) is a closed symplectic manifold, and in fact it is the only symplectic sphere: Any sphere $\mathbb{S}^{2n}$ with $n \geq 2$ can not be a symplectic manifold. This is a straightforward consequence of the fact mentioned above that ω^n as a cohomology class is non-zero, while $H^k(\mathbb{S}^n;\mathbb{R}) = 0$ for $k > n$ and thus $H^{2n}(\mathbb{S}^n;\mathbb{R}) = 0$ for $n \geq 2$.

Definition 1.2. *A diffeomorphism F between symplectic manifolds (M_1,ω_1) and (M_2,ω_2) is called a symplectomorphism if $F^*\omega_2 = \omega_1$. A symplectic isotopy is an isotopy through symplectomorphisms.*

If F is a local diffeomorphism (i.e. defined on a open set or open neighborhood of an arbitrary point) satisfying the condition locally (i.e. symplectic forms restricting to corresponding open sets or neighborhoods), then it is naturally called local symplectomorphism.

The theorem below is very important and it is the starting point of symplectic geometry and topology.

Theorem 1.3 (Darboux)**.** *Let (M, ω) be a symplectic manifold. For each point $x \in M$, there is a coordinate chart $(U, (x_1, \dots, x_n, y_1, \dots, y_n))$ around that point on which $\omega = \sum_{i=1}^{n} dx_i \wedge dy_i$.*

The proof of Darboux's theorem uses Moser's trick to build a symplectic isotopy to bring any symplectic structure to the standard one, c.f. [10] Chapter 7 and 8.

Due to Darboux's theorem, we can naturally interpret symplectic structure on a $2n$-manifold as an integrable G-structure with $G = \mathrm{Sp}(n, \mathbb{R})$. The closedness $d\omega$ is actually the integrability condition, and dropping this we will attain the *almost symplectic structure* as a general $\mathrm{Sp}(n, \mathbb{R})$-structure (c.f. [7] §2 Example 2.8).

The most significant outcome of this theorem is that all symplectic manifolds locally looks exactly the same: They are all locally symplectomorphic to $(\mathbb{R}^{2n}, \omega_{\mathrm{std}})$. In other words, symplectic manifolds are model on the standard symplectic vector space $(\mathbb{R}^{2n}, \omega_{\mathrm{std}})$. The standard symplectic structure on Euclidean space is therefore said to be the normal form of general symplectic structures. There is no local geometric invariant, such as curvature, for symplectic manifolds.

1.2 Contact manifolds

Definition 1.4. *Suppose M is a real $2n+1$ dimensional smooth manifold. A contact structure on M is a smooth hyperplane distribution $\xi \subset TM$ such that its locally defining 1-form θ called contact form (i.e. $\xi = \ker(\theta)$ locally for some 1-form θ around each x) yields that $d\theta|_\xi$ is non-degenerate (i.e. ξ is a symplectic vector bundle over M). The pair (M, ξ) is then called a contact manifold.*

Similarly, the non-degeneracy of $d\theta|_\xi$ enforces the dimension of underlying manifold to be odd: At each point $x \in M$, we can split the tangent bundle into $T_xM = \ker(\theta_x) \oplus \ker(d\theta_x)$, since $d\theta_x|_{\xi_x}$ is non-degenerate, by linear algebra $\dim \xi_x = 2n$ and $(d\theta_x)^n|_{\xi_x} \neq 0$, hence $\theta_x \wedge (d\theta_x)^n \neq 0$ and $\dim M = 2n+1$. By Frobenius theorem (in terms of differential forms[1], this hyperplane distribution is maximally non-integrable, whereas "maximally" is a literal meaning for non-degeneracy of $d\theta|_\xi$.

If θ is a global 1-form, then $\theta \wedge (d\theta)^n \neq 0$ is a volume form on M. A global contact form exists if and only if the quotient bundle TM/ξ, which is isomorphic to conormal bundle of the hyperplane distribution

$$N^*(\xi) = \bigsqcup_{x \in M} \{\nu \in T_x^*M \,|\, \xi_x \subseteq \ker(\nu)\},$$

[1]Recall (c.f. [9] Chapter 1 and [1] Chapter 7) for a k-dimensional distribution on a n-dimensional smooth manifold M denoted as $\mathscr{D} \subset TM$ $(k \leq n)$, the subring

$$\mathcal{I} := \bigoplus_{l \in \mathbb{N}_0} \{\omega \in \Omega^l(M) \,|\, \omega(X_1, \dots, X_l) = 0 \; \forall X_1, \dots, X_l \in C^\infty(M, \mathscr{D})\} \subset \Omega^\bullet(M)$$

is in fact an ideal. Locally it is spanned by $n-k$ independent 1-forms $\alpha^1, \dots, \alpha^{n-k}$, i.e. $\mathscr{D}_x = \bigcap_{i=1}^{n-k} \ker(\alpha^i(x)) \; \forall x \in M$. If $\mathscr{D}$ is involutive (hence integrable by Frobenius theorem), then $\mathcal{I}$ is a differential ideal, i.e. $d\mathcal{I} \subset \mathcal{I}$, and locally there is a coordinate system $(x^1, \dots, x^n)$ such that $\mathcal{I}$ is spanned by $dx^1, \dots, dx^{n-k}$. Hence $d\alpha^i \equiv 0 \bmod \{\alpha^1, \dots, \alpha^{n-k}\}$ or equivalently there exist 1-forms $\{\omega_j^i\}$ such that $d\alpha^i = \sum_{j=1}^{n-k} \omega_j^i \wedge \alpha^i$, $1 \leq i \leq n-k$.

is orientable. Due to dimension reason, it is a real line bundle and orientability means it is a trivial line bundle, then a global section of $N^*(\xi)$ is a global contact form. The orientation of this real line bundle then orients M as well. In this case, we say the contact manifold is coorientable.

Examples.

(1) On Euclidean space $\mathbb{R}^{2n+1}$, there is a standard contact form which reads $\theta_{\text{std}} = dz - \sum_{i=1}^{n} x_i dy_i$ under the coordinate $(x_1, \ldots, x_n, y_1, \ldots y_n, z) \in \mathbb{R}^{2n+1}$. The standard contact structure is $\xi_{\text{std}} = \ker(\theta_{\text{std}})$. In 3-dimensional case, we have $(\mathbb{R}^3, \xi_{\text{std}} = \ker(xdy + z))$ and $\theta_{\text{std}} \wedge d\theta_{\text{std}} = dx \wedge dy \wedge dz \neq 0$ is the standard volume form on $\mathbb{R}^3$. The contact hyperplane at a point (x, y, z) is

$$\xi_{\text{std}}|_{(x,y,z)} = \left\{ X = a\frac{\partial}{\partial x} + b\frac{\partial}{\partial y} + c\frac{\partial}{\partial z} \,\middle|\, \theta_{\text{std}}(X) = bx + c = 0 \right\}$$

If we take cylindrical coordinate $(r_1 \cos\vartheta_1, \ldots, r_n \cos\vartheta_n, r_1 \sin\vartheta_1, \ldots, r_n \sin\vartheta_n, z) \in \mathbb{R}^{2n+1}$, then the standard contact form becomes the rotation one $\theta_{\text{rot}} = dz - \sum_{i=1}^{n}(y_i dx_i - x_i dy_i) = dz + \sum_{i=1}^{n} r_i^2 d\vartheta_i$. The coordinate change preserves the contact structure.

(2) For every $k \in \mathbb{N}$, $(\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3, \xi_k = \ker(\theta_k))$ is a contact 3-manifold, where $\theta_k = \cos(ky)dz - \sin(ky)dx$. When $k = 1/2$, $(\mathbb{T}^3, \ker(\theta_{1/2}))$ is an example of non-coorientable contact manifolds.

Definition 1.5. *Let (M_1, ξ_1) and (M_2, ξ_2) be two contact manifolds. A diffeomorphism $F : M_1 \to M_2$ is a contactomorphism if $F_*(\xi_1) = \xi_2$. A contact isotopy is an isotopy through contactomorphisms, i.e. a smooth 1-parameter family of contactomorphisms F_t so that $F_0 = id$ and $(F_t)_*\xi_1 = \xi_2$ for all t.*

We see from above example (1) that the coordinate change in $\mathbb{R}^{2n+1}$ is a contactomorphism.

A smooth vector field $X \in C^\infty(M, TM)$ is called a *contact vector field* if $[X, Y] \in C^\infty(M, \xi)$ for all $Y \in C^\infty(M, \xi)$. A contact vector field is not a vector field tangent to the contact hyperplane ξ, instead on an open set it always transverses to ξ by the non-degeneracy/non-integrability.

Proposition 1.6. *Let φ_t be a contact isotopy on a contact manifold (M, ξ), then $X_t = \frac{d\varphi_t}{dt} \circ \varphi_t^{-1}$ is a contact vector field $\forall\, t \in [0, 1]$. Conversely, if X_t is a smooth family of contact vector fields, then its flow φ_t is a contact isotopy.*

Proof. Suppose φ_t is a contact isotopy and $Y \in C^\infty(M, \xi)$. Then $(\varphi_t)_* Y_x \in \xi_x$ at each point $x \in M$ and $(\varphi_{t+\varepsilon})_* Y_x - (\varphi_t)_* Y_x \in \xi_x$. Taking limit $\varepsilon \to 0$ we obtain Lie derivative

$$L_{\frac{d\varphi_t}{dt}}(Y \circ \varphi_t) = [\frac{d\varphi_t}{dt}, Y \circ \varphi_t] = [X_t, Y] \circ \varphi_t \in \xi$$

where we put $X_t = \frac{d\varphi_t}{dt} \circ \varphi_t^{-1}$. Thus $[X_t, Y] \in C^\infty(M, \xi)$ implies X_t is a contact vector field.

Conversely assume X_t is a given contact vector field, its flow φ_t is defined by equation

$$\frac{d\varphi_t}{dt} = X_t \circ \varphi_t.$$

At arbitrary point $x \in M$, given a vector field $Y \in C^\infty(M, \xi)$, define $c : [0, 1] \to (TM/\xi)_x$ by $c(t) = (q \circ (\varphi_t)_*)(Y_x)$ where $q : TM \to TM/\xi$ is the bundle quotient map. Then $c(0) = 0$ and $\frac{dc(t)}{dt} = q([X_t, Y]) = 0$. This implies $c(t) = 0$ for all t, hence φ_t is a contact isotopy. □

Applying this proposition, one can show the following theorem:

Theorem 1.7. *$T_{id}Cont(M,\xi)$ is isomorphic to $C^\infty(M,TM/\xi)$ as vector space via the restriction of quotient map $C^\infty(M,TM)\to C^\infty(M,TM/\xi)$.*

We denote $\mathrm{Cont}(M,\xi)$ as the contactomorphism group of contact manifold (M,ξ). Its tangent fiber at id is the space of contact vector fields. In this sense, we also says contact vector fields generate infinitesimal contactomorphisms.

Corollary 1.8 (isotopy extension)**.** *Let (M,ξ) and (N,ξ') be two contact manifolds of same dimension. Let $F_t: N\hookrightarrow M$ be a smooth path of contact embeddings. Then there is a supported contact isotopy φ_t on M such that $\varphi_t\circ F_0 = F_t$, i.e. φ_t is be identity outside any open neighborhood $U_t\subseteq M$ of $\overline{F_t(N)}$.*

Proposition 1.9. *Let (M,ξ) be a contact manifold with contact form θ. A diffeomorphism $F\in Diff(M)$ is a contactomorphism if and only if $F^*\theta = f\theta$ for some nowhere vanishing function $f\in C^\infty(M)$. A vector field $X\in C^\infty(M,TM)$ is a contact vector field if and only if $L_X\theta = g\theta$ for some (possibly zero) function $g\in C^\infty(M)$.*

Proof. Let $Y\in C^\infty(M,\xi)$. Rescaling the contact form θ with $f\not\equiv 0\in C^\infty(M)$ does not change the contact structure ξ. By definition of contactomorphism and $(F^*\theta)(Y) = \theta(F_*Y)$, the first statement is clear.

For the second statement, we compute:

$$0 = L_X(\theta(Y)) = (L_X\theta)(Y) + \theta([X,Y]).$$

X being contact means $\theta([X,Y]) = 0$, it follows that $\xi = \ker(\theta)\subseteq \ker(L_X\theta)$. Therefore we can write $L_X\theta = g\theta$ for some $g\in C^\infty(M)$ allowing $g = 0$. Conversely, the computation would leads to $\theta([X,Y]) = 0$ and by definition X is a contact vector field. □

The contact form θ actually induces an isomorphism $\theta: T_{\mathrm{id}}\mathrm{Cont}(M,\xi)\to C^\infty(M)$ of vector spaces. The *Reeb vector field* $\mathcal{R}$ corresponding to θ is defined to be the contact vector field sent to $1\in C^\infty(M)$ under this isomorphism. So we have the following defining properties for Reeb vector field:

$$\begin{cases}\theta(\mathcal{R}) = 1\\ i_{\mathcal{R}}d\theta = 0\end{cases}$$

Similar to Hamiltonian vector field on symplectic manifold, there is also contact Hamiltonian vector field associated to a smooth function $f\in C^\infty(M)$ given by $X_f = Y_f - f\mathcal{R}$ with Y_f satisfying:

$$\begin{cases}\theta(Y_f) = 0\\ i_{Y_f}d\theta \equiv df \bmod \theta\end{cases}$$

This is clearly a contact vector field by proposition 1.9, and due to the isomorphism above we know that all infinitesimal contactomorphisms can be generated from contact Hamiltonian vector fields.

Theorem 1.10 (Gray's stability)**.** *Let M be a smooth manifold and $\{\xi_t\}_{t\in[0,1]}$ be a compact supported homotopy of contact structures on M, i.e. $\xi_t = \xi_0$ for all t outside a compact subset in M. Then there exists a smooth isotopy $\varphi_t: M\to M$ so that $(\varphi_t)_*\xi_0 = \xi_t$.*

Proof of Gray's stability also involves Moser's trick constructing the isotopy as the flow of a vector field. C.f. [10] Chapter 10 and [11] Chapter 8. Using this theorem, one can show the contact version of Darboux's theorem:

Theorem 1.11 (Darboux's theorem)**.** *Let (M, ξ) be a $2n+1$-dimensional contact manifold with contact form θ. At every point $x \in M$, there is a coordinate chart $(U, (x_1, \ldots, x_n, y_1, \ldots, y_n, z))$ around that point on which the contact form can be expressed as $\theta = dz - \sum_{i=1}^{n} x_i dy_i$.*

Proof. Pick an open neighborhood $U \subseteq M$ of $x \in M$, and find a contact embedding $F : (U, \xi|_U) \hookrightarrow (\mathbb{R}^{2n+1}, \xi_{\text{std}})$, i.e. $F(x) = 0$ and $F^*\theta_{\text{std}}|_x = \theta_x$. Define $\theta_t := (1-t)\theta + tF^*\theta_{\text{std}}$. By making U small enough, we can assume $\ker(\theta_t)$ is a contact structure on U for all $t \in [0,1]$ and $\mathcal{R}_t$ has no closed orbit in U.

Now apply Gray's stability: Suppose we have an smooth isotopy φ_t so that $\varphi_t^*\theta_t = \theta$. Differentiating this gives:

$$\varphi_t^*(L_{X_t}\theta_t + \dot{\theta}_t) = 0 \tag{1}$$

where $X_t = \frac{d\varphi_t}{dt} \circ \varphi_t^{-1}$. By Cartan's formula

$$i_{X_t} d\theta_t + d i_{X_t}\theta_t + \dot{\theta}_t = 0.$$

We put $X_t = Y_t + g_t\mathcal{R}_t$ where Y_t is tangent to ξ and g_t is some function. Then the equation (1) becomes

$$i_{Y_t} d\theta_t + dg_t + \dot{\theta}_t = 0. \tag{2}$$

Inserting $\mathcal{R}_t$ into the 1-forms from left hand side gives

$$\mathcal{R}_t(g_t) + \dot{\theta}(\mathcal{R}_t) = 0$$

Since there is no closed Reeb orbit of $\mathcal{R}_t$ in U, we can solve the equation (2) for g_t by integrating along a Reeb orbit $\gamma : s \mapsto \gamma(s) \in U$ of $\mathcal{R}_t$ ($\frac{d\gamma(s)}{ds}\big|_{s=s_0} = \mathcal{R}_t(\gamma(s_0))$):

$$g_t = -\int_\gamma \dot{\theta}_t(\mathcal{R}_t)\, ds.$$

We see g_t is defined up to constant on all Reeb orbits. Therefore we have freedom to choose that $g_t = 0$ and $(dg_t)_x|_{\ker(\theta_t)} = 0$. By the equation (2), $(dg_t)_x(\mathcal{R}_t) = -(\dot{\theta}_t)_x(\mathcal{R}_t)$, since $\dot{\theta}_t|_x = \theta_x - \theta_x = 0$, we have $(dg_t)_x = 0$ on entire T_xU. Plugging this result back to (2) and restricting the equation to $\ker(\theta_t)$, we know that there exists a unique solution Y_t by the contact condition, and notice that $Y_t|_x = 0$. Thus we find a contact vector field X_t solving

$$L_{X_t}\theta_t + \dot{\theta}_t = 0.$$

We would like to taker the flow φ_t of X_t, but X_t is not complete in general, therefore φ_t is not an isotopy on U. At any point $y \in U$, however, we can still define the integral curve $\varphi_t(y) \in U$ existing for short time $t \in [0, T(y))$. Since $X_t|_x = 0$ for all t, the flow $\varphi_t(x)$ is constant and therefore defined for all $t \in [0,1]$. The set $\{(t,y) : 0 < t < T(y)\} \subseteq [0,1] \times U$ is the domain where the flow is defined, and there exists some open subset $U' \subseteq U$ so that $\varphi_t : U' \to U \subseteq M$ is defined for all $t \in [0,1]$. We put $\psi := F \circ \varphi_1 : U' \to \mathbb{R}^{2n+1}$, this is the desired chart diffeomorphism. □

Remark. The proof of contact Darboux's theorem relies heavily on that there is no closed Reeb orbit in the neighborhood, namely this is a local treatment. So this strategy can not be used to proof Gray's stability, which is a global theorem. Indeed, there is no Gray's stability for contact forms.

Similar to symplectic structure, the standard contact structure on Euclidean space is the normal form of general contact structure. All contact manifold is modeled on $(\mathbb{R}^{2n+1}, \xi_{\text{std}} = \ker(\theta_{\text{std}}))$. Historically, contact structure was discovered by Sophus Lie in 1872 when studying certain differential equation systems. And after the pioneer work of Cartan, contact structure is recognized as one of the only four topological stable distributions on a smooth manifold, whereas the rest of three are line fields, even contact structures and Engel structures. As a common feature, they all do not possess local invariants.

It was shown in 1970's by Martinet that contact structure always exists on closed orientable 3-manifolds, and by Lutz that there is a contact structure in every homotopy class of oriented plane fields. Actually, this contact homotopy representative is a so-called *overtwisted contact structure*: A contact 3-manifold (M, ξ) contains a overtwisted disk, which is a 2-disk $\mathbb{D}_{\text{OT}}$ such that $T_x\mathbb{D}_{\text{OT}} = \xi_x$ at each point $x \in \partial\mathbb{D}_{\text{OT}}$, is called overtwisted; otherwise it is called *tight*. Tight contact structure can not be contactomorphic to overtwisted one. This is a fundamental dichotomy for contact structures in dimension 3. Eliashberg later proved that two overtwisted contact structures are isotopic if and only if they are homotopic as oriented plane fields. Combining these results:

There is a unique (up to isotopy) overtwisted contact structure on a closed oriented 3-manifold in every homotopy class of oriented plane fields.

On higher dimension, there are manifolds, such as Wu manifold SU(3)/SO(3) (a simply connected 5-manifold), that does not admits contact structure.

Contact manifolds are closely related with symplectic manifolds. We discuss next this perspective.

In the study of Hamilton system on a symplectic manifold, contact manifolds appear mostly as some energy hypersurfaces. We called a codimension 1 submanifold Σ of a symplectic manifold (M, ω) a hypersurface if there exits $f \in C^\infty(M)$ with 0 as regular value so that $\Sigma = f^{-1}(0)$. The characteristic line field $\mathscr{L}_\Sigma$ on the hypersurface is determined by the *Hamiltonian vector field* X_f of f ($i_{X_f}\omega = df$):

$$\mathscr{L}_\Sigma := \{\lambda X_f \,|\, \lambda \in \mathbb{R}\}.$$

The following definition is introduced by Alan Weinstein:

Definition 1.12. *A hypersurface Σ in a symplectic manifold (M, ω) is of contact type if there is a 1-form θ on Σ such that*

i) $d\theta = \iota^*\omega$ where $\iota : \Sigma \hookrightarrow M$ is the embedding,

ii) $\theta(X) \neq 0 \ \forall\, X \in \mathscr{L}_\Sigma \backslash \{0\}$.

Weinstein also proves that:

Lemma 1.13. *A hypersurface $\Sigma \subset M$ is of contact type if and only if there exists a Liouville vector field X ($L_X\omega = \omega$) defined on a neighborhood of Σ and transversal to Σ.*

Liouville vector fields are conformally symplectic, i.e. its flow is a symplectic dilation $\varphi_t^*\omega = e^t\omega$. If further $\omega = d\lambda$ (e.g. cotangent bundles), then $\varphi_t^*\lambda = e^t\lambda$. The Hamiltonian vector field restricted to contact type hypersurface proportional to the Reeb vector field on it. That is saying up to normalization it is the Hamiltonian dynamics is equivalent to Reeb dynamics on the contact type hypersurface.

Examples.

(1) The unit sphere $\mathbb{S}^{2n-1} \overset{\iota}{\hookrightarrow} \mathbb{R}^{2n}$ (defined by Hamiltonian function $\|x\|^2 - 1$ for $x \in \mathbb{R}^{2n}$) has contact form $\theta = \iota^*\lambda_{\text{rot}}$ where $\lambda_{\text{rot}} = \frac{1}{2}\sum_{i=1}^n (x_i dy_i - y_i dx_i)$ is a primitive of ω_{std} on $\mathbb{R}^{2n}$, which defines a Liouville vector field $X = \frac{1}{2}\sum_{i=1}^n (x_i \frac{\partial}{\partial x_i} + y_i \frac{\partial}{\partial y_i})$ transversal to $\mathbb{S}^{2n-1}$. The Reeb vector field is $\mathcal{R} = 2\sum_i (x_i \frac{\partial}{\partial y_i} - y_i \frac{\partial}{\partial x_i})$ whose orbits of its flow are the circle fibers of Hopf fibration.

(2) On cotangent bundle $(T^*Q, \omega_{\text{std}} = d\lambda_{\text{std}})$, the Liouville 1-form λ_{std} determines a Liouville vector field X which is fiberwise radial, in coordinate $(q_1, \dots, q_n, p_1, \dots, p_n)$ it is written as $X = \sum_{i=1}^n p_i \frac{\partial}{\partial q_i}$. Pick a metric on Q, the unit sphere cotangent bundle $\mathbb{S}T^*Q := \{(q,p) \in T^*Q : \|p\| = 1\}$ (the Hamiltonian function is $\|p\|^2 - 1$) with $\theta_{\text{std}} = \lambda_{\text{std}}|_{\mathbb{S}T^*Q}$ is a contact manifold. The Reeb flow is equivalent to the geodesic flow on Q.

(3) We define the full *symplectization* of a contact manifold (M, ξ):

$$S(M,\xi) := \{(x,\alpha) : x \in M \ , \ \alpha \in T_x^*M \ , \ \xi_x = \ker(\alpha)\} \subseteq T^*M.$$

Pick a specific contact form θ, then $\alpha = r\theta_x$ for some $r \in \mathbb{R}^\times$. Therefore $S(M,\xi)$ locally diffeomorphic to $U \times \mathbb{R}^\times$ on some open neighborhood $U \subseteq M$, while globally it is a $\mathbb{R}^\times$-bundle over M and isomorphic to $(TM/\xi)^\times$ (i.e. subtracting zero section). Locally $\lambda_{\text{std}}|_{S(M,\xi)} = r\theta$ for some specified contact form θ and $\omega_{\text{std}}|_{S(M,\xi)} = d\lambda_{\text{std}}|_{S(M,\xi)} = d(r\theta) = dr \wedge \theta + r d\theta$, $(\omega_{\text{std}}|_{S(M,\xi)})^n = r^{n-1} dr \wedge \theta \wedge (d\theta)^{n-1} \neq 0$. So $S(M,\xi)$ with $\omega_{\text{std}}|_{S(M,\xi)}$ is indeed a symplectic manifold. The contact manifold M can be viewed as a contact type hypersurface in $S(M,\xi)$ via the Hamiltonian function $f : x \mapsto \alpha = r\theta_x \mapsto r \in \mathbb{R}$ (here $r \in \mathbb{R}^\times$ is extended over 0), one sees $\iota^*\omega_{\text{std}} = \iota^*(r\theta) = d\theta$ and $\theta(X) \neq 0$ for all $X \in \mathscr{L}_M \backslash \{0\}$, the Liouville vector field in coordinate (x, r) is $X = r\frac{\partial}{\partial r}$. When (M,ξ) is coorientable, $S(M,\xi)$ is the disjoint union of two components: $S_+(M,\xi) \cong M \times \mathbb{R}_+$ and $S_-(M,\xi) \cong M \times \mathbb{R}_-$, which are called the positive and negative symplectization. If (M,ξ) is not coorientable, then it is connected.

Definition 1.14. *Let (M,ξ) be a cooriented closed contact manifolds of dimension $2n-1$ and (W,ω) be a $2n$-dimensional compact symplectic manifold.*

• *(W,ω) is called a weak symplectic filling of (M,ξ) if $\partial W = M$ as oriented manifold and $\omega^{n-1}|_\xi > 0$.*

• *(W,ω) is called a strongly symplectic filling of (M,ξ) if $\partial W = M$ and there exists a Liouville vector field X defined near ∂W (i.e. in a collar neighborhood) pointing transversely outwards along ∂W and satisfying $\xi = \ker(i_X\omega|_{TM})$. In such case, we say (M,ξ) is ω-convex boundary of (W,ω).*

Definition 1.15. *Let $(M_\pm, \xi_\pm)$ be a cooriented closed contact manifolds of dimension $2n-1$. A (strong) symplectic cobordism from (M_-,ξ_-) to (M_+,ξ_+) is a compact $2n$-dimensional symplectic manifold (W,ω) such that:*

i) $\partial W = M_+ \sqcup -M_-$, where $-M_-$ stands for M_- with reverse orientation.

ii) In a collar neighborhood of ∂W, there exists a Liouville vector field X for ω transversely pointing outwards along M_+ and inwards along M_-.

iii) The 1-form $\theta := i_X\omega$ restricts to $M_\pm$ as a contact form for $\xi_\pm$.

Then M_+ is called the convex boundary and M_- the concave boundary of cobordism W.

Given two (strong) symplectic cobordism with a common boundary, we can glue them together along the common boundary to obtain a new symplectic cobordism. Every closed contact manifold is symplectically cobordant to itself via the trivial symplectic cobordism $([0,1] \times M, d(e^t\theta))$ with Liouville vector field $\frac{\partial}{\partial t}$.

In terms of symplectic cobordism, we can describe a strong symplectic filling of a contact manifold (M,ξ) as a symplectic cobordism from $\varnothing$ to (M,ξ), this is a convex filling ("symplectic cup"). A concave filling ("symplectic cap") is then a symplectic cobordism from (M,ξ) to $\varnothing$.

A closed contact manifold (M, ξ) is said to be strongly/weakly symplectically fillable if it admits strong/weak symplectic filling. The following is due to Eliashberg and Gromov:

Theorem 1.16 (Eliashberg-Gromov)**.** *A weakly symplectically fillable contact 3-manifold is tight.*

This is a remarkable result in 3-dimensional contact topology. The key technique in the original proof in 1989 is Gromov's h-principle. A more modern proof was accomplished by Ozbagci and Stipsicz in 2004 involving fancy techniques such as Weinstein handle attachment in contact surgery and Seiberg-Witten invariant for symplectic 4-manifolds.

Remark. The converse of this theorem is false, the first counter example was discovered by John B. Etnyre and Ko Honda, and soon after plenty counter examples was constructed by Lisca and Stipsicz using Heegaard Floer homology.

Examples.

(1) A trivial but standard example is that $(\mathbb{S}^3, \xi_{\text{std}})$, where ξ_{std} is in fact the unique tight contact structure on it, is strongly (hence weakly) symplectically filled as $(\mathbb{D}^4, \omega_{\text{std}})$.

(2) Shown by Eliashberg in 1996, $\mathbb{T}^3$ carries tight contact structures which are weakly (but not strongly) symplectically fillable.

There are some oriented closed 3-manifolds that do not admit any symplectic filling. For example: Lisca shows that the Brieskorn sphere $-\Sigma(2,3,n)$ does not admit any symplectically fillable contact structures for $n = 3, 4, 5$.

The symplectic fillability of contact manifolds is an active research area of contact topology in both low and high dimension nowadays.

1.3 Complex manifolds

Definition 1.17. *On a $2n$-dimensional smooth manifold M, an almost complex structure is a smooth field of complex structure on tangent spaces, i.e. endomorphisms $J_x : T_xM \to T_xM$ s.t. $J_x^2 = -Id$. The pair (M, J) is then called an almost complex manifold. A submanifold Σ of almost complex manifold (M, J) is an almost complex submanifold when $JT\Sigma \subseteq T\Sigma$.*

For every almost complex manifold (M, J), we have the splitting of its complexified tangent bundle $T_{\mathbb{C}}M := TM \otimes \mathbb{C}$ into "holomorphic" and "anti-holomorphic" tangent spaces:

$$\begin{aligned} T_{\mathbb{C}}M &\cong T^{1,0}M \oplus T^{0,1}M \\ T^{1,0}M &:= \{Z \in T_{\mathbb{C}}M \,|\, JZ = +\mathbf{i}Z\} = \{X \otimes 1 - JX \otimes \mathbf{i} \,|\, X \in TM\} \\ T^{0,1}M &:= \{Z \in T_{\mathbb{C}}M \,|\, JZ = -\mathbf{i}Z\} = \{X \otimes 1 + JX \otimes \mathbf{i} \,|\, X \in TM\} \end{aligned}$$

Similar holds for complexified cotangent bundle $T^*_{\mathbb{C}}M := T^*M \otimes \mathbb{C} \cong T^{*1,0}M \oplus T^{*0,1}M$, and hence the space of complex-value differential forms $\Omega^k(M; \mathbb{C}) \cong \bigoplus_{i+j=k} \Omega^{i,j}(M)$. This gives rise to the Dolbeault complex and cohomology along with the splitting exterior differential operator $d = \partial + \bar{\partial}$.

A complex manifold by its standard definition (see [10] Definition 15.1) is a even dimensional smooth manifold with a maximal complex/holomorphic atlas. It is easy to show that every complex manifold admits a canonical almost complex structure defined pointwise: At a point $z \in M$ with complex coordinate $z_i = x_i + \mathbf{i}y_i$ we put

$$\begin{cases} J_z(\frac{\partial}{\partial x_i}) = \frac{\partial}{\partial y_i} \\ J_z(\frac{\partial}{\partial y_i}) = -\frac{\partial}{\partial x_i} \end{cases}$$

and we can show that on the overlap of any two charts, such locally defined almost complex structures coincide. This implies the almost complex structure is actually globally well-defined, and hence. A necessarily condition for an almost complex manifold to become complex is that

$$[T^{1,0}M, T^{1,0}M] \subset T^{1,0}M.$$

Consequently, complex manifold is always oriented by its complex structure. Such property is called the integrability of almost complex structure. Now we define the Nijenhuis tensor for an almost complex structure J as

$$\mathcal{N}_J(X,Y) := [JX, JY] - J[X, JY] - J[JX, Y] - [X, Y] \qquad \forall X, Y \in C^\infty(M, TM).$$

The integrability of J is then means $\mathcal{N}_J \equiv 0$. An almost complex structure induced by complex structure is of course integrable. However, the converse is a rather difficult problem.

Theorem 1.18 (Newlander-Nirenberg)**.** *Integrable almost complex structure is complex structure.*

In terms of Nijenhuis tensor, this can be equivalently phrased as:

Theorem 1.19 (Newlander-Nirenberg)**.** *An almost complex manifold (M, J) is a complex manifold if and only if $\mathcal{N}_J \equiv 0$.*

The proof of this deep theorem involves large amount of analysis of partial differential equations. As a consequence, one can compute $\mathcal{N}_J(X, JX)$ on orientable surface and deduce that $\mathcal{N}_J \equiv 0$, namely every orientable surface is a complex manifold.

Almost complex structure on real $2n$-dimensional manifold can be regarded as a $\mathrm{GL}(n, \mathbb{C})$-structure where we think of $\mathrm{GL}(n, \mathbb{C}) \subset \mathrm{GL}(2n, \mathbb{R})$. An almost complex structure is integrable $\mathrm{GL}(n, \mathbb{C})$-structure if and only if it comes from a complex structure, and the integrability condition is the vanishing of Nijenhuis tensor which happens to be an intrinsic torsion of the G-structure ($G = \mathrm{GL}(n, \mathbb{C})$-structure) that we have not defined (it involves the Spencer complex). C.f. [7] §2 Example 2.4.

Examples.

(1) The complex space $\mathbb{C}^n = (\mathbb{R}^{2n}, J)$ is a trivial complex manifold where $J(x_1, y_1, \ldots, x_n, y_n) = (-y_1, x_1, \ldots, -y_n, x_n)$ and correspondingly $J(z_1, \ldots, z_n) = \mathbf{i}(z_1, \ldots, z_n)$. Hence people often denote in this case simply $J = \mathbf{i}$.

(2) $\mathbb{S}^2$ is an almost complex manifold and a complex manifold. $\mathbb{S}^4$ is proved by Ehresmann and Hopf not an almost complex manifold. $\mathbb{S}^6$ is an almost complex manifold, whether it is complex is still an open question. $\mathbb{S}^8$ and higher spheres are not almost complex manifolds.

(3) The complex torus $\mathbb{C}^n/\mathbb{Z}^{2n}$ is a complex manifold. In 1-dimensional case, $\mathbb{C}/\Lambda$ where $\Lambda = \mathbb{Z} + \tau\mathbb{Z}$ is a lattice with $\tau \in \mathbb{H} := \{z \in \mathbb{C} \,|\, \mathrm{Im}(z) > 0\}$, this is called an elliptic curve and it is a complex manifold. Elliptic curves with different parameter τ are not biholomorphic.

(4) Complex Lie groups such as $\mathrm{GL}(n, \mathbb{C}), \mathrm{SL}(n, \mathbb{C}), \mathrm{Sp}(n, \mathbb{C})$ are complex manifolds.

The interaction of complex manifolds and symplectic manifolds leads us to the following type of manifolds.

Definition 1.20. *Let (M, ω) be a symplectic manifold. An almost complex structure on M is said to be ω-compatible if $g(X, Y) = \omega(X, JY)$ $\forall X, Y \in TM$ is a Riemannian metric on M. If there exits a integrable ω-compatible almost complex structure J on M, then the triple (M, ω, J) is called a Kähler manifold and ω is called a Kähler form.*

A well-known basic fact: The space of ω-compatible almost complex structures $\mathcal{J}(M,\omega)$ is non-empty and contractible. On a Kähler manifold, the Kähler form has general form in coordinate chart $(U, z_i = x_i + \mathbf{i}y_i)$:

$$\omega = \frac{\mathbf{i}}{2}\sum_{i,j=1}^{n} h_{ij} dz_i \wedge d\bar{z}_j \quad , \quad h_{ij} \in C^\infty(U, \mathbb{C})$$

which is a $(1,1)$-form, and non-degeneracy is equivalent to $\det(h_{ij}) \neq 0$. Let ω be a closed $(1,1)$-form on a complex manifold M, then using holomorphic version of Poincaré's lemma one can show that at each point there is a neighborhood and a local real function $\rho \in C^\infty(U, \mathbb{R})$ such that $\omega = \frac{\mathbf{i}}{2}\partial\bar{\partial}\rho$, the function ρ is called a local Kähler potential. Gluing them together via partition of unity, we obtain a global Kähler potential, and if further ρ (now defined globally) is *strictly plurisubharmonic* (i.e. on each local complex chart its Hessian $\frac{\partial^2\rho}{\partial z_i \partial \bar{z}_j}$ is positive-definite) then the Kähler form is said to be positive. For any complex submanifold of a Kähler manifold, the strictly plurisubharmonic Kähler potential pullback to a plurisubharmonic function on the submanifold along the embedding. So this implies that any complex submanifold of a Kähler manifold is also Kähler submanifold.

Examples.

(1) On $\mathbb{C}^n$ with complex coordinates $z = (z_1, \ldots, z_n)$, there is function $\rho(z) = \sum_i |z_i|^2$. Then $h_{ij} = \frac{\partial^2\rho}{\partial z_i \partial \bar{z}_j} = \delta_{ij}$, i .e. ρ is strictly plurisubharmonic. The corresponding Kähler form is $\omega = \frac{\mathbf{i}}{2}\partial\bar{\partial}\rho = \frac{\mathbf{i}}{2}\sum_i dz_i \wedge d\bar{z}_i$. Another Kähler form called the Fubini-Study form is given by $\omega_{\mathrm{FS}} = \frac{\mathbf{i}}{2}\partial\bar{\partial}(1 + |z|^2)$ where $1 + |z|^2$ is also a strictly plurisubharmonic function.

(2) The complex projective space $\mathbb{CP}^n := (\mathbb{C}^{n+1}\backslash\{0\})/\sim$, $(z_0, \ldots, z_n) \sim \lambda(z_0, \ldots, z_n)\ \forall\, \lambda \in \mathbb{C}^\times$ has also Fubini-Study form as its Kähler form via pullback of ω_{FS} on $\mathbb{C}^n$ along complex chart homeomorphisms $[z_0 : \cdots : z_n] \mapsto (\frac{z_0}{z_i}, \cdots, \frac{z_{i-1}}{z_i}, \frac{z_{i+1}}{z_i}, \ldots, \frac{z_n}{z_i})$.

(3) Every Riemann surface (1-dimensional complex manifold) is a Kähler manifold with Kähler form proportional to volume (area) form $dz \wedge d\bar{z}$.

Plurisubharmonic functions give rise to an important type of submanifolds in complex manifolds which is our main studying interest in this thesis. Eliashberg's lecture note [13] and his book with Kai Cieliebak [14] are by far the best references on the subject of symplectic geometry of plurisubharmonic functions. So we will follow the descriptions therein. Let (M, J) be a complex manifold of dimension $2n$, and let Σ be a real $2n+1$-dimensional hypersurface which is cooriented by $\rho = 0$ (ρ is for instance just a real-value function on M) so that the coorientation is given by the inward transversal direction to the domain $\{\rho \leq 0\}$. Denote $\xi := T\Sigma \cap JT\Sigma \subset T\Sigma$ as the J-invariant subbundle, then $\xi = \{X \in T\Sigma \,|\, d^{\mathbb{C}}\rho(X) = 0\}$ where $d^{\mathbb{C}}\rho(X) := d\rho(JX) = -\mathbf{i}(\partial - \bar{\partial})\rho(X)$ ($\bar{\partial} = \frac{1}{2}(d + \mathbf{i} \circ J \circ d)$). The skew-symmetric form $dd^{\mathbb{C}}\rho$ defines a symmetric bilinear form on ξ:

$$\begin{aligned} L : \xi \times \xi &\to \mathbb{R} \\ (X, Y) &\mapsto dd^{\mathbb{C}}\rho|_\xi(X, JY) \end{aligned}$$

which is called the *Levi form* of real hypersurface Σ. The Levi form L is independent of the choice of ρ up to a positive function.

Definition 1.21. *The real hypersurface Σ is called strictly pseudo-convex or strictly J-convex if its Levi form is positive definite. It is called pseudo-convex or J-convex if $L(X,X) \geq 0\ \forall\, X \in \xi$. If $L \equiv 0$, then Σ is called Levi-flat.*

Remark.

• The strictly pseudo-convexity can be equivalently described by the property: In a neighborhood of any point on Σ, the real hypersurface can be made convex by a biholomorphism. C.f. [13] Proposition 1.1. This means pseudo-convexity is a biholomorphic invariant notion.

• If (M, J) is an almost complex manifold, above definitions can still be carried out, however, $d\theta(X, JX)$ will not necessarily be symmetric in this general case. The Levi form then should be defined by symmetrization $L(X, Y) = -\frac{1}{2}(d\theta(X, JY) + d\theta(Y, JX))$.

Now write $\theta := d^{\mathbb{C}}\rho$, we see already that $\xi = \ker(\theta)$. If $d\theta > 0$ on ξ, i.e. strictly pseudo-convex case, then θ meets the definition of contact form and hence ξ is actually a contact structure on Σ. The skew-symmetric form $dd^{\mathbb{C}}\rho$ can also define an Hermitian form on (M, J):

$$H_\rho(X, Y) := dd^{\mathbb{C}}\rho(JX, Y) + \mathbf{i}dd^{\mathbb{C}}\rho(X.Y) \qquad \forall\, X, Y \in TM,$$

then we have a Kähler form and the strictly plurisubharmonicity or strictly J-convexity of ρ as discussed earlier. As we know if the symplectic form on symplectic manifold (M, ω) is given by a plurisubharmonic function, then it is a Kähler manifold. So we cam deduce via local arguments that: If Σ is a contact type hypersurface in a symplectic manifold, then there exists a ω-compatible complex structure J (i.e. integrable almost complex structure) such that Σ is strictly pseudo-convex.

The following proposition ([13] Proposition 1.3) tells us that the strictly J-convexity for functions ρ and for hypersurfaces Σ are in the same relation as the notions of convexity for functions and hypersurfaces.

Proposition 1.22. *A hypersurface Σ is strictly J-convex if and only if it can be defined as $\rho^{-1}(0)$ for a strictly J-convex function ρ on a neighborhood of Σ. if all the level-sets $\rho^{-1}(c)$ of a function ρ are strictly J-convex, then there exists a diffeomorphism on $r : \mathbb{R} \to \mathbb{R}$ such that $r \circ \rho$ is strictly J-convex. The function $\rho \in C^\infty(M, \mathbb{R})$ is strictly J-convex if the hypersurface $\{Im(z) = \rho(x) : x \in M\,,\, z \in \mathbb{C}\} \subset M \times \mathbb{C}$ is strictly J-convex.*

Examples.

(1) In $\mathbb{C}^n$, a domain $\Omega = \{\rho \leq 0\} \subset \mathbb{C}^n$ in above sense with the contact form

$$\theta = -\mathbf{i}\sum_{i=1}^{n}\left(\frac{\partial\rho}{\partial z_i}dz_i - \frac{\partial\rho}{\partial\bar{z}_i}d\bar{z}_i\right) = \sum_{i=1}^{n}\left(-\frac{\partial\rho}{\partial y_i}dx_i + \frac{\partial\rho}{\partial x_i}dy_i\right)$$

is called a strictly pseudo-convex. Its boundary is the strictly pseudo-convex hypersurface $\Sigma = \partial\Omega$. If we consider $\mathbb{C}^n$ as Kähler manifold, such domain is also Kähler manifold with Kähler form $\omega = \mathbf{i}\partial\bar{\partial}\ln K(z, z)$ and Kähler metric $h = \sum_{i,j=1}^{n}\frac{\partial^2 \ln K(z,z)}{\partial z_i \partial\bar{z}_j}dz_i \otimes d\bar{z}_j$ where $K(z, z')$ is called the Bergman kernel. This Kähler metric blows up at the boundary, and such domain is known as a domain of holomorphy. As a concrete example, we recall the unit sphere $\mathbb{S}^{2n-1} \subset \mathbb{C}^n$ is given by the zero locus of $\rho = |z|^2 - 1$, which is clearly plurisubharmonic. Visually, it is the convex boundary of a convex domain, the unit disk $\mathbb{D}^n$. Consider another function $\rho(z_1, \ldots, z_n) = \sum_{i=1}^{n}(x_i^2 - \lambda_i y_i^2)$ with $\lambda_i \in [0, 1]$ for all $i = 1, \ldots n$. This is also a strictly plurisubharmonic function, and so all its level-sets are strictly pseudo-convex.

(2) Recall that a totally real submanifold Σ in a complex manifold (M, J) is a submanifold without tangent complex lines, i.e. $T\Sigma \cap JT\Sigma = 0$. The boundary of a sufficiently small tubular neighborhood of a totally real submanifold is strictly pseudo-convex. This is true for any choice of Riemannian metric.

The strictly pseudo-convex hypersurfaces are the most frequently studied examples of Cauchy-Riemann manifolds, which will be defined and extensively discussed in the next section. We will go through some similar concepts above in an intrinsic way, namely not as submanifolds of complex manifolds. In fact, it is crucial to ask whether a Cauchy-Riemann manifold can be realized as some submanifold in a complex manifold.

Another important problem is the (local) equivalence between CR manifolds. As people have discovered, symplectic manifolds and contact manifolds has no local invariants. But CR manifolds do have local invariants, which brings the whole point of studying CR geometry via connections and curvatures like the Riemannian geometry.

2 Cauchy-Riemann manifolds

2.1 CR structures and pseudo-Hermitian structures

Definition 2.1. *A CR manifold M is a real $(2n+k)$-dimensional smooth manifold equipped with a CR structure (ξ, J) where $\xi \subset TM$ is a rank-$2n$ subbundle and $J \in C^\infty(M, End(\xi))$ is a complex structure assigned to ξ such that for any $X, Y \in C^\infty(M, \xi)$ the following conditions hold*

i) $[X,Y] - [JX, JY] \in C^\infty(M, \xi)$

ii) $\mathcal{N}_J(X,Y) = J([X,Y] - [JX,JY]) - ([JX,Y] + [X,JY]) = 0$ *(i.e. J is formally integrable)*

We say (M, ξ, J) is a CR codimension k manifold.

ξ in general can be either integrable or non-integrable. If we only assume condition i) above, (M, ξ, J) is then called an *almost CR manifold*. These terminologies are in analog to the complex manifolds: Codimension zero case is almost complex manifold (condition i) in this case is satisfied automatically), by Newlander-Nirenberg theorem, if the torsion of the almost complex structure (i.e. the Nijenhuis tensor $\mathcal{N}_J$) vanishes, the almost complex manifold is actually a complex manifold.

The above definition can be formulated in an alternative equivalent way: Let (M, ξ, J) be a CR manifold in above sense. There is a subbundle of complexified tangent bundle $T_{\mathbb{C}}M = TM \otimes_{\mathbb{R}} \mathbb{C}$ of complex rank-n defined as $\xi^{1,0}_x = \{X - \mathbf{i}JX : X \in \xi_x\}$ and similarly $\xi^{0,1}_x = \overline{\xi^{1,0}_x} = \{X + \mathbf{i}JX : X \in \xi_x\}$ (they are actually the $\mathbf{i}$ and $-\mathbf{i}$ eigenspaces of J respectively). $\xi^{1,0}$ is called the *holomorphic tangent bundle*, it is formally integrable, i.e.

$$[\xi^{1,0}, \xi^{1,0}] \subset \xi^{1,0}. \tag{3}$$

Note that $\xi^{1,0} \cap \xi^{0,1} = \{0\}$, $\xi^{1,0} \oplus \xi^{0,1} = \xi_{\mathbb{C}}$, $\mathrm{Re}(\xi^{1,0} \oplus \xi^{0,1}) = \xi$. Let $Z := X - \mathbf{i}JX$ and $\bar{Z} := X + \mathbf{i}JX$ for some $X \in \xi$. The complex structure acts on $\xi^{1,0}$ as $J \,\mathrm{Re}\, Z = -\,\mathrm{Im}\, Z$, $J \,\mathrm{Im}\, Z = \mathrm{Re}\, Z$, i.e. $JZ = \mathbf{i}Z$ for any $Z \in \xi^{1,0}$. Thus CR manifold can also be defined via the pair $(M, \xi^{1,0})$. It is important to notice that a priori $[\xi^{1,0}, \xi^{1,0}] \subset \xi^{1,0}$ does not imply the existence of any integral submanifolds since $\xi^{1,0}$ consists of complex vector fields where the usual Frobenius theorem does not apply. If we drop the formal integrability condition $[\xi^{1,0}, \xi^{1,0}] \subset \xi^{1,0}$, then we come to the almost CR structure.

From now on, we consider CR manifolds of CR codimension 1, which are contact manifolds by definition. A great amount of CR manifolds appears as certain pseudo-convex hypersurfaces. To make a proper formal notion of this for general abstract CR manifolds, we introduce the Levi form: Consider the conormal bundle $N^*(\xi)$ of the hyperplane distribution ξ. Since ξ is oriented by its complex structure, then M is orientable if and only if $N^*(\xi)$ is orientable, namely $N^*(\xi)$ admits a global non-zero section and hence is a trivial real line bundle. This implies a nowhere vanishing 1-form $\theta \in C^\infty(M, N^*(\xi))$ such that $\ker(\theta) = \xi$. Any such θ is referred as a *pseudo-Hermitian structure* on M. The Levi form is defined as

$$L_\theta(Z, \bar{Z}') := -\mathbf{i}d\theta(Z, \bar{Z}') \qquad \forall\, Z \in \xi^{1,0}, \bar{Z}' \in \xi^{0,1} \tag{4}$$

which is a Hermitian form on $\xi_{\mathbb{C}}$. This can be identified as $\mathbf{i}[Z, \bar{Z}'] \bmod \xi_{\mathbb{C}}$ on oriented CR manifold $(M, \xi^{1,0})$. It is clear that any two $\theta, \tilde{\theta}$ is related by $\tilde{\theta} = f\theta$ for some nowhere vanishing real-value function $f \in C^\infty(M)$, and one can show that $L_{\tilde{\theta}} = fL_\theta$. Note that L_θ and the $\mathbb{C}$-linear extension of bilinear form $d\theta(\cdot, J\cdot)$ on ξ to $\xi_{\mathbb{C}}$ coincide on $\xi^{1,0} \oplus \xi^{0,1}$. We do not make different terminologies and simply call them both the Levi form since one determines the other uniquely.

Definition 2.2. *If the Levi form is non-degenerate, the CR structure is called non-degenerate. If the Levi form is further positive or negative definite, then the CR structure is called strictly pseudo-convex or strictly J-convex.*

The non-zero Levi form implies the non-integrability of ξ, and hence the strictly pseudo-convex (we always pick positive definiteness as convention, while on the other case namely negative definiteness it is called pseudo-concave) CR manifold coincides with the definition of a (2n+1)-dimensional (coorientable) contact manifold: θ is the contact form while ξ is the contact structure. In particular for 3-dimensional case, the integrability condition is fulfilled automatically: Pick X, Y as a local basis spanning ξ so that $JY = X$, $JX = -Y$ implying $\mathcal{N}_J = 0$, hence condition $i)$ and $ii)$ can be neglected. Or equivalently, $\xi^{1,0}$ is a complex line bundle locally spanned by $Z = X - \mathbf{i}JX$, hence $[\xi^{1,0}, \xi^{1,0}] \subset \xi^{1,0}$. Fixing a contact form θ on the contact manifolds, recall from last section we know there is a unique nowhere zero vector field $\mathcal{R}$ called the Reeb vector field or characteristic direction on M transverse to ξ and satisfying $d\theta(\mathcal{R}, \cdot) = 0$, $\theta(\mathcal{R}) = 1$. Thus $TM = \xi \oplus \mathbb{R}\mathcal{R}$ and $T_{\mathbb{C}}M = \xi^{1,0} \oplus \overline{\xi^{1,0}} \oplus \mathbb{C}\mathcal{R}$.

Example.

(1) The most simple space is $\mathbb{R}^3$, we can regard it as a real submanifold in $\mathbb{C}^2$ determined by

$$\left\{ (z_1, z_2) = (x_1 + \mathbf{i}y_1, x_2 + \mathbf{i}y_2) \in \mathbb{C}^2 \;\middle|\; \operatorname{Im} z_2 = \frac{\mathbf{i}}{2}(z_2 - \bar{z}_2) = 0 \right\}$$

and J is the induced from standard complex structure in $\mathbb{C}^2$ via multiplication by $\mathbf{i}$. A complex vector field $\bar{Z}$ tangent to $\mathbb{R}^3$ in $\mathbb{C}^2$ should satisfy $\bar{Z}(\frac{\mathbf{i}}{2}(z_2 - \bar{z}_2)) = 0$ (similarly for Z). It is clear that $\bar{Z} = \frac{\partial}{\partial \bar{z}}$ is such one. $\xi = \operatorname{span}_{\mathbb{R}}\{\operatorname{Re}\frac{\partial}{\partial \bar{z}}, \operatorname{Im}\frac{\partial}{\partial \bar{z}}\} = \operatorname{span}_{\mathbb{R}}\{\frac{\partial}{\partial x_1}, \frac{\partial}{\partial y_1}\}$. We can also see this as $\xi = T\mathbb{R}^3 \cap JT\mathbb{R}^3$ with $T\mathbb{R}^3 = \operatorname{span}_{\mathbb{R}}\{\frac{\partial}{\partial x_1}, \frac{\partial}{\partial y_1}, \frac{\partial}{\partial x_2}\}$ and $JT\mathbb{R}^3 = \operatorname{span}_{\mathbb{R}}\{\frac{\partial}{\partial x_1}, \frac{\partial}{\partial y_1}, \frac{\partial}{\partial y_2}\}$. In this case, $[\xi, \xi] \subset \xi$, i.e. ξ is integrable.

(2) For unit $2n+1$-sphere $\mathbb{S}^{2n+1} \subset \mathbb{C}^{n+1}$, as discussed in the examples from section 1.3, we know that J is again induced from standard complex structure in $\mathbb{C}^{n+1}$, the standard contact form reads $\theta = \frac{\mathbf{i}}{2}\sum_{i=1}^{n+1}(z_i d\bar{z}_i - \bar{z}_i dz_i)$, the standard contact structure is $\xi = \ker(\theta) = T\mathbb{S}^{2n+1} \cap JT\mathbb{S}^{2n+1}$, and the Reeb vector field is obtained as $\mathcal{R} = \mathbf{i}\sum_{i=1}^{n+1}(z_i\partial_{z_i} - \bar{z}_i\partial_{\bar{z}_i}) = J\sum_{i=1}^{n+1}(z_i\partial_{z_i} + \bar{z}_i\partial_{\bar{z}_i})$ where $\sum_{i=1}^{n+1}(z_i\partial_{z_i} + \bar{z}_i\partial_{\bar{z}_i})$ is the unit normal vector field to $\mathbb{S}^{2n+1}$. In particular, on the unit 3-sphere $\mathbb{S}^3 \subset \mathbb{C}^2$, one finds complex tangent vector field

$$\bar{Z} = z_2\frac{\partial}{\partial \bar{z}_1} - z_1\frac{\partial}{\partial \bar{z}_2}$$

and

$$\begin{aligned}\xi &= \operatorname{span}_{\mathbb{R}}\{\operatorname{Re}\bar{Z}, \operatorname{Im}\bar{Z}\} \\ &= \operatorname{span}_{\mathbb{R}}\left\{ x_2\frac{\partial}{\partial x_1} - y_2\frac{\partial}{\partial y_1} - x_1\frac{\partial}{\partial x_2} + y_1\frac{\partial}{\partial y_2}\,,\; y_2\frac{\partial}{\partial x_1} + x_2\frac{\partial}{\partial y_1} - y_1\frac{\partial}{\partial x_2} - x_1\frac{\partial}{\partial y_2} \right\}\end{aligned}$$

(3) A hyperquadric in $\mathbb{C}^{n+1}$ is defined as

$$\mathbf{Q}_{p,q} = \left\{ (z_1, \ldots, z_{n+1}) \in \mathbb{C}^{n+1} \;\middle|\; \operatorname{Im} z_{n+1} = \sum_{k=1}^{p} |z_k|^2 - \sum_{j=p+1}^{p+q=n} |z_j|^2 \right\} \tag{5}$$

There is a global non-vanishing 1-form

$$\theta = \frac{1}{2}dz_{n+1} - \mathbf{i}\left(\sum_{k=1}^{p} \bar{z}_k dz_k - \sum_{j=p+1}^{p+q=n} \bar{z}_j dz_j \right) \tag{6}$$

Note that this 1-form is complex on $\mathbb{C}^{n+1}$ in general, however it turns out to be real when restricting on $\mathbf{Q}_{p,q}$, and $\theta\wedge(d\theta)^n \neq 0$ on the entire hyperquadric. This implies θ is a contact form and $\xi = \ker(\theta)$ is non-integrable, i.e. $[\xi,\xi] \not\subset \xi$. The compatible complex structure obtained by restricting the standard one from $\mathbb{C}^{n+1}$ to the hyperquadric. We have complex tangent vectors

$$\bar{Z}_k = \frac{\partial}{\partial \bar{z}_k} - \mathbf{i} z_k \frac{\partial}{\partial x_{n+1}} \quad , \quad k = 1,\dots,p$$
$$\bar{Z}_j = \frac{\partial}{\partial \bar{z}_j} + \mathbf{i} z_j \frac{\partial}{\partial x_{n+1}} \quad , \quad j = p+1,\dots,p+q = n.$$

(4) The Heisenberg group $\mathbf{H}_n = \mathbb{C}^n \times \mathbb{R}$ is a Lie group whose the underlying space is endowed with the group law: $(z,t)\cdot(z',t') = (z+z', t+t'+2\,\mathrm{Im}\langle z,z'\rangle)$ where $\langle z,z'\rangle = \sum_{i=1}^n z_i\bar{z}'_i$. There are complex tangent vector fields

$$Z_i = \frac{\partial}{\partial z_i} + \mathbf{i}\bar{z}_i \frac{\partial}{\partial t} \quad , \quad 1 \leq i \leq n, \tag{7}$$

which $\mathbb{C}$-linearly span the holomorphic tangent bundle $\xi^{1,0}$. And $[Z_i, Z_j] = 0$ for all i,j, thus $[\xi^{1,0},\xi^{1,0}] \subset \xi^{1,0}$, i.e. this is a CR structure of $\mathbf{H}_n$. There is a global non-vanishing 1-form

$$\theta = dt + \frac{\mathbf{i}}{2}\sum_{i=1}^n (z_i d\bar{z}_i - \bar{z}_i dz_i), \tag{8}$$

and $\theta\wedge(d\theta)^n \neq 0$. The corresponding Reeb vector field is $\mathcal{R} = \frac{\partial}{\partial t}$. The contact form gives the Heisenberg group a pseudo-Hermitian structure, thereby makes it a strictly pseudo-convex CR hypersurface in $\mathbb{C}^{n+1}$. The Heisenberg group can be actually identified with the boundary of the so-called Siegel domain

$$\Omega_{n+1} := \{(z,w) \in \mathbb{C}^n \times \mathbb{C} : \mathrm{Im}\, w \geq \|z\|^2\}$$

in $\mathbb{C}^{n+1}$ under the map $(z,t) \mapsto (z, t+\mathbf{i}\|z\|^2)$.

In literatures, those complex tangent vector fields (tangent to $\xi_{\mathbb{C}}$) which span $\xi^{1,0}$ or $\xi^{0,1}$ are often called *Lewy operators*.

2.2 CR diffeomorphisms and biholomorphisms

Definition 2.3 (CR maps and CR equivalence)**.**

• *A smooth map $F : M_1 \to M_2$ is a CR map with respect to CR structures $(M_1, \xi_1^{1,0})$, $(M_2, \xi_2^{1,0})$ if $dF(\xi_1^{1,0}) \subset \xi_2^{1,0}$.*

• *If for any point $p \in M_1$, there is a neighborhood $U_p \subseteq M_1$ and on U_x a diffeomorphism $F : U_p \subseteq M_1 \to V_y \subseteq M_2$, $y = F(x)$ satisfies $dF(\xi_1^{1,0}|_{U_x}) = \xi_2^{1,0}|_{V_y}$, then $(M_1, \xi_1^{1,0})$, $(M_2, \xi_2^{1,0})$ are CR equivalent. If such diffeomorphism exists globally, then it is called CR diffeomorphism.*

Proposition 2.4. *A smooth map F between CR manifolds (M_1, ξ_1, J_1), (M_2, ξ_2, J_2) is a CR map if and only if $dF(\xi_1|_x) \subset \xi_2|_{F(x)}$ and $dF \circ J_1 = J_2 \circ dF$ on ξ_x for every $x \in M$.*

Proof. For $Z \in \xi_1^{1,0}$, $dF(Z+\bar{Z}) = dF(Z) + dF(\bar{Z}) = dF(Z) + \overline{dF(Z)} \in \xi_2$, $dF(Z) \in \xi_2^{1,0}$. $dF(J_1(Z+\bar{Z})) = dF(\mathbf{i}Z - \mathbf{i}\bar{Z}) = \mathbf{i}[dF(Z) + \overline{dF(Z)}] = J_2[dF(Z) + \overline{dF(Z)}] = J_2(dF(Z+\bar{Z}))$, this implies $dF \circ J_1 = J_2 \circ dF$ on ξ.

Conversely, since any element $Z \in \xi_1^{1,0}$ can be written as $Z = X - \mathbf{i}J_1X$ where $X \in \xi$, we have $dF(Z) = dF(X) - \mathbf{i}dF(J_1X) = dF(X) - \mathbf{i}J_2dF(X)$. By assumption $dF(X) \in \xi_2$, thus $dF(X) - \mathbf{i}J_2dF(X) \in \xi_2^{1,0}$, namely $dF(Z) \in \xi_2^{1,0}$. □

Examples.

(1) We can easily tell that in the examples from previous section, $\mathbb{R}^3$ and the real 3-dimensional hyperquadric $\mathbf{Q} := \mathbf{Q}_{1,0}$ as CR manifolds in $\mathbb{C}^2$ are not locally CR diffeomorphic due to the fact ξ on $\mathbb{R}^3$ is integrable while on $\mathbf{Q}$ it is non-integrable.

(2) The Cayley transformation

$$\begin{aligned}\mathbb{S}^{2n+1}\backslash\{(0,\dots,0,1)\} &\to \mathbf{H}_n\\ (z_1,\dots,z_{n+1}) &\mapsto \left(\frac{z_1}{1+z_{n+1}},\dots,\frac{z_n}{1+z_{n+1}},\mathbf{i}\frac{1-z_{n+1}}{1+z_{n+1}}\right)\end{aligned}$$

is a (global) CR diffeomorphism from sphere with a point removed to Heisenberg group.

(3) For $\delta > 0$, the dilation map on $\mathbf{H}_n$ given by $(z,t) \mapsto (\delta z, \delta^2 t)$ is a group homomorphism and a CR isomorphism. Pullback the canonical contact form along dilation will gain a scaling factor δ^2.

Definition 2.5. *A complex-valued function f on $(M, \xi^{1,0})$ is a CR function if $\bar{Z}(f) = 0$ for any section $\bar{Z} : M \to \xi^{0,1}$. Or it is a local CR function if for any point $x \in M$, there is a neighborhood U_p such that $\bar{Z}(f) = 0$ for any local section $\bar{Z} : U_x \to \xi^{0,1}$. This is equivalent to $\bar{\partial}_M f := pr_{\xi^{0,1}}(df) = 0$ on M.*

On CR hypersurfaces in complex spaces, CR maps/functions are closely related to holomorphic maps/functions.

Lemma 2.6. *Let $M \subset \mathbb{C}^{n+1}$ be a hypersurface with induced CR structure. If f is a holomorphic function on a open set $U \subseteq \mathbb{C}^{n+1}$, then $f|_{M\cap U}$ is a CR function.*

Proof. Any $\bar{Z} \in \xi^{0,1}$ defined on $M \cap U$ extends to an anti-holomorphic tangent vector field in $T^{0,1}\mathbb{C}^{n+1}$ on the same open domain. Since f is locally holomorphic, it is annihilated by all anti-holomorphic tangent vector fields defined locally. Restricting $\bar{Z}(f) = 0$ back to $(M \cap U, \xi^{0,1}|_{M\cap U})$ yields $\bar{Z}(f|_{M\cap U}) = 0$. By definition 2.5, $f|_{M\cap U}$ is a CR function. □

Consequently, if a holomorphic function on $U\backslash(M\cap U)$ extends smoothly to M, then the extension is a CR function on U. In general, such extension is not holomorphic. However, when the hypersurface and the CR function are both real analytic, then there exits holomorphic extension.

Theorem 2.7 ([17] §3 Theorem 1)**.** *Let $M \subset \mathbb{C}^{n+1}$ be a real analytic CR hypersurface with induced CR structure and f is a locally real analytic complex-valued CR function on M. Then it is locally holomorphic function.*

Remark. As a consequence, we see the (local) CR diffeomorphisms between real analytic CR hypersurfaces in $\mathbb{C}^{n+1}$ can be typically given by (local) biholomorphisms. Comparing with the first point of the remark below definition 1.21, the result here is somewhat stronger since it says all real analytic CR hypersurfaces not just pseudo-convex (the latter case only consists of those underlying manifolds which are contact) are biholomorphic invariant.

2.3 Embeddability and fillability of CR manifolds

Definition 2.8. *If a $2n+1$-dimensional CR manifold M is CR equivalent to a hypersurface in $\mathbb{C}^{n+1}$ with induced CR structure, then M is called embeddable or realizable.*

Theorem 2.9 ([17] Lemma 8)**.** *Let $F : M_1 \to M_2 \subset \mathbb{C}^{n+1}$ be a realization of M_1. Then each component in $F = (f_1, \ldots, f_{n+1})$ is a (locally) CR functions. Conversely if $F = (f_j)$ with all components (locally) CR functions on M_1, then M_1 is CR equivalent to a hypersurface $M_2 = F(M_1) \subset \mathbb{C}^{n+1}$, hence realizable.*

The embeddability of CR manifolds is a very rigid property and it is one of the core studying topic in CR geometry. We outline some general facts mentioned at the beginning of section 1.1.6 in [18] in the following.

Theorem 2.10 (Andreotti-Hill)**.** *Any real analytic CR manifold $(M, \xi^{1,0})$ (i.e. M is real analytic manifold and $\xi^{1,0} \subset T_{\mathbb{C}}M$ is a real analytic subbundle) of type $(n, k), k \geq 1$ is locally embeddable, namely there is a neighborhood U of any point such that $(U, \xi^{1,0}|_U)$ is CR isomorphic via a real analytic CR map to a real analytic hypersurface in $\mathbb{C}^{n+k}$.*

Theorem 2.11 (Andreotti-Fredricks)**.** *Real analytic CR manifolds are globally embeddable, i.e. it is globally CR isomorphic to a CR submanifold of some complex manifold.*

Theorem 2.12 (M. Kuranishi, T. Akahori, S. Webster)**.** *Each strictly pseudo-convex CR manifolds of real dimension $2n + 1 \geq 7$ is locally embeddable in $\mathbb{C}^{n+1}$.*

The general embedding problem in 5-dimensional case is still open. The strongest result shown by Boutet de Monvel states that any compact 5-dimensional strictly pseudo-convex CR manifold admits a global CR embedding into $\mathbb{C}^n$ for some n. There exists counterexample in dimension 3. David E. Barrett in 1988 constructed concretely on 3-torus CR structures that can be CR immersed in $\mathbb{C}^2$ and also those which are not locally CR embeddable. More generally, H. Jacobowitz and F. Trèves found that analytically small perturbations of 3-dimensional embeddable strictly pseudo-convex CR manifolds are non-embeddable. The embeddability of CR manifold, e.g. the results above, is often related to solvability of certain PDEs produced from Lewy operators.

Definition 2.13. *A closed contact manifold (M, ξ) is said to be holomorphically fillable if M can be realized as the J-convex boundary of a compact complex manifold (W, J) and $\xi_x = T_xM \cap JT_xM$. In such case, we say the manifold M carries a holomorphically fillable CR structures.*

The boundaries of bounded strictly pseudo-convex domain in $\mathbb{C}^{n+1}$ or more generally in a Kähler manifold are examples of holomorphic fillable contact manifolds/CR manifolds.

Theorem 2.14 ([13] Corollary 4.2)**.** *The space of holomorphically fillable pseudo-convex CR structures on $\mathbb{S}^3$ is connected and contractible.*

Theorem 2.15 ([14] Corollary 6.27)**.** *Holomorphically fillable contact structures are weakly symplectically fillable.*

Similar to symplectic filling and symplectic cobordism, one can also discuss the relation between complex cobordism and holomorphic filling.

Theorem 2.16 (Epstein-Henkin). *If two strictly pseudo-convex CR manifolds (M_1, ξ_1, J_1) and (M_2, ξ_2, J_2) are Stein cobordant, and (M_1, ξ_1, J_1) is holomorphically fillable, then (M_2, ξ_2, J_2) is also holomorphically fillable. In particular, assuming the Stein cobordism respects the complex structure, if (M_1, ξ_1) is not Stein fillable, then (M_2, ξ_2, J_2) can not be holomorphically fillable, even if (M_2, ξ_2) is Stein fillable.*

We will not give precise definitions of Stein domain, Stein filling and Stein cobordism here. The discussion of these concepts can be found in [14]. Stein cobordism is a special case of general complex cobordism. De Oliveira presented some examples of complex cobordism (but not Stein) from non-fillable CR structures to fillable ones. This implies it is crucial to have Stein cobordism as priori setting in above statement.

2.4 Hyperquadrics and normal forms

As mentioned in section 1.2, there is no local invariant for contact structures and all contact structures have normal form which is standard contact structure of Euclidean space or the standard contact structure of unit sphere (people usually consider sphere for compactness reason). However, CR structures are not simply all equivalent to the standard one on the unit sphere, which we will call it the spherical CR structure. We will discuss spherical CR structures in more details later. In this section, we study an important example of CR manifold which is hyperquadrics and its automorphism group. The hyperquadrics in approximation sense can be understood as normal form of all strictly pseudo-convex hypersurfaces according to the seminal paper by Chern and Moser [21]. We mainly focus on 3-dimensional case. For general dimensions we refer to [21]. One can find more computational details therein, which is essentially the same comparing to the 3-dimensional case.

Consider a real hyperquadric $\mathbf{Q}$ in $\mathbb{C}^2$ determined by the equation

$$-\frac{\mathbf{i}}{2}(z_2 - \bar{z}_2) - z_1\bar{z}_1 = 0\ , \quad \text{i.e.} \quad \operatorname{Im} z_2 = |z_1|^2$$

with a nowhere vanishing global 1-form $\theta = \frac{1}{2}dz_2 - \mathbf{i}\bar{z}_1 dz_1$ (recall example (3) from section 2.1). The holomorphic tangent bundle $\xi^{1,0}_{(z_1,z_2)}$ is spanned by $\bar{Z} = \frac{\partial}{\partial \bar{z}_1} - 2\mathbf{i}z_1\frac{\partial}{\partial \bar{z}_2}$ and the contact hyperplane is

$$\xi_{(x_1,y_1,x_2,y_2)} = \operatorname{span}_{\mathbb{R}}\{\operatorname{Re}\bar{Z}, \operatorname{Im}\bar{Z}\} = \operatorname{span}_{\mathbb{R}}\left\{\frac{\partial}{\partial x_1} + 2y_1\frac{\partial}{\partial x_2}, \frac{\partial}{\partial y_1}, -2x_1\frac{\partial}{\partial x_2}\right\}.$$

This can also be given by $\xi_x = T_x\mathbf{Q} \cap JT_x\mathbf{Q}$ where $x = (z_1, z_2)$ and $J : \mathbb{C}^2 \to \mathbb{C}^2$, $(z_1, z_2) \mapsto (\mathbf{i}z_1, \mathbf{i}z_2)$. We know $\mathbf{Q}$ is a strictly pseudo-convex CR 3-manifold with contact form θ.

Now consider a unit disk $\mathbb{D}^3 = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 \leq 1\}$ and the filling of hyperquadric $\mathbb{H} = \{(z_1, z_2) \in \mathbb{C}^2 : \operatorname{Im} z_2 \geq |z_1|^2\}$, this is called the complex hyperbolic 3-space. Since $\mathbb{H}$ is non-compact, we denote $\overline{\mathbb{H}}$ for the corresponding compact object by including a complex line at infinity. This is similar to that $\mathbb{CP}^1$ is the compactification of $\mathbb{C}$ by adding a point at infinity, $\mathbb{CP}^2$ compactifies $\mathbb{C}^2$ by adding a infinity complex projective line $\mathbb{CP}^1$ to it. It is decent to consider the corresponding compactified objects in projective space, for instance think of two lines in $\mathbb{C}^2$ and merely adding points at infinity to each will not make them intersect, while in projective space any two lines intersects at one point, we will see this is useful later. The boundary is denoted as $\partial\overline{\mathbb{H}} = \overline{\mathbf{Q}}$.

Lemma 2.17. *$\mathbb{H}$ is biholomorphic to $\overset{\circ}{\mathbb{D}}{}^3$ and $\overline{\mathbf{Q}}$ is CR diffeomorphic to $\mathbb{S}^3$.*

Proof. Construct a map

$$f : \mathbb{C}^2 \to \mathbb{C}^2$$
$$(z_1, z_2) \mapsto \left(\frac{z_1}{z_2 - \mathbf{i}}, \frac{1 - \mathbf{i}z_2}{z_2 - \mathbf{i}} \right)$$

In the interior domain $\overset{\circ}{\mathbb{D}}{}^3$ and $\mathbb{H}$, this map is a biholomorphism. Extending to the boundary, there are $f(0, \mathbf{i}) = (\infty, \infty)$ and $f(0, -\mathbf{i}) = (0, 0)$. Clearly $f|_{\partial\mathbb{D}^3}$ is a diffeomorphism onto $\partial\overline{\mathbb{H}}$ and $df \circ \mathbf{i} = \mathbf{i} \circ df$. Thus f is a CR diffeomorphism between compact CR 3-manifolds $\mathbb{S}^3$ and $\overline{\mathbf{Q}}$. □

By introducing homogeneous coordinates $[\zeta_0 : \zeta_1 : \zeta_2]$ for $\mathbb{CP}^2$ and via identification $z_1 = \zeta_1/\zeta_0, z_2 = \zeta_2/\zeta_0$ one can embed $\mathbb{C}^2$ as an open subset into $\mathbb{CP}^2$ and therefore $\mathbf{Q}$ can be reexpress in terms of coordinates $(\zeta_0, \zeta_1, \zeta_2)$ of $\mathbb{C}^3 \backslash \{0\}$ as

$$\zeta_1 \bar{\zeta_1} + \frac{\mathbf{i}}{2}(\zeta_2 \bar{\zeta_0} - \zeta_0 \bar{\zeta_2}) = 0.$$

We have a complex vector $Z = (\zeta_0, \zeta_1, \zeta_2)^T$ in $\mathbb{C}^3 \backslash \{0\}$, according to above equation one can define a Hermitian inner product so that $(Z, Z) = 0$ is exactly the equation determining $\mathbf{Q}$ such that $[Z]$ is a point on $\mathbf{Q}$ except for a point $[0 : 0 : 1]$ lying in the infinite complex projective line. Including this infinity point, $\overline{\mathbf{Q}}$ is regarded as the closure or compactification of $\mathbf{Q}$ in $\mathbb{CP}^2$.

This gives rise a Hermitian structure on $\mathbb{C}^3$:

$$C = \begin{pmatrix} 0 & 0 & -\frac{\mathbf{i}}{2} \\ 0 & 1 & 0 \\ \frac{\mathbf{i}}{2} & 0 & 0 \end{pmatrix}$$

so that $(Z, Z) = Z^T C \bar{Z}$. The transformations preserving the Hermitian structure forms a group

$$\mathrm{U}(2, 1) = \{A \in \mathrm{GL}(3, \mathbb{C}) : \bar{A}^T C A = C\}.$$

The action on $\mathbb{C}^3$ given by $(\zeta_0, \zeta_1, \zeta_2)^T \mapsto A(\zeta_0, \zeta_1, \zeta_2)^T$ is transitive and induces an action on $\mathbb{C}^2 \subset \mathbb{CP}^2$ via the homogeneous coordinates. The isotropy subgroup of origin in $\mathbb{C}^2$ i.e. $\mathrm{Stab}_{\mathrm{U}(2,1)}([1 : 0 : 0])$ consists of the matrices (c.f. [17]):

$$\begin{pmatrix} \lambda & -\mathbf{i}\lambda\bar{a} & r\lambda - \mathbf{i}\lambda|a|^2 \\ 0 & \mu & \mu a \\ 0 & 0 & 1/\bar{\lambda} \end{pmatrix} \ , \ \lambda \in \mathbb{C}^* \ , \ a \in \mathbb{C} \ , \ |\mu| = 1 \ , \ r \in \mathbb{R}.$$

Clearly this subgroup does not act effectively on $\mathbb{C}^2$ because each $\lambda\mathbb{I}$ fixes every point in $\mathbb{C}^2$. Then $\mathrm{Stab}_{\mathrm{U}(2,1)}([1 : 0 : 0])/\mathbb{C}^*$ acts on $\mathbb{C}^2$ effectively and maps $\mathbf{Q}$ to itself biholomorphically with singularity at infinity on $\mathbf{Q}$ or we can say it is the automorphism group of $\overline{\mathbf{Q}}$.

Another possible way to get effective action from $\mathrm{U}(2, 1)$ on $\mathbb{C}^2$ is restricting to

$$\mathrm{SU}(2, 1) = \{A \in \mathrm{GL}(3, \mathbb{C}) : \bar{A}^T C A = C \, , \, \det(A) = 1\}.$$

However this group still does not act effectively, since A and εA with $\varepsilon^3 = 1 \, , \, \varepsilon \neq 1$ are different elements of $\mathrm{SU}(2, 1)$ but they act the same on $\mathbb{C}^2$. Let K be group of transformation $Z \mapsto \varepsilon \cdot Z$ where $\varepsilon^3 = 1$ ("$\cdot$" means the multiplication action on each component), this is a normal subgroup of $\mathrm{SU}(2, 1)$ and $\mathrm{SU}(2, 1)/K$ acts effectively on $\mathbb{C}^2$. Since $K = Z(\mathrm{U}(2, 1)) \cap \mathrm{SU}(2, 1)$, the group $\mathrm{SU}(2, 1)/K$ by definition of projective unitary group (for finite dimensional Hermitian space) is

$\mathrm{PSU}(2,1)$, which is identical to $\mathrm{PU}(2,1) = \mathrm{U}(2,1)/\mathrm{U}(1)$ due to the second isomorphism theorem for groups. The hyperquadric itself can be viewed as a Lie subgroup $\mathrm{Stab}_{\mathrm{SU}(2,1)}([Z_2])$, c.f. (1.26)-(1.28) in [21]. Knowing that $\mathbb{S}^3$ is a Lie group, the CR isomorphism in lemma 2.17 is then also a Lie group isomorphism.

We define the so-called **Q**-frame namely complex vectors $\{Z_0, Z_1, Z_2\}$ in $\mathbb{C}^3$ satisfying the following equations

$$(Z_1, Z_1) = 1 \ , \ (Z_0, Z_2) = -(Z_2, Z_0) = -\frac{\mathbf{i}}{2} \tag{9}$$

with the rest vanish and $\det(Z_0, Z_1, Z_2) = 1$. The group $\mathrm{SU}(2,1)$ can be identified with the space of all **Q**-frames via $A = (Z_0, Z_1, Z_2)$. It is obvious that there is unique matrix in $\mathrm{SU}(2,1)$ taking one **Q**-frame to another. This implies that SU(2,1) acts transitively on $\overline{\mathbf{Q}}$ making the hyperquadric a homogeneous space whose coordinate transformations take values in $\mathrm{PU}(2,1)$. The pair $(\overline{\mathbf{Q}}, \mathrm{PU}(2,1)) \cong (\mathbb{S}^3, \mathrm{PU}(2,1))$ is understood as a geometric structure in the sense of Ehresmann (also known as Ehresmann structure).

We would like to understand the group transformation from geometric point of view. $[Z_0], [Z_2]$ are points on $\overline{\mathbf{Q}}$. A complex 2-plane in $\mathbb{C}^3$ projects to some $\mathbb{CP}^1$ in $\mathbb{CP}^2$, and it can be further identified as complex line (closure with infinity) in $\mathbb{C}^2$. Let $[Z]$ be a point on $\overline{\mathbf{Q}} \subset \mathbb{CP}^2$ and L a complex line L tangent at $[Z]$ (here in $\mathbb{CP}^2$ this line is represented as a copy of $\mathbb{CP}^1$), another point $[W]$ is on L if and only if $(Z, W) = 0$. So this implies $[Z_1]$ is the unique intersection point of the complex tangent lines at $[Z_0]$ and $[Z_2]$, i.e. $[Z_1] = L_0 \cap L_2$.

Theorem 2.18 ([17] Theorem 5)**.** *Let $x, x' \in \overline{\mathbf{Q}} \subset \mathbb{CP}^2$ be distinct two points and x'' be the intersection of the complex tangent lines at x, x'. Let $Z_0 \in \mathbb{C}^3$ be some vector with $[Z_0] = x$. There are unique vectors $Z_1, Z_2 \in \mathbb{C}^3$ such that $[Z_2] = x'$, $[Z_1] = x''$ and $A = (Z_0, Z_1, Z_2) \in SU(2,1)$. Conversely if $A = (Z_0, Z_1, Z_2) \in SU(2,1)$ then $[Z_0], [Z_1] \in \overline{\mathbf{Q}}$ are distinct points and $[Z_1]$ is the point of intersection of the corresponding complex tangent lines.*

The **Q**-frames component Z_0, Z_1 actually span the holomorphic tangent space of $\overline{\mathbf{Q}}$ at $[Z_0]$ which is complex 1-dimensional, i.e. $\mathrm{span}_{\mathbb{C}}\{Z_0, Z_1\} = \xi^{1,0}_{[Z_0]} \subset T_{\mathbb{C}}\overline{\mathbf{Q}}|_{[Z_0]}$. This picture holds similarly for the higher dimensional cases, c.f. the last paragraph of §1 in [21]. Therefore the effective automorphism group $\mathrm{PU}(2,1)$ is actually the CR automorphism group of $\overline{\mathbf{Q}}$ as well as $\mathbb{S}^3$ with standard CR structure. The isotropy subgroup in $\mathrm{SU}(2,1)$ at a point $[Z_0]$ is the group of transformations of complex frames for the hyperquadric.

Lemma 2.19 ([21] (1.15))**.** *The isotropy subgroup $H \subset SU(2,1)$ leaving a point $[Z_0]$ in $\overline{\mathbf{Q}}$ fixed is a group of matrices*

$$\begin{pmatrix} t & 0 & 0 \\ \bar{t}\tau & \bar{t}t^{-1} & 0 \\ t(r + \mathbf{i}|\tau|^2) & 2\mathbf{i}\bar{\tau} & \bar{t}^{-1} \end{pmatrix} \ , \ t \in \mathbb{C}^\times, \tau \in \mathbb{C}, r \in \mathbb{R}$$

Furthermore H_1 denotes a subgroup of H with $|t| = 1$.

Referring to [26], the Lie algebra of $\mathrm{SU}(2,1)$ is

$$\mathfrak{su}(2,1) = \left\{ \begin{pmatrix} a & b & r \\ -2\mathbf{i}\bar{c} & \bar{a} - a & 2\mathbf{i}\bar{b} \\ s & c & -\bar{a} \end{pmatrix} \ : \ a, b, c \in \mathbb{C} \ , \ r, s \in \mathbf{R} \right\}.$$

Let $E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$ and define $\mathfrak{g}_k := \{A \in \mathfrak{su}(2,1) : [E,A] = kA\,,\, k \in \mathbb{Z}\}$. There is a decomposition into subspaces

$$\mathfrak{su}(2,1) = \mathfrak{g}_{-2} + \mathfrak{g}_{-1} + \mathfrak{g}_0 + \mathfrak{g}_1 + \mathfrak{g}_2$$

with $[\mathfrak{g}_k, \mathfrak{g}_l] \subset \mathfrak{g}_{k+l}$ turning $\mathfrak{su}(2,1) = \sum_k \mathfrak{g}_k$ into a graded Lie algebra. The Lie algebra of H_1 as well as H_1/K is

$$\mathfrak{h} = \left\{ \begin{pmatrix} a & 0 & 0 \\ -2\mathbf{i}\bar{c} & \bar{a}-a & 0 \\ s & c & -\bar{a} \end{pmatrix} : a, c \in \mathbb{C}\,,\, s \in \mathbb{R} \right\} = \mathfrak{g}_{-2} + \mathfrak{g}_{-1} + \mathfrak{g}_0.$$

For higher dimension case, namely $\mathrm{SU}(p,q)$ with $p+q = 2n+1$, the Lie algebra has similar graded structure, c.f. [24]. In fact, $\mathfrak{su}(p,q)$ is semisimple for $p+q \geq 2$ and closed under conjugate transpose, thus it is reductive Lie algebra. Consequently, there exist Killing form on the Lie algebra respecting the decomposition, and the sum becomes direct sum.

Finally, we present the normal form result in 3-dimensional case:

Theorem 2.20 ([22] first lemma)**.** *Let $M \subset \mathbb{C}^2$ be a strictly pseudo-convex hypersurface. There exists a biholomorphism mapping M to the following form*

$$\{(z_1, z_2) \in \mathbb{C}^2 \mid \operatorname{Im} z_2 = |z_1|^2 + f(z_1, \bar{z}_1, \operatorname{Re} z_2)\,,\, f \in C^\infty\,,\, D^k f(0) = 0 \text{ for } 0 \leq |k| \leq 3\}$$

with a based point $x \in M$ mapped to $0 \in \mathbb{C}^2$.

One can tell that the zero order form is exactly the real hyperquadric. The result for general dimensional case is proved in [21] §2 and §3, e.g. Theorem 3.5. Such normal form is therefore called Chern-Moser normal form. In general, the space of such normalizing transformations is parameterized by $\mathfrak{h}$ the Lie algebra of isotropy subgroup in $\mathrm{Aut}(\mathbf{Q})$ at 0, which is always 5-dimensional. Moreover, each finite jet of a normalizing transformation is parameterized by $\mathfrak{h}$ rationally.

3 Connections and curvatures for pseudo-Hermitian structures

The main goal of this section is to develop Riemannian connection and curvature theory for strictly pseudo-convex CR manifolds following [18] Chapter 1 section 1.2-1.3.

First of all, we need a certain proper metric on CR manifolds. Fortunately, we already have such notion for strictly pseudo-convex CR manifolds, namely given by the pseudo-Hermitian structure: Let (M, ξ, J) be a strictly pseudo-convex CR manifold of $2n+1$-dimensional, and θ is a fixed contact form/pseudo-Hermitian structure on it. One may extend J to a bundle endomorphism $J : TM \to TM$ by putting $J\mathcal{R} = 0$. In this case, there is an induced Riemannian metric called *Webster metric* such that

$$g_\theta(X, Y) = d\theta(X, JY)\,,\ g_\theta(X, \mathcal{R}) = 0\,,\ g_\theta(\mathcal{R}, \mathcal{R}) = 1 \qquad \forall\, X, Y \in \xi.$$

The CR manifold becomes a Riemannian manifold (when the CR manifold is not strictly pseudo-convex but merely non-degenerate, the metric g_θ is semi-Riemannian). The data $(J, \mathcal{R}, \theta, g_\theta)$ is called a *contact metric structure* on (M, ξ).

Suppose ∇ is a affine connection on M and T^∇ is its torsion tensor given by

$$T^\nabla(X, Y) := \nabla_X Y - \nabla_Y X - [X, Y] \qquad \forall\, X, Y \in TM$$

and naturally extends to $T_{\mathbb{C}}M$. The torsion is said to be *pure* if it satisfies the conditions: For all $Z, Z' \in \xi^{1,0}$,

$$\begin{aligned} T^\nabla(Z, Z') &= 0 \\ T^\nabla(Z, \bar{Z}') &= 2d\theta(Z, \bar{Z}')\mathcal{R} \\ \tau \circ J + J \circ \tau &= 0 \end{aligned}$$

where $\tau : TM \to TM$ is defined by $\tau(X) = T^\nabla(\mathcal{R}, X)$ for any $X \in TM$. On a strictly pseudo-convex, there is a canonical affine connection compatible with the complex structure J and the Levi form. This is stated precisely as the following:

Theorem 3.1 ([18] Theorem 1.3)**.** *Let $(M, \xi^{1,0})$ be a strictly pseudo-convex CR manifold and θ is a fixed pseudo-Hermitian structure. J is a complex structure extended to TM as above, and g_θ is the corresponding Webster metric. There is a unique affine connection ∇ on M such that:*

i) ξ is parallel with respect to ∇, i.e. $\nabla_X C^\infty(M, \xi) \subseteq C^\infty(M, \xi)\ \forall\, X \in TM$.

ii) $\nabla J = 0$ and $\nabla g_\theta = 0$ (namely J and g_θ are also ∇-parallel).

iii) T^∇ is pure.

Remark.

• The theorem holds generally for any non-degenerate CR manifold. Such connection ∇ is called the *Tanaka-Webster connection* of $(M, \xi^{1,0}, \theta)$, and the corresponding τ is called the *pseudo-Hermitian torsion* of ∇.

• Since ξ and J are ∇-parallel, the Tanaka-Webster connection ∇ parallelizes $\xi^{1,0}$ and $\xi^{0,1}$. Hence ∇ is reducible to a connection in $\xi^{1,0}$. Moreover, $\nabla\mathcal{R} = 0$ and $\nabla d\theta = 0$.

• The purity of torsion implies that $\mathrm{pr}_{\xi^{1,0}} T^\nabla(Z, \bar{X}) = 0\ \forall\, Z \in \xi^{1,0}, X \in T_{\mathbb{C}}M$.

A contact metric structure on M is said to be *normal* if and only if the Tanaka–Webster connection has vanishing pseudo-Hermitian torsion and such a normal contact metric structure is called a *Sasakian structure*.

The proof of theorem 3.1 presented in [18], which was first carried out by N. Tanaka in 1967, is in a global way. Later in 1978, S. M. Webster used Cartan's moving frame method to prove this result, c.f. [20]. Webster's approach is of our main interest. In the light of theorem 3.1, we now discuss the local computations via local frame and coframe fields: Pick a local frame field $\{Z_1, \ldots, Z_n\}$ of $\xi^{1,0}$. Since Tanaka-Webster connection parallelizes $\xi^{1,0}$, there are unique complex 1-forms $\{\omega^\alpha_\beta\}_{1\leq\alpha,\beta\leq n}$ defined locally with the same domain as $\{Z_\alpha\}_{1\leq\alpha\leq n}$ such that

$$\nabla Z_\beta = \omega^\alpha_\beta \otimes Z_\alpha. \tag{10}$$

Similar holds for $\xi^{0,1}$, and we put $Z_{\bar{\alpha}} = \bar{Z}_\alpha$, $\alpha = 1, \ldots, n$. The Christoffel symols is given as usual by:

$$\Gamma^\alpha_{\gamma\beta} = \omega^\alpha_\beta(Z_\gamma) \quad , \quad \Gamma^\alpha_{\bar{\gamma}\beta} = \omega^\alpha_\beta(Z_{\bar{\gamma}}),$$

thus the covariant derivatives are given as

$$\nabla_{Z_\gamma} Z_\beta = \Gamma^\alpha_{\gamma\beta} Z_\alpha \quad , \quad \nabla_{Z_{\bar{\gamma}}} Z_\beta = \Gamma^\alpha_{\bar{\gamma}\beta} Z_\alpha \quad , \quad \nabla_{\mathcal{R}} Z_\beta = \Gamma^\alpha_{0\beta} Z_\alpha.$$

Locally the pseudo-Hermitian structure yields

$$L_\theta(Z_\alpha, Z_{\bar{\beta}}) = h_{\alpha\bar{\beta}} \quad , \quad d\theta(Z_\alpha, Z_{\bar{\beta}}) = -\mathbf{i} h_{\alpha\bar{\beta}}. \tag{11}$$

$h_{\alpha\bar{\beta}}$ is Hermitian, i.e. $h_{\alpha\bar{\beta}} = \overline{h_{\beta\bar{\alpha}}} = h_{\bar{\beta}\alpha}$. We put $(h^{\bar{\alpha}\beta}) = (h_{\alpha\bar{\beta}})^{-1}$ so that $h^{\bar{\alpha}\beta} h_{\beta\bar{\gamma}} = \delta^{\bar{\alpha}}_{\bar{\gamma}} = \delta^\alpha_\gamma$. One can derive that

$$\begin{aligned}
\Gamma^\alpha_{\gamma\beta} &= h^{\bar{\sigma}\alpha}[Z_\gamma(h_{\beta\bar{\sigma}}) - g_\theta(Z_\beta, [Z_\gamma, Z_{\bar{\sigma}}])] \\
\Gamma^\alpha_{\bar{\gamma}\beta} &= h^{\bar{\sigma}\alpha} g_\theta([Z_{\bar{\gamma}}, Z_\beta], Z_{\bar{\sigma}}) \\
\Gamma^\alpha_{0\alpha} &= h^{\bar{\sigma}\alpha} g_\theta([\mathcal{R}, Z_\beta], Z_{\bar{\sigma}}).
\end{aligned}$$

Using these Christoffel symbols expression together with the purity condition of torsion and Webster metric compatibility

$$\nabla g_\theta = 0 \Leftrightarrow X(g_\theta(Y,Z)) = g_\theta(\nabla_X Y, Z) + g_\theta(Y, \nabla_X Z) \qquad \forall\, X, Y, Z \in TM,$$

one can show:

Lemma 3.2 ([18] Lemma 1.3, Lemma 1.4)**.** *Let ∇ be the Tanaka-Webster connection of a strictly pseudo-convex CR manifold with a fixed pseudo-Hermitian structure (M, θ). Then the torsion tensor T^∇ is given by*

$$T^\nabla = 2(\theta \wedge \tau - d\theta \otimes \mathcal{R}).$$

Moreover, the Levi-Civita connection ∇^{LC} of the Riemannian manifold (M, g_θ) is related to ∇ by

$$\nabla^{LC} = \nabla + (d\theta - A) \otimes \mathcal{R} + \tau \otimes \theta + 2\theta \odot J,$$

where A is a symmetric tensor defined by $A(X,Y) := g_\theta(\tau X, Y)$ $\forall\, X, Y \in TM$.

In terms of local frame field $\{Z_\alpha, Z_{\bar{\alpha}}, \mathcal{R}\}$, we have

$$\tau Z_\beta = A^{\bar{\alpha}}_\beta Z_{\bar{\alpha}} \quad , \quad A^{\bar{\beta}}_\alpha = -h^{\sigma\bar{\beta}} g_\theta([\mathcal{R}, Z_\alpha], Z_\sigma),$$

and hence

$$\tau = \begin{pmatrix} 0 & A_{\alpha}^{\bar{\beta}} & 0 \\ A_{\bar{\alpha}}^{\beta} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Next, we want to derive connection and curvature using Cartan's moving frame method: For a local frame field $\{Z_\alpha, Z_{\bar{\alpha}}, \mathcal{R}\}$, there is unique local coframe field $\{\theta^\alpha, \theta^{\bar{\alpha}}, \theta\}$ such that

$$\theta^\alpha(Z_\beta) = \delta^\alpha_\beta \quad , \quad \theta^\alpha(Z_{\bar{\beta}}) = \theta^\alpha(\mathcal{R}) = 0, \tag{12}$$

and we label $\mathcal{R}$ as well as θ with index 0. The formal integrability of CR structure (3) leads us to the pseudo-Hermitian structure equation:

$$d\theta = \mathbf{i} h_{\alpha\bar{\beta}} \theta^\alpha \wedge \theta^{\bar{\beta}} + \theta \wedge (\eta_\alpha \theta^\alpha + \eta_{\bar{\alpha}} \theta^{\bar{\alpha}}), \tag{13}$$

where $\overline{\eta_\alpha} = \eta_{\bar{\alpha}}$. We make a remark that the integrability condition prevents the term $\theta^\alpha \wedge \theta^\beta$ appearing since $2d\theta(Z_\alpha, Z_\beta) = -\theta([Z_\alpha, Z_\beta]) = 0$, similar holds for $\theta^{\bar{\alpha}} \wedge \theta^{\bar{\beta}}$ terms. In fact, the formal integrability condition (3) is equivalent to (c.f. [18] Theorem 1.14):

$$\begin{aligned} d\theta &\equiv 0 \bmod \{\theta, \theta^\beta\} \\ d\theta^\alpha &\equiv 0 \bmod \{\theta, \theta^\beta\}. \end{aligned}$$

In order to study the pseudo-Hermitian structure equivalence problem (considering the CR diffeomorphisms $F : M \to M$ such that $F^*\theta = \theta$, which form a compact Lie group denoted as $\mathrm{Psh}(M, \theta)$, c.f. [18] Theorem 1.9 or originally [20]), Webster reduces the structure equation (13) to

$$d\theta = \mathbf{i} h_{\alpha\bar{\beta}} \theta^\alpha \wedge \theta^{\bar{\beta}} \tag{14}$$

by choosing so-called admissible coframes (namely those satisfying structure equation (14) and requiring $h_{\alpha\bar{\beta}} = \delta_{\alpha\bar{\beta}}$ with signature (p, q) (c.f. [20] Page 27). These local 1-forms $\{\theta, \theta^\alpha, \theta^{\bar{\alpha}}\}$ obeying the above equation encode the data of pseudo-Hermitian structure on M. In fact, this is the procedure of interpreting the pseudo-Hermitian structure as a integrable G-structure where $G = \mathrm{U}(p, q)$ is a reduction from $\mathrm{GL}(n, \mathbb{C})$. From the structure equation (14) we know that the integrability of the G-structure directly comes from the formal integrability of the CR structure explained in section 2.1. In this understanding, the connection matrix (ω^α_β) of ∇ determines a principal connection on the pseudo-Hermitian structure as a G-structure. The pseudo-Hermitian torsion can be expressed as

$$\tau = \tau^\alpha \otimes Z_\alpha + \tau^{\bar{\alpha}} \otimes Z_{\bar{\alpha}}, \tag{15}$$

where local 1-forms $\tau^\alpha = A^\alpha_{\bar{\beta}} \theta^{\bar{\beta}}$ and $\tau^{\bar{\alpha}} = \overline{\tau^\alpha}$. We computes $d\theta^\alpha(X, Y)$ for any $X, Y \in TM$ straightforwardly:

$$\begin{aligned} 2(d\theta^\alpha)(X, Y) &= X\theta^\alpha(Y) - Y\theta^\alpha(X) - \theta^\alpha([X, Y]) \\ &= [\nabla_X(\theta^\alpha(Y)) - \theta^\alpha(\nabla_X Y)] - [\nabla_Y(\theta^\alpha(X)) - \theta^\alpha(\nabla_Y X)] - \theta^\alpha(T^\nabla(X, Y)) \\ &= (\nabla_X \theta^\alpha)(Y) - (\nabla_Y \theta^\alpha)(X) - \theta^\alpha(T^\nabla(X, Y)). \end{aligned}$$

This gives rise the Cartan's first structure equation:

$$d\theta^\alpha = \theta^\beta \wedge \omega^\alpha_\beta + \theta \wedge \tau^\alpha. \tag{16}$$

So we have extracted connection from local coframe field. Further differentiating the pseudo-Hermitian structure equation (13) and using $\nabla g_\theta = 0$ in local expression namely

$$dh_{\alpha\bar{\beta}} = h_{\alpha\bar{\gamma}} \omega^{\bar{\gamma}}_{\bar{\beta}} + \omega^\gamma_\alpha h_{\gamma\bar{\beta}}, \tag{17}$$

we can show that $A_{\beta\sigma} = A_{\sigma\beta}$, i.e. A is indeed symmetric tensor.

Finally, we come to derive the curvature 2-form using Cartan's second stricture equation

$$\Omega_\alpha^\beta = d\omega_\alpha^\beta - \omega_\alpha^\gamma \wedge \omega_\gamma^\beta.$$

After some computations, Webster shows that:

Theorem 3.3 ([18] Theorem 1.7). *The curvature 2-form is expressed as*

$$\Omega_\alpha^\beta = R_\alpha{}^\beta{}_{\rho\bar{\sigma}}\theta^\rho \wedge \theta^{\bar{\sigma}} + W_{\alpha\rho}^\beta \theta^\rho \wedge \theta - W_{\alpha\bar{\sigma}}^\beta \theta^{\bar{\sigma}} \wedge \theta + \mathbf{i}\theta_\alpha \wedge \tau^\beta - \mathbf{i}\tau_\alpha \wedge \theta^\beta,$$

where $R_\alpha{}^\beta{}_{\rho\bar{\sigma}}$ is called the pseudo-Hermitian curvature tensor and $W_{\alpha\bar{\sigma}}^\beta = h^{\bar{\gamma}\beta}A_{\bar{\sigma}\bar{\gamma},\alpha}$, $W_{\alpha\rho}^\beta = h^{\bar{\gamma}\beta}A_{\beta\rho,\bar{\gamma}}$ the Webster curvature tensor.

The pseudo-Hermitian curvature tensor is given by

$$\begin{aligned} R_{\alpha\bar{\beta}\rho\bar{\sigma}} &= g_\theta(R(Z_\rho, Z_{\bar{\sigma}})Z_\alpha, Z_{\bar{\beta}}) = h_{\gamma\bar{\beta}}R_\alpha{}^\gamma{}_{\rho\bar{\sigma}} \\ R(X,Y)Z &= (\nabla_X T^\nabla)(Y,Z) + T^\nabla(T^\nabla(X,Y),Z) \qquad \forall\, X,Y,Z \in TM. \end{aligned}$$

The pseudo-Hermitian curvature tensor is symmetric in the following sense ([18] Theorem 1.8):

$$R_{\rho\bar{\beta}\sigma\bar{\alpha}} = R_{\sigma\bar{\beta}\rho\bar{\alpha}}.$$

In above theorem, we have also adopted a simplified notation for covariant derivatives:

$$A_{\alpha\beta,\gamma} = (\nabla_{Z_\gamma}A)(Z_\alpha, Z_\beta) \quad , \quad \nabla_{Z^\gamma}A = \nabla_{Z_\gamma}A_{\alpha\beta}\theta^\alpha \otimes \theta^\beta,$$

any one of indices α, β, γ can also be replaced by its complex conjugation. We know $A_{\alpha\beta}$ is a symmetric $(0,2)$-tensor. For any (k,l)-tensor

$$T = T^{i_1\cdots i_n}_{j_1\cdots j_l} Z_{i_1} \otimes \cdots \otimes Z_{i_k} \otimes \theta^{j_1} \otimes \cdots \otimes \theta^{j_l}, \quad i_1, \ldots, i_k, j_1, \ldots, j_l \in \{0, 1, \ldots, n, \bar{1}, \ldots, \bar{n}\},$$

expressing through Christoffel symbol, the first covariant derivative of T along Z_γ is:

$$\begin{aligned} \nabla_{Z_\gamma}T =& Z_\gamma(T^{i_1\cdots i_k}_{j_1\cdots j_l})Z_{i_1} \otimes \cdots \otimes Z_{i_k} \otimes \theta^{j_1} \otimes \cdots \otimes \theta^{j_l} \\ &+ \sum_{s=1}^{k} T^{i_1\cdots\delta\cdots i_k}_{j_1\cdots j_l}\Gamma^{i_s}_{\gamma\delta} Z_{i_1} \otimes \cdots \otimes Z_{i_{s-1}} \otimes Z_{i_{s+1}} \otimes \cdots \otimes Z_{i_k} \otimes Z_{i_s} \otimes \theta^{j_1} \otimes \cdots \otimes \theta^{j_l} \\ &- \sum_{s=1}^{l} T^{i_1\cdots i_k}_{j_1\cdots\delta\cdots j_l}\Gamma^{\delta}_{\gamma j_s} Z_{i_1} \otimes \cdots \otimes Z_{i_k} \otimes \theta^{j_1} \otimes \cdots \otimes \theta^{j_{s-1}} \otimes \theta^{j_{s+1}} \otimes \cdots \otimes \theta^{j_l} \otimes \theta^{j_s}. \end{aligned}$$

For simplicity, the basis elements is usually omitted and we only write the coefficients representing the tensor, then the first derivative is abbreviated as:

$$T^{i_1\cdots i_k}_{j_1\cdots j_l,\gamma} = \nabla_{Z_\gamma}T^{i_1\cdots i_k}_{j_1\cdots j_l} = Z_\gamma(T^{i_1\cdots i_k}_{j_1\cdots j_l}) + \sum_{s=1}^{k} T^{i_1\cdots\delta\cdots i_k}_{j_1\cdots j_l}\Gamma^{i_s}_{\gamma\delta} - \sum_{s=1}^{l} T^{i_1\cdots i_k}_{j_1\cdots\delta\cdots j_l}\Gamma^{\delta}_{\gamma j_s}.$$

We can also upper the covariant derivative index by contacting with $h^{\alpha\bar{\beta}}$:

$$T^{i_1\cdots i_k\,\gamma}_{j_1\cdots j_l,} = h^{\gamma\bar{\sigma}}T^{i_1\cdots i_k}_{j_1\cdots j_l,\bar{\sigma}} = h^{\gamma\bar{\sigma}}\nabla_{Z_{\bar{\sigma}}}T^{i_1\cdots i_k}_{j_1\cdots j_l}.$$

Second order covariant derivatives are written according to order as:

$$\begin{aligned} T^{i_1\cdots i_k}_{j_1\cdots j_l,\gamma\delta} &= \nabla_{Z_\sigma}\nabla_{Z_\gamma}T^{i_1\cdots i_k}_{j_1\cdots j_l} \\ T^{i_1\cdots i_k\,\gamma\delta}_{j_1\cdots j_l,} &= h^{\delta\bar{\rho}}h^{\gamma\bar{\sigma}}\nabla_{Z_{\bar{\rho}}}\nabla_{Z_{\bar{\sigma}}}T^{i_1\cdots i_k}_{j_1\cdots j_l} \\ T^{i_1\cdots i_k\ \ \delta}_{j_1\cdots j_l,\gamma} &= h^{\delta\bar{\rho}}\nabla_{Z_{\bar{\rho}}}\nabla_{Z_\gamma}T^{i_1\cdots i_k}_{j_1\cdots j_l} \\ T^{i_1\cdots i_k\ \delta}_{j_1\cdots j_l,\ \ \gamma} &= h^{\delta\bar{\rho}}\nabla_{Z_\gamma}\nabla_{Z_{\bar{\rho}}}T^{i_1\cdots i_k}_{j_1\cdots j_l}. \end{aligned}$$

Higher order covariant derivatives are written analogously.

In the end, we list some useful lemmas below.

Lemma 3.4 ([23] Lemma 2.1)**.** *In a neighborhood of of any point $x \in M$, there exits an admissible coframe field (θ^α) with $\omega_\alpha^\beta = 0$ at x.*

The key to the proof is that we can extend any coframe at a point via parallel transport given by Tanaka-Webster connection along the geodesic. Then the standard theory of affine connection implies that the connection 1-forms vanish at this point.

Remark. This lemma is often used to simplify the calculation involving covariant derivative. Under such coframe, the connection vanishes, thus the covariant derivatives are identical to the ordinary derivatives. This help to simplify the verification for any tensorial identity.

Lemma 3.5 ([23] Lemma 2.2)**.** *The pseudo-Hermitian curvature and torsion satisfy:*

$$\begin{aligned}
A_{\alpha\rho,\gamma} &= A_{\alpha\gamma,\rho} \\
R_{\alpha\bar\beta\rho\bar\sigma,\gamma} - R_{\alpha\bar\beta\gamma\bar\sigma,\rho} &= \mathbf{i}(A_{\alpha\gamma,\bar\beta}h_{\rho\bar\sigma} + A_{\alpha\gamma,\bar\sigma}h_{\rho\bar\beta} - A_{\alpha\rho,\bar\beta}h_{\gamma\bar\sigma} - A_{\alpha\rho,\bar\sigma}h_{\gamma\bar\beta}) \\
R_{\alpha\bar\beta\rho\bar\sigma,0} &= A_{\alpha\rho,\bar\beta\bar\sigma} + A_{\bar\beta\bar\sigma,\alpha\rho} + \mathbf{i}(h_{\alpha\bar\sigma}A_{\rho\gamma}A^\gamma_{\bar\beta} - h_{\rho\bar\beta}A_{\alpha\gamma}A^\gamma_{\bar\sigma}) \\
A_{\alpha\rho,\bar\beta\gamma} - A_{\alpha\gamma,\bar\beta\rho} &= \mathbf{i}(h_{\rho\bar\beta}A_{\alpha\gamma,0} - h_{\gamma\bar\beta}A_{\alpha\rho,0}) + R_{\alpha\bar\beta\rho\bar\sigma}A^{\bar\sigma}_\gamma - R_{\alpha\bar\beta\gamma\bar\sigma}A^{\bar\sigma}_\rho.
\end{aligned}$$

These are called the Bianchi identities.

Lemma 3.6 ([23] Lemma 2.3)**.** *The second covariant derivatives of a scalar function u and a $(1,0)$-form $v_\alpha\theta^\alpha$ satisfy the following commutation relations:*

$$\begin{aligned}
u_{,\alpha\bar\beta} - u_{,\bar\beta\alpha} &= \mathbf{i}h_{\alpha\bar\beta}u_{,0} \\
u_{,\alpha\beta} &= u_{,\beta\alpha} \\
u_{,0\alpha} - u_{,\alpha 0} &= A_{\alpha\beta}u_{,}{}^{\beta} \\
v_{\alpha,\beta\gamma} - v_{\alpha,\gamma\beta} &= \mathbf{i}(A_{\alpha\gamma}v_\beta - A_{\alpha\beta}v_\gamma) \\
v_{\alpha,\bar\beta\bar\gamma} - v_{\alpha,\bar\gamma\bar\beta} &= \mathbf{i}(h_{\alpha\bar\beta}A_{\bar\gamma\bar\rho}v^{\bar\rho} - h_{\alpha\bar\gamma}A_{\bar\beta\bar\rho}v^{\bar\rho}) \\
v_{\alpha,\beta\bar\gamma} - v_{\alpha,\bar\gamma\beta} &= \mathbf{i}h_{\beta\bar\gamma}v_{\alpha,0} + R_{\alpha}{}^{\rho}{}_{\beta\bar\gamma}v_\rho \\
v_{\alpha,0\beta} - v_{\alpha,\beta 0} &= v_{\alpha,}{}^{\gamma}A_{\gamma\beta} - v_\gamma A_{\alpha\beta,}{}^{\gamma} \\
v_{\alpha,0\bar\beta} - v_{\alpha,\bar\beta 0} &= v_{\alpha,\gamma}A^\gamma_{\bar\beta} + v_\gamma A^\gamma_{\bar\beta,\alpha}.
\end{aligned}$$

The proof is simply computing the differential of

$$\begin{aligned}
du &= u_{,0}\theta + u_{,\alpha}\theta^\alpha + u_{,\bar\beta}\theta^{\bar\beta} \\
dv_\alpha &= v_{\alpha,0}\theta + v_{\alpha,\gamma}\theta^\gamma + v_{\alpha,\bar\beta}\theta^{\bar\beta} + v_\gamma\omega^\gamma_\alpha,
\end{aligned}$$

and comparing similar terms.

4 Connections and curvatures for CR structures

In previous section we have introduced Tanaka-Webster connection and curvature using Cartan's moving frame method, which are some local geometric invariants that can be used to distinguished the pseudo-Hermitian structures up to local equivalence. We will investigate in this section particularly the strictly pseudo-convex CR 3-manifolds in terms of these quantities. Both the pseudo-Hermitian structure and CR structure can be described in the framework of G-structure. To this end, Shiing-Shen Chern and Jürgen Moser encode CR structure in a principal bundle via moving frame method in order to solve the CR equivalence problem. We study the Chern-Moser construction for 3-dimensional case, and regard the connection therein as Cartan connection. By relating the principle bundles for pseudo-Hermitian structure and CR structures, we can represented the Cartan connection as well as curvature for CR structure in terms of pseudo-Hermitian data, i.e. Tanaka-Webster connection, pseudo-Hermitian curvature and Webster curvature. The Chern-Moser construction for general dimensional case is referred to [21].

4.1 Pseudo-Hermitian structure bundle

For strictly pseudo-convex CR 3-manifold $(M, \xi^{1,0})$, on some open neighborhood $U \subset M$ the local forms $\theta, \theta^1, \theta^{\bar{1}}$ form a basis for $T^*_{\mathbb{C}}M|_U$. By integrability of CR structure, we have (recall (13)):

$$d\theta = \mathbf{i}h_{1\bar{1}}\theta^1 \wedge \theta^{\bar{1}} + \theta \wedge \varphi,$$

where θ, φ are both real 1-forms on U (φ should annihilate the entire $TM|_U$) and $h_{1\bar{1}}$ can be fixed as 1. We see there is a natural orientation on every CR 3-manifold making use only of CR information locally. Further by fixing a global contact form θ and requiring $\varphi = 0$ due to $i_{\mathcal{R}}d\theta = 0$, we look for the admissible coframes $\{\theta^1, \theta^{\bar{1}}\}$ of $\xi_{\mathbb{C}} = \ker(\theta)_{\mathbb{C}}$ satisfying structure equation

$$d\theta = \mathbf{i}\theta^1 \wedge \theta^{\bar{1}} \tag{18}$$

on M. The collection of all these admissible coframes form a $\mathbb{C}^\times$-bundle. The structure group is given by transformations

$$\begin{pmatrix} \theta'^1 \\ \theta'^{\bar{1}} \end{pmatrix} = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda^{-1} \end{pmatrix} \begin{pmatrix} \theta^1 \\ \theta^{\bar{1}} \end{pmatrix}$$

with $\lambda \in \mathbb{C}^\times$. We consider the pseudo-Hermitian norm preserving transformations, then $\lambda \in \mathrm{U}(1)$ and the structure group reduces to $\mathrm{U}(1)$ producing a $\mathrm{U}(1)$-bundle denoted as $\mathscr{X}$. This bundle is called the pseudo-Hermitian structure bundle since it determines the pseudo-Hermitian structure $h_{1\bar{1}}$ completely.

Proposition 4.1. *Given coframe $\theta^1, \theta^{\bar{1}}$, there are uniquely determined 1-forms θ^1_1, τ^1 satisfying the structure equation*

$$d\theta^1 = \theta^1 \wedge \theta^1_1 + \theta \wedge \tau^1$$

with normalization $\theta^1_1 + \theta^{\bar{1}}_{\bar{1}} = 0$ and $\tau^1 \wedge \theta^{\bar{1}} = 0$.

Proof. By (16), we can formally write down the equation

$$d\theta^1 = \theta^1 \wedge \theta^1_1 + \theta \wedge \tau^1.$$

Observe that

$$0 = d(d\theta) = \mathbf{i}(d\theta^1 \wedge \theta^{\bar{1}} - \theta^1 \wedge d\theta^{\bar{1}})$$

and obtain

$$0 = -\mathbf{i}\theta^1 \wedge \theta^{\bar{1}} \wedge (\theta_1^1 + \theta_{\bar{1}}^{\bar{1}}) + \mathbf{i}\theta \wedge (\tau^1 \wedge \theta^{\bar{1}} - \tau^{\bar{1}} \wedge \theta^1) = -d\theta \wedge (\theta_1^1 + \theta_{\bar{1}}^{\bar{1}}) + \mathbf{i}\theta \wedge (\tau^1 \wedge \theta^{\bar{1}} - \tau^{\bar{1}} \wedge \theta^1).$$

The simplest normalization is $\theta_1^1 + \theta_{\bar{1}}^{\bar{1}} = 0$ and $\tau^1 \wedge \theta^{\bar{1}} = \tau^{\bar{1}} \wedge \theta^1 = 0$. Since the pseudo-Hermitian structure equation (18) is invariant under the local frames transformation taking value in $\mathrm{U}(1)$, the 1-forms θ_1^1, τ^1 are uniquely determined by the structure equation with respect to coframe $\theta^1, \theta^{\bar{1}}$ and normalization in the statement.

On the other hand, all $\theta^1, \theta^{\bar{1}}$ can be generally expressed in terms of a fixed reference frame $\tilde{\theta}^1, \tilde{\theta}^{\bar{1}}$, i.e. $\theta^1 = \lambda\tilde{\theta}^1, \theta^{\bar{1}} = \lambda^{-1}\tilde{\theta}^{\bar{1}}$ for some $\lambda \in \mathrm{U}(1)$. We then compute

$$\begin{aligned}
d\theta^1 = d\lambda \wedge \tilde{\theta}^1 + \lambda d\tilde{\theta}^1 &= d\lambda \wedge \tilde{\theta}^1 + \lambda(\alpha\tilde{\theta}^1 \wedge \tilde{\theta}^{\bar{1}} + \beta\tilde{\theta}^1 \wedge \theta + \gamma\tilde{\theta}^{\bar{1}} \wedge \theta) \\
&= \frac{d\lambda}{\lambda} \wedge \theta^1 + \lambda\alpha\theta^1 \wedge \theta^{\bar{1}} + \beta\theta^1 \wedge \theta + \lambda^2\gamma\theta^{\bar{1}} \wedge \theta \\
&= \theta^1 \wedge (-\frac{d\lambda}{\lambda} + \lambda\alpha\theta^{\bar{1}}) + \theta \wedge (-\beta\theta^1 - \lambda^2\gamma\theta^{\bar{1}}) \\
d\theta^{\bar{1}} = d\lambda^{-1} \wedge \tilde{\theta}^1 + \lambda^{-1} d\tilde{\theta}^1 &= -\frac{d\lambda}{\lambda^2} \wedge \tilde{\theta}^{\bar{1}} + \frac{1}{\lambda}(\bar{\alpha}\tilde{\theta}^{\bar{1}} \wedge \tilde{\theta}^1 + \bar{\beta}\tilde{\theta}^{\bar{1}} \wedge \theta + \bar{\gamma}\tilde{\theta}^1 \wedge \theta) \\
&= -\frac{d\lambda}{\lambda} \wedge \theta^{\bar{1}} + \lambda^{-1}\bar{\alpha}\theta^{\bar{1}} \wedge \theta^1 + \bar{\beta}\theta^{\bar{1}} \wedge \theta + \lambda^{-2}\bar{\gamma}\theta^1 \wedge \theta \\
&= \theta^{\bar{1}} \wedge (\frac{d\lambda}{\lambda} + \lambda^{-1}\bar{\alpha}\theta^1) + \theta \wedge (-\bar{\beta}\theta^{\bar{1}} - \lambda^{-2}\bar{\gamma}\theta^1)
\end{aligned}$$

We can put $\theta_1^1 = -\frac{d\lambda}{\lambda} - \lambda^{-1}\bar{\alpha}\theta^1 + \lambda\alpha\theta^{\bar{1}}$ so that it satisfies the normalization $\theta_1^1 + \theta_{\bar{1}}^{\bar{1}} = 0$. Similarly put $\tau^1 = -\beta\theta^1 - \lambda^2\gamma\theta^{\bar{1}}$, by $\tau^1 \wedge \theta^{\bar{1}} = \tau^{\bar{1}} \wedge \theta^1 = 0 \Rightarrow \beta = 0$, hence $\tau^1 = -\lambda^2\gamma\theta^{\bar{1}} = A_{\bar{1}}^1\theta^{\bar{1}}$ with $A_{\bar{1}}^1 = \lambda^2\gamma$. Now θ_1^1, τ^1 satisfy the structure equation and the normalization in the statement for the corresponding coframe $\theta^1, \theta^{\bar{1}}$. This finishes the proof. □

We know that θ_1^1 is the pseudo-Hermitian connection 1-form. We will use this symbol instead of "ω_1^1" for connection 1-form on CR 3-manifold from now on.

Differentiating the structure equation (18) and using proposition 4.1, we obtain

$$0 = \theta^1 \wedge d\theta_1^1 + \theta \wedge (d\tau^1 - \tau^1 \wedge \theta_1^1),$$

therefore we can write

$$d\theta_1^1 = R\theta^1 \wedge \theta^{\bar{1}} + W\theta^1 \wedge \theta - \overline{W}\theta^{\bar{1}} \wedge \theta \tag{19}$$

which is compatible with $0 = d(\theta_1^1 + \theta_{\bar{1}}^{\bar{1}}) = d\theta_1^1 + d\theta_1^{\bar{1}}$ with R real and W complex.

The Tanaka-Webster connection ∇ on $T_{\mathbb{C}}M$ in terms of local frame $Z_1.Z_{\bar{1}}, \mathcal{R}$ is expressed as

$$\nabla Z_1 = \theta_1^1 \otimes Z_1 \quad , \quad \nabla Z_{\bar{1}} = \theta_{\bar{1}}^{\bar{1}} \otimes Z_{\bar{1}} \quad , \quad \nabla\mathcal{R} = 0.$$

Then

$$W = \nabla_{Z_1} A_1^{\bar{1}} \quad , \quad \overline{W} = \nabla_{Z_{\bar{1}}} A_{\bar{1}}^1.$$

R and $W, \overline{W}$ are the pseudo-Hermitian curvature and Webster curvature, while $A_{\bar{1}}^1$ is the Webster torsion.

Examples.

(1) On $\mathbb{S}^3 = \{z = (z_1, z_2) \in \mathbb{C}^2 : |z|^2 = |z_1|^2 + |z_2|^2 = 1\}$, the standard contact form is given by

$$\theta = \frac{\mathbf{i}}{2}(\bar\partial - \partial)|z|^2 = \frac{\mathbf{i}}{2}(z_1 d\bar z_1 + z_2 d\bar z_2 - \bar z_1 dz_1 - \bar z_2 dz_2),$$

the corresponding Reeb vector field is

$$\mathcal{R} = \mathbf{i}(z_1\partial_{z_1} - \bar z_1\partial_{\bar z_1} + z_2\partial_{z_2} - \bar z_2\partial_{\bar z_2})$$

and the standard complex structure on $\xi = \ker(\theta)$ is induced by restricting the standard one from $\mathbb{C}^2$. Then the standard pseudo-Hermitian or CR structure can be given as $\theta^1 = z_2 dz_1 - z_1 dz_2$ and $d\theta = \mathbf{i}\theta^1 \wedge \theta^{\bar 1}$. Further computation yields $d\theta^1 = \theta^1 \wedge \theta^1_1$ with $\theta^1_1 = -2\mathbf{i}\theta$. In an alternative way, we know there is a U(1)-action: $\lambda \cdot (z_1, z_2) = (\lambda z_1, \lambda z_2)\ \forall\, (z_1, z_2) \in \mathbb{S}^3$, $\lambda \in \mathrm{U}(1)$. It gives rise the Hopf fibration, and the generator of this action is the Reeb vector field. The base space of the U(1)-bundle is the space of complex line through origin, namely the projective space $\mathbb{CP}^1$. Let (ζ_1, ϑ) be the local coordinates for $\mathbb{S}^3$, where ϑ is the fiber coordinates and $\zeta_1 = z_1/z_2$ is the homogeneous coordinates for $\mathbb{CP}^1$. The Reeb vector field is $\mathcal{R} = \frac{\partial}{\partial\vartheta}$ and the contact form can be given as

$$\theta = d\vartheta + \frac{\mathbf{i}}{2}\frac{1}{1+|\zeta^1|^2}(\zeta^1 d\zeta^{\bar 1} - \zeta^{\bar 1} d\zeta^1),$$

which is also viewed as the connection 1-form of the U(1)-bundle. A local framing for $\xi^{1,0}$ is given by $\frac{\partial}{\partial\zeta^1} + \frac{\mathbf{i}}{2}\frac{1}{1+|\zeta^1|^2}\mathcal{R}$, from principal bundle point of view, it is then the horizontal lift of $\frac{\partial}{\partial\zeta^1}$. The dual frame field is thus $d\zeta^1$. So we obtain structure equation

$$d\theta = \frac{\mathbf{i}}{1+|\zeta^1|} d\zeta^1 \wedge d\zeta^{\bar 1} = \mathbf{i}\theta^1 \wedge \theta^{\bar 1}$$

by setting $\theta^1 = \frac{d\zeta^1}{(1+|\zeta^1|^2)^{1/2}}$. In either way, we can easily see that $R = 2$, $A^{\bar 1}_1 = 0$, and $W = 0$.

If we consider $\mathbb{S}^3$ as the Lie group SU(2), we can observe the left-invariant coframe: Let $\gamma^1, \gamma^2, \gamma^3$ be the standard SU(2)-invariant 1-forms on SU(2) (i.e. the dual of left-invariant vector fields who form the standard basis of Lie algebra $\mathfrak{su}(2)$), then by the Maurer–Cartan equation $d\gamma^i = -\frac{1}{2}c^i_{jk}\gamma^j \wedge \gamma^k$ with structure constants $c^1_{23} = c^2_{31} = c^3_{12} = 2$, and setting $\theta = \gamma^3, \theta^1 = -\frac{\mathbf{i}}{\sqrt{2}}(\gamma^1 - \mathbf{i}\gamma^2)$, we have:

$$d\theta = \mathbf{i}\theta^1 \wedge \theta^{\bar 1} \quad , \quad d\theta^1 = \mathbf{i}\theta \wedge \theta^1.$$

This is actually the standard pseudo-Hermitian structure equation, and the curvatures are the same as above. One can also check the coframe in terms of (z_1, z_2) are SU(2)-invariant by taking action of SU(2) on $(z_1, z_2) \in \mathbb{S}^3$ given by matrices

$$\mathrm{SU}(2) = \left\{ \begin{pmatrix} a & -\bar b \\ b & \bar a \end{pmatrix} \,\middle|\, a, b \in \mathbb{C}\ ,\ |a|^2 + |b|^2 = 1 \right\}.$$

We can construct a family of invariant coframes $\theta^1_s = \frac{1}{\sqrt{1-s^2}}(\theta^1 + s\theta^{\bar 1})$ with $s \in (-1, 1)$, satisfying the same pseudo-Hermitian structure equations:

$$\begin{aligned} d\theta &= \mathbf{i}\theta^1_s \wedge \theta^{\bar 1}_s \\ d\theta^1_s &= \theta^1_s \wedge \left(-\mathbf{i}\frac{1+s^2}{1-s^2}\theta\right) + \theta \wedge \left(-\mathbf{i}\frac{2s}{1-s^2}\theta^{\bar 1}_s\right) \end{aligned}$$

where we see $\theta^1_{1\,s} = -2\mathbf{i}\frac{1+s^2}{1-s^2}\theta$ and $\tau^1_s = -\mathbf{i}\frac{4s}{1-s^2}\theta^{\bar{1}}_s$. Thus we obtain $d\theta^1_{1\,s} = 2(\frac{1+s^2}{1-s^2})\theta^1_s \wedge \theta^{\bar{1}}_s$. In conclusion, we find $R = 2\frac{1+s^2}{1-s^2}$, $A^1_{\bar{1}} = -\mathbf{i}\frac{4s}{1-s^2}$, and $W = 0$. These produce a family of SU(2)-invariant CR structures.

(2) One can easily computes that the pseudo-Hermitian curvature and Webster curvature of the Heisenberg group $(\mathbf{H}_n, \theta = dt + \frac{\mathbf{i}}{2}\sum_{i=1}^n z_i d\bar{z}_i - \bar{z}_i dz_i)$ vanish. This can be shown alternatively by computing the Christoffel symbols in terms of the Lewy operators, c.f. [18] (1.55). In fact they all vanish, the Tanaka-Webster connection is flat. The Heisenberg group can be thought of as the "Euclidean space" in the realm of CR geometry. Similar to $\mathbb{S}^3$, this standard pseudo-Hermitian structure on Heisenberg group determines also an invariant CR structure (c.f. [30] §4).

4.2 CR structure bundle

Now we construct the Chern-Moser connection (originally introduced in [21]) on a so-called CR structure bundle $\mathscr{Y}$ to solve CR structure equivalence problem. We keep focusing on CR 3-manifold. Since the general CR diffeomorphisms can dilate the contact form unlike the pseudo-Hermitian structure equivalence problem, this motivates to work on the symplectization of the contact manifold: There exits a global contact form θ on (M, ξ), the positive symplectization or the symplectic cone of the contact manifold (M, ξ) defined as

$$S_+(M,\xi) := \{(x, u\theta_x) : x \in M\,,\, \ker(\theta_x) = \xi_x\,,\, u \in \mathbb{R}_+\} \subset T^*M$$

is a trivial real half-line bundle over M. This is actually the collection of all the contact 1-forms on M. There is naturally a globally well-defined tautological 1-form $\Theta \in T^*S_+(M,\xi)$ such that

$$\Theta(X) = u\theta(\mathrm{pr}_* X) \qquad \forall\, X \in T_{(x,u\theta_x)}S_+(M,\xi).$$

Under local trivialization $S_+(M,\xi) \supset \mathrm{pr}^{-1}(U) \to U \times \mathbb{R}_+$ ($U \subset M$ open set) one can identify this tautological 1-form restricted on U as $\Theta = u\theta$. Then on a trivializing neighborhood of $T^*_{\mathbb{C}}S_+(M,\xi)$:

$$\begin{aligned} d\Theta = d(u\theta) &= ud\theta + du \wedge \theta = u(\mathbf{i}\theta^1 \wedge \theta^{\bar{1}} + \theta \wedge \varphi) - \theta \wedge du \\ &= \mathbf{i}u\theta^1 \wedge \theta^{\bar{1}} + u\theta \wedge \Big(-\frac{du}{u} + \varphi\Big). \end{aligned}$$

Now we can define similarly some other local forms $\Theta^1, \Theta^{\bar{1}}, \phi$ on U also via the local trivialization such that $\Theta^1 = \sqrt{u}\theta^1, \Theta^{\bar{1}} = \sqrt{u}\theta^{\bar{1}}, \phi = -\frac{du}{u} + \varphi$. Then we have the so-called CR structure equation

$$d\Theta = \mathbf{i}\Theta^1 \wedge \Theta^{\bar{1}} + \Theta \wedge \phi \tag{20}$$

where $\Theta, \Theta^1, \Theta^{\bar{1}}, \phi$ are linearly independent local forms on $\mathrm{pr}^{-1}(U)$ forming a basis for $T^*_{\mathbb{C}}S_+(M,\xi)|_{\mathrm{pr}^{-1}(U)}$. The reason why it is called the CR structure equation is that rescaling contact form does not affect the contact structure ξ, and the CR structure in terms of $\xi^{1,0}$ or $\xi^{0,1}$ is completely determined by this equation. If we perform any local CR diffeomorphism, which is clearly local contactomorphism, then this is simply rescaling the contact form θ and correspondingly the local frame $\theta^1, \theta^{\bar{1}}$ as above, and we see the structure equitation (20) does not change of course.

Next we consider all the local coframe fields $\{\Theta, \Theta^1, \Theta^{\bar{1}}, \phi\}$ satisfying the structure equation (20) so that the collection of all these coframes forms a subbundle $\mathscr{Y}$ of coframe bundle $\mathrm{Fr}(T^*_{\mathbb{C}}S_+(M,\xi))$, which is a $\mathrm{GL}(4,\mathbb{C})$-principal bundle, called the CR structure bundle for (M,ξ). The admissible

(preserving the CR structure equation) coframe transformations have the general form

$$\begin{aligned}\Theta' &= \Theta\\ \Theta^{1'} &= u_1^1\Theta^1 + v^1\Theta\\ \Theta^{\bar{1}'} &= u_{\bar{1}}^{\bar{1}}\Theta^{\bar{1}} + v^{\bar{1}}\Theta\\ \phi' &= \phi + s\Theta + w\Theta^1 + \bar{w}\Theta^{\bar{1}}\end{aligned}$$

with $u_1^1, u_{\bar{1}}^{\bar{1}}, v^1, v^{\bar{1}}, w, \bar{w} \in \mathbb{C}$ and $s \in \mathbb{R}$. Insert into the restriction equation then

$$d\Theta = \mathbf{i}|u_1^1|^2\Theta^1\wedge\Theta^{\bar{1}} + \Theta\wedge\left(\phi + (-\mathbf{i}u_1^1v^{\bar{1}} + w)\Theta^1 + (\mathbf{i}u_{\bar{1}}^{\bar{1}}v^1 + \bar{w})\Theta^{\bar{1}}\right)$$

By requiring $|u_1^1| = 1$ i.e. $u_1^1 = e^{\mathbf{i}\alpha}$ ($\alpha \in \mathbb{R}$) and $w = \mathbf{i}u_1^1v^{\bar{1}}$, $\bar{w} = -\mathbf{i}u_{\bar{1}}^{\bar{1}}v^1$, we have the generic transformation matrix

$$\begin{pmatrix}\Theta'\\ \Theta'^1\\ \Theta'^{\bar{1}}\\ \phi'\end{pmatrix} = \begin{pmatrix}1 & 0 & 0 & 0\\ v^1 & e^{\mathbf{i}\alpha} & 0 & 0\\ v^{\bar{1}} & 0 & e^{-\mathbf{i}\alpha} & 0\\ s & \mathbf{i}e^{\mathbf{i}\alpha}v^{\bar{1}} & -\mathbf{i}e^{-\mathbf{i}\alpha}v^1 & 1\end{pmatrix}\begin{pmatrix}\Theta\\ \Theta^1\\ \Theta^{\bar{1}}\\ \phi\end{pmatrix}$$

One easily check that the set of these matrices is closed under multiplication therefore forms a subgroup of $\mathrm{GL}(4,\mathbb{C})$ denoted as G_1. This yields a reduction from $\mathrm{GL}(4,\mathbb{C})$-bundle to G_1-bundle or a G_1-structure over $S_+(M,\xi)$. The formal integrability of the CR structure in terms of $\xi^{1,0}$ described in section 2.1 leading to CR structure equation (20) implies directly that this G_1-structure is integrable. The group G_1 is a real 4-dimensional Lie group, hence $\mathscr{Y}$ is a real 8-dimensional manifold.

Proposition 4.2. *There are 1-forms $\phi_1^1, \phi_{\bar{1}}^{\bar{1}}, \phi^1, \phi^{\bar{1}}, \psi$ together with $\Theta, \Theta^1, \Theta^{\bar{1}}, \phi$ such that these 1-forms satisfy the structure equations:*

$$d\Theta = \mathbf{i}\Theta^1\wedge\Theta^{\bar{1}} + \Theta\wedge\phi \tag{21}$$

$$d\phi = \mathbf{i}\Theta^1\wedge\phi^{\bar{1}} + \mathbf{i}\phi^1\wedge\Theta^{\bar{1}} + \Theta\wedge\psi \tag{22}$$

$$d\Theta^1 = \Theta^1\wedge\phi_1^1 + \Theta\wedge\phi^1 \tag{23}$$

$$d\phi_1^1 = \mathbf{i}\Theta^{\bar{1}}\wedge\phi^1 + 2\mathbf{i}\Theta^1\wedge\phi^{\bar{1}} + \frac{1}{2}\Theta\wedge\psi \tag{24}$$

$$d\phi^1 = (\phi_1^1 + \phi_{\bar{1}}^{\bar{1}})\wedge\phi^1 + \phi^1\wedge\phi_1^1 + \frac{1}{2}\Theta^1\wedge\psi + Q_{\bar{1}}^1\Theta^{\bar{1}}\wedge\Theta \tag{25}$$

$$d\psi = (\phi_1^1 + \phi_{\bar{1}}^{\bar{1}})\wedge\psi + 2\mathbf{i}\phi^1\wedge\phi^{\bar{1}} + (R_1\Theta^1 + R_{\bar{1}}\Theta^{\bar{1}})\wedge\Theta \tag{26}$$

with normalization $\phi_1^1 + \phi_{\bar{1}}^{\bar{1}} - \phi = 0$ and $R_1, Q_{\bar{1}}^1$ are some complex-valued functions. The eight 1-forms $\Theta, \Theta^1, \Theta^{\bar{1}}, \phi, \phi_1^1, \phi^1, \phi^{\bar{1}}, \psi$ can be regarded as real linearly independent forms on $\mathscr{Y}$ tautologically.

Proof. The integrability of holomorphic tangent bundle $\xi^{1,0}$ implies:

$$d\Theta^1 = \Theta^1\wedge\phi_1^1 + \Theta\wedge\phi^1$$

with some complex 1-forms ϕ_1^1 , ϕ^1. With a light derivation:

$$\begin{aligned}0 = d(d\Theta) &= d(\mathbf{i}\Theta^1\wedge\Theta^{\bar{1}} + \Theta\wedge\phi)\\ &= \mathbf{i}\Theta^1\wedge\Theta^{\bar{1}}\wedge(-\phi_1^1 - \phi_{\bar{1}}^{\bar{1}} + \phi) + \Theta\wedge(\mathbf{i}\Theta^1\wedge\phi^{\bar{1}} + \mathbf{i}\phi^1\wedge\Theta^{\bar{1}}) - \Theta\wedge d\phi\\ &= d\Theta\wedge(-\phi_1^1 - \phi_{\bar{1}}^{\bar{1}} + \phi) + \Theta\wedge(\mathbf{i}\Theta^1\wedge\phi^{\bar{1}} + \mathbf{i}\phi^1\wedge\Theta^{\bar{1}} - d\phi),\end{aligned}$$

this implies $\phi^1_1 + \phi^{\bar{1}}_{\bar{1}} \equiv \phi \bmod\{\Theta^1, \Theta^{\bar{1}}\}$, we can settle down the equality with normalization

$$\phi = \phi^1_1 + \phi^{\bar{1}}_{\bar{1}}$$

and similarly

$$\begin{aligned} d\phi &\equiv \mathbf{i}\Theta^1 \wedge \phi^{\bar{1}} + \mathbf{i}\phi^1 \wedge \Theta^{\bar{1}} \bmod \Theta \\ &= \mathbf{i}\Theta^1 \wedge \phi^{\bar{1}} + \mathbf{i}\phi^1 \wedge \Theta^{\bar{1}} + \Theta \wedge \psi, \end{aligned}$$

where ψ is another real 1-form on $S_+(M, \xi)$. We further get

$$\begin{aligned} d\phi^1_1 + d\phi^{\bar{1}}_{\bar{1}} = d\phi &= \mathbf{i}\Theta^1 \wedge \phi^{\bar{1}} + \mathbf{i}\phi^1 \wedge \Theta^{\bar{1}} + \Theta \wedge \psi \\ \Rightarrow d\phi^1_1 &= \mathbf{i}\Theta^{\bar{1}} \wedge \phi^1 + 2\mathbf{i}\Theta^1 \wedge \phi^{\bar{1}} + \frac{1}{2}\Theta \wedge \psi, \end{aligned}$$

$$\begin{aligned} 0 = d(d\Theta^1) &= \Theta \wedge (\phi^1 \wedge \phi^1_1 + \phi \wedge \phi^1 + \frac{1}{2}\Theta^1 \wedge \psi - d\phi^1) \\ \Rightarrow d\phi^1 &\equiv \phi \wedge \phi^1 + \phi^1 \wedge \phi^1_1 + \frac{1}{2}\Theta^1 \wedge \psi \bmod \Theta \\ &= (\phi^1_1 + \phi^{\bar{1}}_{\bar{1}}) \wedge \phi^1 + \phi^1 \wedge \phi^1_1 + \frac{1}{2}\Theta^1 \wedge \psi + Q^1_{\bar{1}}\Theta^{\bar{1}} \wedge \Theta, \end{aligned}$$

$$\begin{aligned} 0 = d(d\phi) \Rightarrow d\psi &\equiv \phi \wedge \psi + 2\mathbf{i}\phi^1 \wedge \phi^{\bar{1}} \bmod \Theta \\ &= (\phi^1_1 + \phi^{\bar{1}}_{\bar{1}}) \wedge \psi + 2\mathbf{i}\phi^1 \wedge \phi^{\bar{1}} + (R_1\Theta^1 + R_{\bar{1}}\Theta^{\bar{1}}) \wedge \Theta. \end{aligned}$$

The 1-forms $\phi^1_1, \phi^{\bar{1}}_{\bar{1}}, \phi^1, \phi^{\bar{1}}, \psi$ are uniquely determined via all above relations with respect to coframe $\Theta, \Theta^1, \Theta^{\bar{1}}, \phi$. This is due to the same reason as in the proof of proposition 4.1: Changing coframes leaves the structure equation (20) invariant, then all above derivations out of this equation will remain invariant. If we pullback $\Theta, \Theta^1, \Theta^{\bar{1}}, \phi, \phi^1_1, \phi^1, \phi^{\bar{1}}, \psi$ along the projection $\mathscr{Y} \to S_+(M, \xi)$ we then obtain eight tautological 1-forms on $\mathscr{Y}$ and they only evaluate on the tangent vectors from $TS_+(M, \xi)$ not those from TG_1. Abusing notations, we may say these are eight real linearly independent 1-forms on $\mathscr{Y}$. □

The connection 1-forms in the context of Cartan's moving frame arising from above structure equations are called Chern-Moser connection 1-forms. In particular, $Q^1_{\bar{1}}$, $Q^{\bar{1}}_1$ are called the Cartan curvature.

We see that the CR structure on M is completely determined by the integrable G_1-structure on $S_+(M, \xi)$ namely the CR structure bundle $\mathscr{Y}$. The CR equivalence between CR 3-manifolds induces CR structure bundles isomorphism and vice versa. Since the CR structure bundle is completely determined by the set of 1-forms $\Theta, \Theta^1, \Theta^{\bar{1}}, \phi, \phi^1_1, \phi^{\bar{1}}_{\bar{1}}, \phi^1, \phi^{\bar{1}}, \psi$, we have following consequence:

Corollary 4.3. *$F : M \to M'$ is a CR equivalence, then there is G_1-equivariant bijection*

$$\{\Theta, \Theta^1, \Theta^{\bar{1}}, \phi, \phi^1_1, \phi^{\bar{1}}_{\bar{1}}, \phi^1, \phi^{\bar{1}}, \psi\} \underset{(F^{-1})^*}{\overset{F^*}{\longleftrightarrow}} \{\Theta', \Theta'^1, \Theta'^{\bar{1}}, \phi', \phi'^1_1, \phi'^{\bar{1}}_{\bar{1}}, \phi'^1, \phi'^{\bar{1}}, \psi'\}$$

In this sense, we say the Chern-Moser connection 1-forms are CR invariant.

4.3 Cartan connection on CR structure bundle

We know the space of **Q**-frames are identified with $\mathrm{SU}(2,1)$. The Ehresmann connections for these **Q**-frames coincide with Maurer-Cartan forms of $\mathrm{SU}(2,1)$ and are therefore flat (no curvature form). The first Cartan structure equation is given by

$$dZ_a = \sum_b \pi_a^b \wedge Z_b$$

while the second Cartan's structure equation is given by

$$d\pi_a^b = \sum_c \pi_a^c \wedge \pi_c^b$$

where $a, b, c, = 0, 1, 2$. Performing exterior differential on the inner product (9) we obtain:

$$\begin{aligned}
&\pi_1^1 + \pi_{\bar{1}}^{\bar{1}} = 0 \\
&\pi_0^2 = \pi_{\bar{0}}^{\bar{2}}\ ,\ \pi_2^0 = \pi_{\bar{2}}^{\bar{0}} \\
&\pi_{\bar{0}}^{\bar{0}} + \pi_2^2 = 0 \\
&\frac{\mathbf{i}}{2}\pi_{\bar{1}}^{\bar{0}} + \pi_2^1 = 0 \\
&\pi_{\bar{1}}^{\bar{2}} + 2\mathbf{i}\pi_0^1 = 0 \\
&\pi_0^0 + \pi_1^1 + \pi_2^2 = 0.
\end{aligned}$$

To obtain a Cartan connection on the hyperquadric $\overline{\mathbf{Q}}$ regarded as closed Lie subgroup of $\mathrm{SU}(2,1)$, we pick some **Q**-frames such that $[Z_0] = [t(1, z_1, z_2)]$ lies on $\overline{\mathbf{Q}}$. Then

$$(dZ_0, Z_0) = \frac{\mathbf{i}}{2}\pi_0^2 = |t|^2(\frac{\mathbf{i}}{2}dz_2 + z_{\bar{1}}dz_1) = -\mathbf{i}u\theta$$

by setting $|t|^2 = u$. This implies $\Theta = \frac{1}{2}\pi_0^2$. Together with similar computations for also

$$\pi_0^0 = 2\mathbf{i}(dZ_0, Z_2) \quad , \quad \pi_0^1 = (dZ_0, Z_1)$$

and comparing the structure equations for the Maurer-Cartan forms to the structure equations for the Chern-Moser connections constructed previously on $\mathscr{Y}$ (see proposition 4.2), we can put

$$\begin{aligned}
&\Theta^1 = \pi_0^1 \\
&\phi = -\pi_0^0 + \pi_2^2 = \pi_0^0 - \pi_{\bar{0}}^{\bar{0}} \\
&\phi^1 = 2\pi_2^1 \\
&\phi_1^1 = \pi_1^1 - \pi_0^0 \\
&\psi = -4\pi_2^0.
\end{aligned}$$

Conversely we get

$$\begin{aligned}
\pi_0^2 &= 2\Theta \\
\pi_0^0 &= -\frac{1}{3}(\phi_1^1 + \phi) \\
\pi_0^1 &= \Theta^1 \\
\pi_1^2 &= 2\mathbf{i}\Theta^{\bar{1}} \\
\pi_2^1 &= \frac{1}{2}\phi^1 \\
\pi_1^1 &= \phi_1^1 + \pi_0^0 \\
\pi_2^0 &= -\frac{1}{4}\psi.
\end{aligned}$$

The structure equation for the Maurer-Cartan is equivalent to the structure equation of the Chern-Moser connections with $R_1 = R_{\bar{1}} = Q_{\bar{1}}^{\bar{1}} = Q_1^{\bar{1}} = 0$. Such G-structure (here G is G_1) is flat. In fact, it follows from Frobenius theorem that every real analytic flat G_1-structure is locally equivalent to the one arises from a nondegenerate real hyperquadric in $\mathbb{C}^{n+1}$ [21] (here in 3-dimensional case $n = 1$).

In order to relate the structure group G_1 of $\mathscr{Y}$ over $\overline{\mathbf{Q}}$ with $\mathrm{SU}(2,1)$, we now consider H_1-action on $\mathbf{Q}$-frames induces action on Maurer-Cartan forms and it further induces an action on Chern-Moser connections. Then comparing Chern-Moser forms $\Theta, \Theta^1, \Theta^{\bar{1}}, \phi$ changed under original G_1-action, this gives rise the following:

Proposition 4.4 ([21] (5.13a), [26] §3)**.** *There is a group homomorphism*

$$\mathfrak{F} : H \to G_1$$
$$\begin{pmatrix} t & 0 & 0 \\ \bar{t}\tau & \bar{t}t^{-1} & 0 \\ t(r + \mathbf{i}|\tau|^2) & 2\mathbf{i}\bar{\tau} & \bar{t}^{-1} \end{pmatrix} \mapsto \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2\bar{t}^2\tau & \bar{t}^3 & 0 & 0 \\ 2t^2\bar{\tau} & 0 & t^3 & 0 \\ 4r & 2\mathbf{i}\bar{t}\bar{\tau} & -2\mathbf{i}t\tau & 1 \end{pmatrix}$$

where $t \in \mathbb{C}^\times, \tau \in \mathbb{C}, r \in \mathbb{R}$. The kernel of this homomorphism is K. This leads to isomorphism $H_1/K \cong G_1$ (denote this induced isomorphism as $\overline{\mathfrak{F}} : H_1/K \xrightarrow{\cong} G_1$) .

In conclusion, the Cartan connections for particular set of $\mathbf{Q}$-frames, whose transformation group is $H_1/K \cong G_1$, are the Maurer-Cartan forms of this group. With these transformation matrices given, the Cartan connections on $\overline{\mathbf{Q}}$ are uniquely determined by the Chern-Moser forms.

On the real hyperquadric, the Cartan connections are flat hence satisfies the Maurer-Cartan equation. However the relations between those Cartan connections and Chern-Moser forms can also be well established in the exact same way for any strictly pseudo-convex CR 3-manifold whose CR structure bundle might not be flat. So we can define an $\mathfrak{su}(2,1)$-valued form on $\mathscr{Y}$ with entries of Chern-Moser forms:

$$\pi = \begin{pmatrix} \pi_0^0 & \pi_0^1 & \pi_0^2 \\ \pi_1^0 & \pi_1^1 & \pi_1^2 \\ \pi_2^0 & \pi_2^1 & \pi_2^2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{3}(\phi_1^1 + \phi) & \Theta^1 & 2\Theta \\ -\mathbf{i}\phi^{\bar{1}} & \frac{1}{3}(\phi_1^1 - \phi_{\bar{1}}^{\bar{1}}) & 2\mathbf{i}\Theta^{\bar{1}} \\ -\frac{1}{4}\psi & \frac{1}{2}\phi^1 & \frac{1}{3}(\phi + \phi_{\bar{1}}^{\bar{1}}) \end{pmatrix}. \tag{27}$$

By showing that π is indeed a Cartan connection, we realize that the formula (27) is indeed the correct one for any strictly pseudo-convex CR 3-manifold and it is unique (up to G_1 or H_1/K action) by corollary 4.3.

Proposition 4.5. *π defined uniquely a Cartan connection on $\mathscr{Y}$.*

Proof. By proposition 0.19, to show that π gives rise a globally defined Cartan connection, one it suffices to check its local transformation properties: Let $g : U \times \mathbb{R}_+ \cong \mathrm{pr}^{-1}(U) \subset S_+(M,\xi) \to G_1$ be a transition function on $\mathscr{Y} \xrightarrow{p} S_+(M,\xi)$ and this induces $h := \overline{\mathfrak{F}}^{-1} \circ g : U \times \mathbb{R}_+ \cong \mathrm{pr}^{-1}(U) \to H_1/K$ by setting parameters

$$t^3 = e^{\mathbf{i}\alpha} \quad , \quad \tau = \frac{v^1\bar{t}}{2} \quad , \quad r = \frac{s}{4}.$$

For transformation of local section $\sigma : U\times\mathbb{R}_+ \cong \mathrm{pr}^{-1}(U) \to \mathscr{Y}$ given by $\sigma'(x,u) = (\Theta', \Theta'^1, \Theta'^{\bar{1}}, \phi')_{(x,u)} = (\Theta, \Theta^1, \Theta^{\bar{1}}, \phi)_{(x,u)} \cdot g(x,u) = \sigma(x,u) \cdot g(x,u)$, we define the right H_1/K-action on the local section naturally as $\sigma \cdot h := \sigma \cdot g$, then we check that

$$\pi' = \mathrm{Ad}(h^{-1}) \circ \pi + h^{-1}dh.$$

Let $\mathfrak{h}$ be the Lie algebra of $G_1 \cong H_1/K \subset \mathrm{SU}(2,1)/K$. The adjoint action of G_1 on $\mathfrak{su}(2,1)$ preserves $\mathfrak{h}$ and induces action on the complement $\mathfrak{m} \cong \mathfrak{su}(2,1)/\mathfrak{h}$. Now we want to show $\mathrm{pr}_{\mathfrak{m}} \circ \pi : T_{\mathbb{C}}S_+(M,\xi)|_x \to \mathfrak{su}(2,1)/\mathfrak{h}$ is an isomorphism. Recall the decomposition of Lie algebra from section 2.4 and note that

$$\mathfrak{m} = \mathfrak{g}_1 + \mathfrak{g}_2 = \left\{ \begin{pmatrix} 0 & b & r \\ 0 & 0 & 2\mathbf{i}\bar{b} \\ 0 & 0 & 0 \end{pmatrix} \,:\, r \in \mathbb{R}, b \in \mathbb{C} \right\}.$$

Comparing to the entries in π and evaluating on tangent vectors, we find out $\Theta^1 \to b$, $2\Theta \to r$. So there is an isomorphism between these two complex vector spaces. Thus by proposition 0.19, π indeed defines uniquely a Cartan connection of $\mathrm{PU}(2,1)/(H_1/K)$-type on principal H_1/K-bundle $\mathscr{Y} \xrightarrow{p} S_+(M,\xi)$ so that π is obtained via pullback σ^* of this global 1-form on $\mathscr{Y}$ to $\mathrm{pr}^{-1}(U) \subset S_+(M,\xi)$, i.e. $\pi = \sigma^*\omega \in \Omega^1(\mathrm{pr}^{-1}(U); \mathfrak{su}(2,1))$ where $\omega \in \Omega^1(\mathscr{Y}; \mathfrak{su}(2,1))$. □

The curvature 2-form is given by

$$\Pi = d\pi + \pi \wedge \pi = \begin{pmatrix} \Pi_0^0 & 0 & 0 \\ \Pi_1^0 & \Pi_1^1 & 0 \\ \Pi_2^0 & \Pi_2^1 & \Pi_2^2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ -\mathbf{i}Q_1^{\bar{1}}\Theta^1 \wedge \Theta & 0 & 0 \\ -\frac{1}{4}(R_1\Theta^1 + R_{\bar{1}}\Theta^{\bar{1}}) \wedge \Theta & \frac{1}{2}Q_{\bar{1}}^1\Theta^{\bar{1}} \wedge \Theta & 0 \end{pmatrix}. \tag{28}$$

Correspondingly there is curvature 2-form $\Omega = d\omega + \omega \wedge \omega \in \Omega^2(\mathscr{Y}; \mathfrak{su}(2,1))$ so that $\Pi = \sigma^*\Omega \in \Omega^2(\mathrm{pr}^{-1}(U); \mathfrak{su}(2,1))$.

Theorem 4.6. *There is a uniquely defined torsion-free $\mathfrak{su}(2,1)$-valued Cartan connection ω on $\mathscr{Y}$ with $tr\Omega = 0$.*

Proof. The torsion 2-form is the projection of Ω on $\mathfrak{su}(2,1)/\mathfrak{h}$. We can see that Π takes value in $\mathfrak{g}_{-2} + \mathfrak{g}_{-1} \subset \mathfrak{h}$ so the projection on $\mathfrak{m} = \mathfrak{g}_1 + \mathfrak{g}_2$ is empty. This implies the torsion form vanishes. The uniqueness follows straightforward from previous proposition. □

By the discussion right above proposition 0.20, we can also view $\mathscr{Y}$ directly as a principal bundle over M:

Theorem 4.7 ([26] Theorem 4.1)**.** *There exists a free H/K-action on $\mathscr{Y}$ turning $\mathscr{Y}$ into a H/K-bundle over M. In particular, this bundle admits a unique Cartan connection of type $PU(2,1)/(H/K)$ determined by the Chern–Moser connection.*

4.4 Inclusion of pseudo-Hermitian structure bundle into CR structure bundle

We know $\mathscr{X} \to M$ is uniquely determined by the fixed θ via structure equation $\theta = \mathbf{i}\theta^1 \wedge \theta^{\bar{1}}$. This means we pick a particular global section $M \to S_+(M,\xi)$, w.o.l.g. this equivalent to fix $u = 1$. On a local trivialization neighborhood $\mathrm{pr}^{-1}(U) \subset S_+(M,\xi)$, the tautological forms $(\Theta, \Theta^1, \Theta^{\bar{1}}, \phi)$ now coincide with original forms $(\theta, \theta^1, \theta^{\bar{1}})$ with $\phi = 0$ ($du = 0, \varphi = 0$). This yields a injective map of a generic point in $\mathscr{X}$ into $\mathscr{Y}$ given by

$$\mathscr{X} \supset \{(\theta^1, \theta^{\bar{1}}) : d\theta = \mathbf{i}\theta^1 \wedge \theta^{\bar{1}}\} \overset{1:1}{\leftrightarrow} \{(\Theta, \Theta^1, \Theta^{\bar{1}}, \phi) : \Theta = \mathrm{pr}^*\theta, u = 1, \phi = 0\} \subset \mathscr{Y}.$$

Proposition 4.8. *There is a bundle inclusion $\iota : \mathscr{X} \to \mathscr{Y}$ with $\iota|_M = id_M$. In particular, it is a homotopy equivalence.*

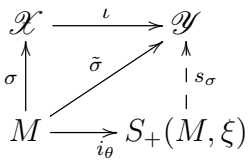
Proof. There is a base inclusion $i_\theta : M \hookrightarrow S_+(M,\xi)$ associated to the given θ. We show that such bundle inclusion exits fiberwise. Pick a reference coframe $(\theta^1, \theta^{\bar{1}})$, starting with the pointwise injective map discussed above, denoted now as ι, we find the corresponding coframe $(\Theta, \Theta^1, \Theta^{\bar{1}}, 0)$ on $S_+(M,\xi)$ with respect to the chosen θ. Next we look for adapted H_1-action on $(\Theta, \Theta^1, \Theta^{\bar{1}}, 0)$ so that the resulted new coframe $(\Theta', \Theta'^1, \Theta'^{\bar{1}}, \phi')$ after transformation still satisfies $\Theta' = \Theta = \theta, \phi' = \phi = 0$ and this implies there is a new coframe $(\theta'^1, \theta'^{\bar{1}}) = \iota^{-1}(\Theta', \Theta'^1, \Theta'^{\bar{1}}, 0)$ on M. Apply the H_1-action on $(\Theta, \Theta^1, \Theta^{\bar{1}}, 0)$ defined previously in the proof of proposition 4.5, and it is clear that r, τ must vanish and by norm preserving requirement $t = \bar{t}^{-1} \in \mathrm{U}(1)$. Thus we have group injective homomorphism:

$$\begin{gathered} \mathrm{U}(1) \hookrightarrow H_1 \\ \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix} \mapsto \begin{pmatrix} t & 0 & 0 \\ 0 & t^{-2} & 0 \\ 0 & 0 & t \end{pmatrix}. \end{gathered}$$

This gives the bundle inclusion $\mathscr{X} \hookrightarrow \mathscr{Y}$.

Furthermore H_1 deformation retracts to the image of the injective group homomorphism above, i.e. $\mathrm{U}(1)$, and $S_+(M,\xi)$ obviously deformation retracts to M, hence this inclusion is a homotopy equivalence. □

This inclusion induces a local section map $\tilde{\sigma} : M \to \mathscr{X} \to \mathscr{Y}$ which factors through the local section $\sigma : M \to \mathscr{X}$ giving local coframe field $(\theta^1, \theta^{\bar{1}})$. Abstractly speaking, there exists correspondingly a unique local section $s_\sigma : S_+(M,\xi) \to \mathscr{Y}$ such that the following diagram commutes:



Therefore one can derive a relation between the Tanaka-Webster connection (in terms of pseudo-Hermitian structure that is obviously not CR invariant since it is not invariant under the full G_1-action) and the Chern-Moser connection (which is CR invariant forms on $S_+(M,\xi)$). These Chern-Moser connection 1-forms are determined by some specific coframes obtained from the inclusion of $\mathscr{X}$ into $\mathscr{Y}$ and in this way one can eventually expressed the Chern-Moser connection and curvature on $S_+(M,\xi)$ in terms of Tanaka-Webster connection and curvature on M. We demonstrate the desired relation below.

Under the inclusion ι, the structure equations for the admissible coframe fields and Chern-Moser connection 1-forms on $S_+(M,\xi)$ turn out to be:

$$\begin{aligned}
d\theta^1 &= \theta^1\wedge\phi^1_1+\theta\wedge\phi^1\\
0 &= \mathbf{i}\theta^1\wedge\phi^{\bar 1}+\mathbf{i}\phi^1\wedge\theta^{\bar 1}+\theta\wedge\psi\\
d\phi^1_1 &= \mathbf{i}\theta^{\bar 1}\wedge\phi^1+2\mathbf{i}\theta^1\wedge\phi^{\bar 1}+\frac{1}{2}\theta\wedge\psi\\
d\phi^1 &= \phi^1\wedge\phi^1_1+\frac{1}{2}\theta^1\wedge\psi+Q^1_{\bar 1}\theta^{\bar 1}\wedge\theta\\
d\psi &= 2\mathbf{i}\phi^1\wedge\phi^{\bar 1}+(R_1\theta^1+R_{\bar 1}\theta^{\bar 1})\wedge\theta
\end{aligned}$$

Comparing with the structure equation in proposition 4.2, we can write $\phi^1_1=\theta^1_1+\mathbf{i}B\theta$ and thus $d\theta^1=\theta^1\wedge\theta^1_1+\theta\wedge(\phi^1-\mathbf{i}B\theta^1)$ with B real. This implies $\phi^1=\mathbf{i}\beta\theta^1+E\theta+\tau^1$. Together with the second equation above we get $\theta\wedge(\psi+\mathbf{i}E\theta^{\bar 1}-\mathbf{i}\bar E\theta^1)=0$ and hence $\psi=G\theta+\mathbf{i}(\bar E\theta^1-E\theta^{\bar 1})$ with G real.

Similarly from the third equation above, we get

$$\begin{aligned}
&d\phi^1_1=d(\theta^1_1+\mathbf{i}B\theta)=\mathbf{i}\theta^{\bar 1}\wedge\phi^1+2\mathbf{i}\theta^1\wedge\phi^{\bar 1}+\frac{1}{2}\theta\wedge\psi\\
&R\theta^1\wedge\theta^{\bar 1}+W\theta^1\wedge\theta-\overline{W}\theta^{\bar 1}\wedge\theta-B\theta^1\wedge\theta^{\bar 1}+\mathbf{i}dB\wedge\theta\\
&\quad=\mathbf{i}\theta^{\bar 1}\wedge(\mathbf{i}B\theta^1+E\theta+\tau^1)+2\mathbf{i}\theta^1\wedge(-\mathbf{i}B\theta^{\bar 1}+\bar E\theta+\tau^{\bar 1})+\frac{1}{2}\theta\wedge(G\theta+\mathbf{i}\bar E\theta^1-\mathbf{i}E\theta^{\bar 1})\\
&\quad=3B\theta^1\wedge\theta^{\bar 1}+\frac{3}{2}\mathbf{i}E\theta^{\bar 1}\wedge\theta+\frac{3}{2}\mathbf{i}\bar E\theta^1\wedge\theta^{\bar 1}
\end{aligned}$$

implying

$$(R-4B)\theta^1\wedge\theta^{\bar 1}+(W-\frac{3}{2}\mathbf{i}\bar E)\theta^1\wedge\theta-(\overline{W}+\frac{3}{2}\mathbf{i}E)\theta^{\bar 1}\wedge\theta+\mathbf{i}dB\wedge\theta=0.$$

This shows that $B=\frac{R}{4}$, we then write $dR=R_{,0}\theta+R_{,1}\theta^1+R_{,\bar 1}\theta^{\bar 1}$ and get

$$(W-\frac{3}{2}\mathbf{i}\bar E+\mathbf{i}\frac{1}{4}R_{,1})\theta^1\wedge\theta-(\overline{W}+\frac{3}{2}\mathbf{i}E-\frac{1}{4}\mathbf{i}R_{,\bar 1})\theta^{\bar 1}\wedge\theta=0$$

which further implies

$$E=\frac{R_{,\bar 1}}{6}+\frac{2\mathbf{i}\overline{W}}{3}.$$

In summary, we have the expression for these admissible Chern-Moser connection forms pulled down to $\mathscr{X}$ in terms of Tanaka-Webster connection as well as torsion forms:

$$\begin{aligned}
\phi^1_1&=\theta^1_1+\frac{\mathbf{i}R}{4}\theta\\
\phi^1&=\frac{\mathbf{i}R}{4}\theta^1+E\theta+\tau^1\\
\psi&=G\theta+\mathbf{i}(\bar E\theta^1-E\theta^{\bar 1})\\
E&=\frac{1}{6}R_{,1}-\frac{2\mathbf{i}}{3}A^{\bar 1}_{1,\bar 1}
\end{aligned}$$

where G as we have seen in above derivation is not important and can be taken as zero. Differentiating the expression of ϕ^1 above and comparing to the structure equation of $d\phi^1$, we get:

$$Q^{\bar 1}_1=\frac{1}{6}R_{,1}{}^{\bar 1}+\frac{\mathbf{i}}{2}RA^{\bar 1}_1-A^{\bar 1}_{1,0}-\frac{2\mathbf{i}}{3}A^{\bar 1}_{1,\bar 1}{}^{\bar 1}. \tag{29}$$

The point of this procedure is to pull down the CR invariant data on $\mathscr{Y} \to S_+(M,\xi)$ to the pseudo-Hermitian invariant data on $\mathscr{X} \to M$. The ultimate goal is to define a global invariant via CR data expressed locally in terms of pseudo-Hermitian data. Maybe for the moment it is not clear whether in such way the global quantity is a CR invariant. However, we will see that the so-called Burns-Epstein global invariant is indeed independent of the choice of $\sigma : M \to \mathscr{X}$, which surprisingly turns out to be a well-defined global CR invariant.

5 Spherical CR structures

Definition 5.1. *A $(2n+1)$-dimensional CR manifold is spherical if it is CR equivalent (i.e. locally CR diffeomorphic) to a unit sphere with standard CR structure.*

At this point, it is good to mention some background of geometric structures to see the motivation of studying spherical CR structures. We have mentioned a few words on Klein geometry, Cartan geometry as well as Ehresmann's geometric structure before. In particular for Cartan geometry, the Cartan connections and curvatures serve as local invariants. In fact, Ehresmann structures are special cases of flat Cartan connections with vanishing curvatures, c.f. [9]. Ehresmann structures modeled on Euclidean space is equivalent to a flat Riemannian metric. More generally, constant curvature Riemannian metrics are can be regarded as Ehresmann structures modeled on the sphere or hyperbolic space and their respective isometric groups. William Thurston proposed his famous geometrization conjecture in terms of Ehresmann structures, in which the relevant geometries are locally homogeneous 3-dimensional Riemannian manifolds with compact isotropy groups, whose universal covering spaces are up to local isometry eight type of local homogeneous spaces with non-compact isotropy groups. A fine survey for this topic is [29]. Among these eight types spaces, the hyperbolic space is of particular interest. In two dimensional case, due to the unifromization theorem, the hyperbolic space provide the local model for all surfaces making the studying of hyperbolic surface a huge industry. In higher dimensional case, it is thus natural to ask which manifold is locally modeled on hyperbolic space and its isometric group. In a Riemannian n-manifold is conformally flat if it is locally modeled on $(\mathbb{S}^n, \mathrm{SO}(n+1,1))$ where $\mathbb{S}^n$ is considered as the boundary of real hyperbolic space. A CR analogy can be made, so one can say a geometric structure is called a spherical CR structure if the local model is provided to be $(\mathbb{S}^{2n+1}, \mathrm{PU}(n+1,1))$ where $\mathbb{S}^{2n+1}$ is the boundary of unit ball in $\mathbb{C}^{n+1}$ (it can be equivalently thought as the boundary of complex hyperbolic space, we have introduced this space in dimension 3 in section 2.4) and $\mathrm{PU}(n+1,1)$ the group of biholomorphisms of the unit ball acting on its boundary by CR automorphisms. There are many parallels between conformally flat structures and spherical CR structures and also big differences. For example, shown in [25] the 3-torus admits a conformally flat structure but not spherical CR structure. There are some uniformization results for spherical CR 3-manifolds in terms of developing map and holonomy group, c.f. [27].

In 3-dimensional case, a necessary and sufficient condition for a Riemannian metric to be conformally flat is that the corresponding Cotton tensor vanishes. In higher dimension case (greater than 4), the condition becomes the vanishing of Weyl tensor (Cotton tensor can be derived from Weyl tensor). Similarly, there are also some curvature quantities that can determine whether a CR structure is spherical or not.

Theorem 5.2 (Cartan, Chern-Moser)**.** *Let (M,θ) be a strictly pseudo-convex pseudo-Hermitian manifold of dimension $2n+1$.*

- *If $2n+1=3$, then M is locally CR spherical if and only if the Cartan tensor*

$$Q_{\alpha\beta} = \mathbf{i}(A_{\alpha\beta,0} - 2\phi_{\alpha,\beta}) + 2P_{\alpha\bar{\nu}}A^{\bar{\nu}}_{\beta}$$

vanishes, where

$$\phi_\alpha = \frac{1}{n+2}\left(\frac{R_\alpha}{2n+2} - \mathbf{i}A_{\alpha\beta,\bar{\nu}}h^{\beta\bar{\nu}}\right)\ ,\ P_{\alpha\bar{\beta}} = \frac{1}{n+2}\left(R_{\alpha\bar{\beta}} - \frac{R}{2n+2}h_{\alpha\bar{\beta}}\right)\ ,\ R = h^{\alpha\bar{\beta}}R_{\alpha\bar{\beta}}\ ,\ R_{\alpha\bar{\beta}} = -R_{\alpha\ \bar{\beta}\sigma}^{\ \ \sigma}.$$

- *If $2n+1 \geq 5$, then M is locally CR spherical if and only if the Chern tensor below vanishes:*

$$C_{\mu\ \bar{\nu}\alpha}^{\ \beta} = -R_{\mu\ \bar{\nu}\alpha}^{\ \beta} - \frac{1}{m+2}(R^\beta_\alpha h_{\mu\bar{\nu}} + R^\beta_\mu h_{\alpha\bar{\nu}} + \delta^\beta_\alpha R_{\mu\bar{\nu}} + \delta^\beta_\mu R_{\alpha\bar{\nu}}) + \frac{R}{(m+1)(m+2)}(\delta^\beta_\alpha h_{\mu\bar{\nu}} + \delta^\beta_\mu h_{\alpha\bar{\nu}}).$$

The Cartan tensor was first introduced by Cartan in [19]. Later in [21], Chern's pseudo-conformal curvature tensor vanishing would also imply the CR manifolds CR equivalent to the standard one on sphere. The Cartan tensor can be extract as a part of Chern's pseudo-conformal curvature tensor. A rather concise exposition on this entire aspect can be found in [40] Chapter 7. In 3-dimensional case, in terms of local coordinate, the Cartan tensor for (M, ξ, J) can be expressed as (c.f. [32]):

$$Q_J = \mathbf{i}Q_{\bar{1}}^{\bar{1}}\theta^1 \otimes Z_{\bar{1}} - \mathbf{i}Q_{\bar{1}}^{1}\theta^{\bar{1}} \otimes Z_1 \tag{30}$$

where $Q_{\bar{1}}^{\bar{1}}$ is derived as (29).

Example.

(1) The simplest spherical CR 3-manifolds are unit 3-sphere $\mathbb{S}^3$ with standard CR structure, and 3-dimensional Heisenberg group $\mathbf{H}_1$ with standard CR structure since it has vanishing curvature (hence the Cartan curvature tensor must vanish). This of course holds for all $2n+1$ dimensional case, i.e. $\mathbb{S}^{2n+1}$ and $\mathbf{H}_n$. In fact, any local diffeomorphism on spheres $\mathbb{S}^{2n+1}$ is a global diffeomorphism[2], so the CR equivalence on spheres is actually global CR diffeomorphism. One concludes in particular that there is only a unique spherical CR structure on $\mathbb{S}^3$ which is the standard one $(\xi_{\text{std}}, \mathbf{i})$ up to CR diffeomorphism. Consequently, $(\mathbb{S}^3, \xi_{\text{OT}})$ does not admit any spherical CR structure, and also no holomorphically fillable CR structure. Similar holds for the Heisenberg groups.

(2) Let $\mathscr{L}$ be a negative holomorphic line bundle over a closed Riemann surface Σ of genus greater then 1 with unique constant curvature $K = -1$ hyperbolic metric g. The volume form $\omega = \frac{d\text{vol}_g}{2\pi\chi(\Sigma)}$ represent a generator in $H^2(\Sigma; \mathbb{Z}) \cong \mathbb{Z}$, the first Cher class is written as $c_1(\mathscr{L}) = -m[\omega] \in H^2(\Sigma; \mathbb{Z})$ with $m \in \mathbb{Z}_+$. There is a unique up to positive constant multiple Hermitian metric $\|\cdot\|$ on $\mathscr{L}$. For a local section s there is $\mathbf{i}\partial_z\partial_{\bar{z}}\ln(\|s(z)\|^2) = 2\pi\omega$. Define the unit circle bundle $\mathbb{S}\mathscr{L} \subset \mathscr{L}$ with respect to this metric. This is a closed CR 3-manifold with contact form θ such that $d\theta = \text{pr}^*\omega$ where $\text{pr} : \mathbb{S}\mathscr{L} \to \Sigma$ is the projection map, the complex structure is induced from L naturally. Pick ω^1, ω^2 be local orthonormal coframe fields with respect to g, and $\eta^1 = \omega^1 + \mathbf{i}\omega^2$, $\eta^{\bar{1}} = \omega^1 - \mathbf{i}\omega^2$. Then $d\text{vol}_g = \omega^1 \wedge \omega^2 = \frac{1}{2}\mathbf{i}\eta^1 \wedge \eta^{\bar{1}}$. Put $\theta^1 = \text{pr}^*\eta^1$, $\theta^{\bar{1}} = \text{pr}^*\eta^{\bar{1}}$, $g_{1\bar{1}} = \frac{1}{2}$, and compute that $R = -1$ and $A_{\bar{1}}^{\bar{1}} = A_{\bar{1}}^{\bar{1}} = 0$, hence $Q_J = 0$ by (29). In fact, any circle bundle over Riemann surface arising in such way is spherical CR 3-manifold regardless it is trivial or not. The negativity here ensures there is no global holomorphic section for $\mathscr{L}$ so that it is non-trivial. More generally, it is shown in §4 [27] that: Any circle bundle over a closed orientable surface of arbitrary genus admits a spherical CR structure. According to constructions, every such circle bundle admits a uniformizable one and among those with genus ≥ 2 can carry also some non-uniformizable ones.

(3) We know $\text{Aut}_{\text{CR}}(\mathbb{S}^3) \cong \text{PU}(2,1)$. Consider a free (no fixed point) finite subgroup Γ. Since $\text{Aut}_{\text{CR}}(\mathbb{S}^3) \subset \text{Cont}(\mathbb{S}^3, \xi_{\text{std}})$, the quotient manifold $\Gamma\backslash\mathbb{S}^3$, where $\mathbb{S}^3$ can be thought as SU(2) with invariant CR structure (the standard one on $\mathbb{S}^3$ is SU(2)-invariant), inherits tight contact structure and spherical CR structure from 3-sphere. The space of such type is called the CR lens space. Burns and Shnider proved in 1976 [28] that the CR lens space is rigid, namely there is no non-trivial CR deformation of the manifold. In other words, one can not smoothly varying spherical CR structure on this manifold. The reason lies purely on algebraic level: By applying some extended version of Kodaira-Spencer-Kuranishi theory, the tangential direction of the deformation space can be studied with Lie algebra cohomology (Hochschild-Serre spectral sequence is thus involved for the computation). Moreover, the CR lens space has positive constant Tanaka-Webster curvature and vanishing torsion.

[2]A local homeomorphism from a compact Hausdorff space to a connected Hausdorff space is a covering map. For a sphere (of any dimension), this is indeed the case. Further for dimension ≥ 2, due to the simpliy connectedness of sphere, it is the universal covering space on its own, so the covering map is a global homeomorphism. Compatible with the smooth structure, local diffeomorphism on sphere yields global diffeomorphism.

In fact, according to R.R. Miner's result, any compact spherical CR $(2n+1)$-manifold with amenable holonomy (a topological group is called amenable if for every continuous group action on a compact metrizable space there exists a invariant probability measure on the space) has finite-sheeted covering space as $\mathbb{S}^{2n+1}$, $\mathbb{S}^1 \times \mathbb{S}^{2n}$ or $\Gamma\backslash\mathbf{H}_n$ where $\mathbf{H}_n$ is the $(2n+1)$-dimensional real Heisenberg group and Γ is a lattice in it. W. Goldman has obtained the same classification despite that in his result the assumption of holonomy is no longer amenable but nilpotent instead. In particular, for the case where the discrete subgroup $\Gamma \subset \mathbf{H}_1$ makes $\Gamma\backslash\mathbf{H}_1$ compact, such quotient manifold can be interpreted geometrically as a unit circle bundle in positive holomorphic line bundle over compact Riemann surface with genus 1 (c.f. [28]).

6 Burns-Epstein invariant

D. M. Burns and C. L. Epstein defined a global biholomorphic invariant for closed strictly pseudo-convex CR 3-manifolds with trivial holomorphic tangent bundle in 1988 [30]. Their idea comes from Chern-Simons conformal invariant on closed oriented Riemannian 3-manifolds. We will discuss the basic construction and properties of this so-called Burns-Epstein invariant. Some simple examples with computations are illustrated. Further we study its generalization (still called Burns-Epstein invariant) by J. S. Cheng and J. M. Lee relaxing the assumption of trivial $\xi^{1,0}$, namely on any closed strictly pseudo-convex CR 3-manifold. In particular the variational aspect of Burns-Epstein invariant is analyzed in detail.

6.1 Chern-Simons form and global CR invariant

We consider for the moment a closed strictly pseudo-convex CR 3-manifold $(M, \xi^{1,0})$ whose holomorphic tangent bundle $\xi^{1,0}$ is trivial. Then all the differential forms (coframes and connections) we discussed previously are globally defined. Using Chern-Weil theory (see appendix A), the second Chern class on $S_+(M,\xi)$ (considered as a 4-dimensional manifold) in terms of Π given by

$$C_2(\Pi) = \frac{1}{8\pi^2}[\mathrm{tr}(\Pi\wedge\Pi) + \mathrm{tr}(\Pi)\wedge\mathrm{tr}(\Pi)] = \frac{1}{8\pi^2}\mathrm{tr}(\Pi\wedge\Pi)$$

where the second term $\mathrm{tr}(\Pi)\wedge\mathrm{tr}(\Pi)$ vanishes because Π is traceless. Moreover, $C_2(\Pi)\equiv 0$ because Π is strictly lower triangular. So the transgression form (refer to Appendix A) for the second Chern class $C_2(\Pi) = dTC_2(\pi)$ is given by the so-called Chern-Simons 3-form:

$$TC_2(\pi) = \frac{1}{8\pi^2}\Big[\mathrm{tr}(\pi\wedge\Pi) + \frac{1}{3}\mathrm{tr}(\pi\wedge\pi\wedge\pi)\Big] = s^*TC_2(\omega)\in H^3(S_+(M,\xi);\mathbb{R}),$$

where $s: S_+(M,\xi)\to\mathscr{Y}$ is any global section. $TC_2(\omega)\in H^3(\mathscr{Y};\mathbb{R})$ is a CR invariant 3-form on $\mathscr{Y}$ since ω is uniquely determined for a CR structure up to CR equivalence (theorem 4.6 and corollary 4.3).

Proposition 6.1. *$[TC_2(\pi)] = s^*[TC_2(\omega)]$ is independent of the choice of section $s: S_+(M,\xi)\to\mathscr{Y}$.*

Proof. Let s and s' be arbitrary two sections so that $s' = s\cdot h$ pointwise with some $h\in H_1/K$.

$$TC_2(\omega) = \frac{1}{8\pi^2}\left[\mathrm{tr}(\omega\wedge\Omega) + \frac{1}{3}\mathrm{tr}(\omega\wedge\omega\wedge\omega)\right] = \frac{1}{24\pi^2}\mathrm{tr}(\omega\wedge\omega\wedge\omega)$$

where we have used $s^*\,\mathrm{tr}(\omega\wedge\Omega) = \mathrm{tr}(\pi\wedge\Pi) = 0$ for the second equality, which can be seen straightforwardly:

$$\begin{aligned}\pi\wedge\Pi &= \begin{pmatrix} -\frac{1}{3}(\phi_1^1+\phi) & \Theta^1 & 2\Theta \\ -\mathbf{i}\phi^{\bar1} & \frac{1}{3}(\phi_1^1-\phi_{\bar1}^{\bar1}) & 2\mathbf{i}\Theta^{\bar1} \\ -\frac{1}{4}\psi & \frac{1}{2}\phi^1 & \frac{1}{3}(\phi+\phi_{\bar1}^{\bar1})\end{pmatrix}\wedge\begin{pmatrix} 0 & 0 & 0 \\ -\mathbf{i}Q_1^{\bar1}\Theta^1\wedge\Theta & 0 & 0 \\ -\frac{1}{4}(R_1\Theta^1+R_{\bar1}\Theta^{\bar1})\wedge\Theta & \frac{1}{2}Q_{\bar1}^1\Theta^{\bar1}\wedge\Theta & 0\end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & 0 \\ -\frac{\mathbf{i}}{3}Q_1^{\bar1}(\phi_1^1-\phi_{\bar1}^{\bar1})\wedge\Theta^1\wedge\Theta - \frac{\mathbf{i}}{2}R_1\Theta^{\bar1}\wedge\Theta^{\bar1}\wedge\Theta & 0 & 0 \\ -\frac{\mathbf{i}}{2}Q_1^{\bar1}\phi^1\wedge\Theta^1\wedge\Theta - \frac{1}{12}(\phi+\phi_{\bar1}^{\bar1})\wedge(R_1\Theta^1+R_{\bar1}\Theta^{\bar1})\wedge\Theta & \frac{1}{6}Q_{\bar1}^1(\phi+\phi_{\bar1}^{\bar1})\wedge\Theta^{\bar1}\wedge\Theta & 0\end{pmatrix}.\end{aligned}$$

Put $\pi' = s'^*\omega$, by the connection transformation formula $\pi' = \mathrm{Ad}(h^{-1})\pi + h^{-1}dh$ and the trace property for product of matrices, we have:

$$\begin{aligned} TC_2(\pi') &= \frac{1}{24\pi^2}\mathrm{tr}(\pi' \wedge \pi' \wedge \pi') \\ &= \frac{1}{24\pi^2}\,\mathrm{tr}(\pi \wedge \pi \wedge \pi + 3h^{-1}dh \wedge h^{-1}\pi \wedge dh + 3h^{-1}\pi \wedge \pi \wedge dh + h^{-1}dh \wedge h^{-1}dh \wedge h^{-1}dh) \\ &= \frac{1}{24\pi^2}\,\mathrm{tr}(\pi \wedge \pi \wedge \pi - 3dh^{-1} \wedge \pi \wedge dh + 3h^{-1}\pi \wedge \pi \wedge dh - h^{-1}dh \wedge dh^{-1} \wedge dh) \end{aligned}$$

With straightforward computation one can show that

$$\begin{aligned} &\mathrm{tr}(h^{-1}dh \wedge dh^{-1} \wedge dh) = 0 \\ &\mathrm{tr}(dh^{-1} \wedge \pi \wedge dh - h^{-1}\pi \wedge \pi \wedge dh) = d\,\mathrm{tr}(h^{-1}\pi \wedge dh). \end{aligned}$$

Inserting into right hand side of above equality, we find

$$TC_2(\pi') - TC_2(\pi) = -\frac{1}{8\pi^2}d\,\mathrm{tr}(h^{-1}\pi \wedge dh),$$

this is an exact form. Hence $[TC_2(\pi)] = s^*[TC_2(\omega)]$ is independent of choice of section s. □

From this proposition, we know that $[TC_2(\pi)]$ is a CR invariant class on $S_+(M,\xi)$. The inclusion $i_\theta : M \hookrightarrow S_+(M,\xi) \cong M \times \mathbb{R}_+$ is a homotopy equivalence, thus $H^3(S_+(M,\xi);\mathbb{R}) \cong H^3(M;\mathbb{R})$. We have hope to further pull down $[TC_2(\pi)]$ to a CR invariant class on M, for instance $i_\theta^*[TC_2(\pi)] \in H^3(M;\mathbb{R})$. Similarly, if we show that this class is independent of choice of contact form θ (we have brought in the inclusion i_θ associated to θ), then we are coming to the standing point of defining a numerical global CR invariant on M. And this is indeed the case: Choosing different θ is evidently equivalent to taking different value $u \in \mathbb{R}_+$ in $S_+(M,\xi) \cong M \times \mathbb{R}_+$, and there is an obvious homotopy between $i_\theta : M \to M \times \mathbb{R}_+$ and $i_{\theta'} : M \to M \times \mathbb{R}_+$ for two distinct contact forms θ, θ'. This induces cochain complex homotopy operator $\mathrm{h} : \Omega^k(M \times \mathbb{R}_+) \to \Omega^{k-1}(M)$, i.e. $i_\theta^* - i_{\theta'}^* = d \circ \mathrm{h} + \mathrm{h} \circ d$. Thus $i_\theta^*[TC_2(\pi)] = i_{\theta'}^*[TC_2(\pi)] \in H^3(M;\mathbb{R})$.

We want to find a CR invariant class utilizing the pseudo-Hermitian data on (M,θ) because they are more commonly studied and flavored Riemannian geometry style quantities on CR manifolds. Due to the discussion in section 4.4, we can pullback the Cartan connection 1-form ω on $\mathscr{Y}$ to M along a map $\tilde{\sigma} : M \to \mathscr{Y}$, where $\tilde{\sigma}$ is the composition of a global section $\sigma : M \to \mathscr{X}$ and the bundle inclusion $\iota : \mathscr{X} \hookrightarrow \mathscr{Y}$, so that it determines a cohomology class $[\tilde{TC}_2(\pi)] := \tilde{\sigma}^*[TC_2(\omega)] \in H^3(M;\mathbb{R})$ on M. By proposition 4.8, the bundle inclusion $\iota : \mathscr{X} \hookrightarrow \mathscr{Y}$ is a homotopy equivalence, and this implies $H^3(\mathscr{Y};\mathbb{R}) \cong H^3(\mathscr{X};\mathbb{R})$, thereby we obtain immediately CR invariant class $\iota^*[TC_2(\omega)] \in H^3(\mathscr{X};\mathbb{R})$ on $\mathscr{X}$. Since $\mathscr{X} \xrightarrow{p} M$ is a 1-sphere bundle, we apply the Gysin sequence ([5] Proposition 14.33):

$$\cdots \xrightarrow{p^*} H^n(\mathscr{X};\mathbb{R}) \xrightarrow{p_*} H^{n-1}(M;\mathbb{R}) \xrightarrow{\cdot \wedge e(\mathscr{X})} H^{n+1}(M;\mathbb{R}) \xrightarrow{p^*} H^{n+1}(\mathscr{X};\mathbb{R}) \xrightarrow{p_*} \cdots$$

where $e(\mathscr{X}) \in H^2(M;\mathbb{R})$ is the Euler class of the oriented 1-sphere bundle $\mathscr{X}$. By the assumption on the triviality of $\xi^{1,0}$, the principal bundle $\mathscr{X}$ admits a nowhere vanishing global section and thus $e(\mathscr{X}) \equiv 0$. By the exactness of the cohomology sequence, when $n = 4$ we have $H^4(\mathscr{X};\mathbb{R}) \cong H^3(M;\mathbb{R})$, while $n = 3$ we can cut out a short exact sequence $H^3(M;\mathbb{R}) \xhookrightarrow{p^*} H^3(\mathscr{X};\mathbb{R}) \xrightarrow{p_*} H^2(M;\mathbb{R})$. So we see $\sigma^* : H^3(\mathscr{X};\mathbb{R}) \to H^3(M;\mathbb{R})$ as a left inverse of p^* such that this short exact sequence splits, i.e. $H^3(\mathscr{X};\mathbb{R}) \cong H^3(M;\mathbb{R}) \oplus H^2(M;\mathbb{R})$ (this can be also obtained by using Künneth formula). Pulling $\iota^*[TC_2(\omega)]$ back to M along σ means dropping the part of $H^2(M;\mathbb{R})$ along with a circle factor from fiber integration p_* of class in $H^1(\mathbb{S}^1;\mathbb{R}) \cong \mathbb{R}$. Once we show that $[\tilde{TC}_2(\pi)] = \sigma^*\iota^*[TC_2(\omega)]$

is independent of the choice of σ, we realize that $[\tilde{T}C_2(\pi)]$ is actually a CR invariant class on M. By the commutative diagram for $(\sigma, \iota, i_\theta, s_\sigma)$ in section 4.4, $[\tilde{T}C_2(\pi)]$ can be expressed uniquely as $[\tilde{T}C_2(\pi)] = i_\theta^* s_\sigma^* [TC_2(\omega)] = i_\theta^*[TC_2(\pi)]$ where $\pi = s_\sigma^*\omega \in \Omega^1(S_+(M,\xi); \mathfrak{su}(2,1))$. Picking two different $\sigma, \sigma' : M \to \mathscr{X}$ inducing correspondingly $s_\sigma, s_{\sigma'} : S_+(M,\xi) \to \mathscr{Y}$, and they are related by $s_{\sigma'} = s_\sigma \cdot h$ for some $h \in H_1/K$. Thanks to the proposition 6.1 we have just proved, $s_\sigma^*[TC_2(\omega)] = s_{\sigma'}^*[TC_2(\omega)] \in H^3(S_+(M,\xi); \mathbb{R})$. Further, we know already $i_\theta^* : H^3(S_+(M,\xi); \mathbb{R}) \to H^3(M; \mathbb{R})$ is a canonical isomorphism independent of θ. In conclusion, we have successfully pulled down a CR invariant class $[\tilde{T}C_2(\pi)]$ on M.

So $\tilde{T}C_2(\pi) = \tilde{\sigma}^* TC_2(\omega)$ is expressed in terms of pseudo-Hermitian data with a canonical representative:

$$[\tilde{T}C_2(\pi)] = \Big[\frac{\mathbf{i}}{8\pi^2}(-\frac{2\mathbf{i}}{3} d\theta_1^1 \wedge \theta_1^1 + \frac{1}{6} R\,\theta \wedge d\theta_1^1 - 2\theta \wedge \tau^1 \wedge \tau^{\bar{1}})\Big] \in H^3(M; \mathbb{R}).$$

The evaluation on the fundamental class

$$\mu(J) = \int_M \tilde{T}C_2(\pi) = \frac{1}{8\pi^2}\left[\int_M \left(\frac{1}{6}R^2 + 2|A_{\bar{1}}^1|^2\right)\theta \wedge d\theta + \int_M \frac{2}{3}\theta_1^1 \wedge d\theta_1^1\right] \tag{31}$$

gives rise a global CR invariant of the CR structure (ξ, J).

Remark. The functional $\mu(J)$ is called the Burns-Epstein invariant. Clearly this global invariant is CR invariant and independent of the choice of contact form. From the theorem above we can also tell the difference comparing to Chern-Simons invariant over oriented closed 3-manifold: The Chern-Simons invariant is a $\mathbb{R}/\mathbb{Z}$-valued conformal invariant (since under change of frames, the integral differs by a number in $\mathbb{Z}$), while the Burns-Epstein CR invariant here takes value in $\mathbb{R}$. On homology 3-sphere, one can show a $\bmod \mathbb{Z}$ relation between them, c.f. [38].

It was pointed out by Kengo Hirachi in [37] that: Recall the complex line bundle $\xi^{1,0}$ is trivial means its first Chern class vanishes, i.e. $c_1(\xi^{1,0}) = 0 \in H^2(M; \mathbb{Z})$. One can generalize the Burns-Epstein invariant to the case where $c_1(\xi^{1,0})$ is a torsion element so that $c_1(\xi^{1,0}) = 0 \in H^2(M; \mathbb{R})$. In this case, the line bundle $\xi^{1,0}$ is flat, so one can choose local coframe such that the transition functions are constants. The connection form θ_1^1 exists globally and does not depend on the choice of the coframe, thus $\tilde{T}C_2$ is still globally well-defined.

By perturbing π (and hence Π), one can easily show that the first differential of $\mu(J)$ with respect to the linear order perturbation $\dot{\pi}$ is

$$-\frac{1}{4\pi^2}\int_M \mathrm{tr}(\dot{\pi} \wedge \Pi),$$

thus the critical point of $\mu(J)$ for a CR structure on M corresponds to $\Pi = 0$. This yields that the Cartan tensor is identically zero and therefore implies the CR structure is spherical, i.e. M is locally CR diffeomorphic to unit 3-sphere $\mathbb{S}^3 \subset \mathbb{C}^2$ with standard contact structure $\hat{\xi}$ and standard complex structure $\hat{J}$. We will discuss later the rigorous variation of Burns-Epstein invariant with respect to the deformation of CR structure J (ξ is fixed up to contactomorphism). In the conformal geometry parallel, similarly the variation of Chern-Simons invariant with respect to deformation of Riemannian metric is carried out in [15] §6.

Examples.

(1) Recall we have studied the pseudo-Hermitian structure for unit sphere $\mathbb{S}^3$ in the example (1) in section 4.1. According to the standard pseudo-Hermitian data $\hat{R} = 2, \hat{A} = 0, \hat{\theta}_1^1 = -2\mathbf{i}\theta$ and the

standard CR structure $\hat{J}$, we can immediately obtain:

$$\mu = \frac{1}{8\pi^2}\left[\int_{\mathbb{S}^3}\frac{4}{6}\theta\wedge d\theta + \frac{2}{3}\int_{\mathbb{S}^3}(-4)\theta\wedge d\theta\right] = \frac{1}{8\pi^2}\int_{\mathbb{S}^3}(-2)\frac{1}{2}d\mathrm{Vol}_{\mathbb{S}^3} = -\frac{2\pi\times 4\pi}{8\pi^2} = -1.$$

In addition, we also construct a SU(2)-invariant family of the standard CR structure encoded in the frames $\theta^1_s, \theta^{\bar{1}}_s$. The pseudo-Hermitian data are $R = 2\frac{1+s^2}{1-s^2}$, $A^1_{\bar{1}} = -\mathbf{i}\frac{4s}{1-s^2}$, then we computes that

$$\begin{aligned}\mu &= \frac{1}{8\pi^2}\int_{\mathbb{S}^3}\left[2\left(\frac{4s}{1-s^2}\right)^2 - 2\left(\frac{1+s^2}{1-s^2}\right)^2\right]\theta\wedge d\theta\\ &= \frac{1}{8\pi^2}\int_{\mathbb{S}^3}(-2)\left[1-3\left(\frac{2s}{1-s^2}\right)^2\right]\frac{1}{2}d\mathrm{Vol}_{\mathbb{S}^3}\\ &= -1+3\left(\frac{2s}{1-s^2}\right)^2.\end{aligned}$$

We see that μ attains its minimum value -1 at $s=0$ under such a family of SU(2)-invariant CR structure.

(2) We next consider the unit holomorphic tangent bundle over a compact Riemann surface Σ. The holomorphic tangent bundle $T^{1,0}\Sigma \xrightarrow{\mathrm{pr}} \Sigma$ is a holomorphic line bundle over Σ, while its unit subbundle $\mathbb{S}T^{1,0}\Sigma$ is a circle bundle over Σ with respect to a Hermitian metric on Σ expressed as $h = h_{1\bar{1}}dz\otimes d\bar{z}$ in terms of local coordinate with $h_{1\bar{1}} > 0$ (it is actually a Kähler metric). The Riemannian metric is given by $\frac{1}{2}(h+\bar{h}) = h_{1\bar{1}}(dx\otimes dx + dy\otimes dy)$. The Gauss curvature is therefore $K = -\frac{\Delta(\ln\sqrt{h_{1\bar{1}}})}{h_{1\bar{1}}}$. There is a canonical coframe $\eta^1 = dz/\sqrt{2h_{1\bar{1}}}$ and a unique torsion free connection form ω^1_1 such that $d\eta^1 = \eta^1\wedge\omega^1_1$. The Gauss equation yields $d\omega^1_1 = K\eta^1\wedge\eta^{\bar{1}}$. Put $\theta = \mathbf{i}\mathrm{sgn}(K)\mathrm{pr}^*\omega^1_1$, $\theta^1 = \sqrt{|K|}\mathrm{pr}^*\eta^1 = \frac{1}{2}\ln|K|\mathrm{pr}^*\eta^1$ so that $d\theta = \mathbf{i}\theta^1\wedge\theta^{\bar{1}}$. Assuming K is nowhere vanishing, this define a pseudo-convex CR structure on $\mathbb{S}T^{1,0}\Sigma$. Since the tangent bundle of $T\Sigma$ is trivial, thus the $\xi_{\mathbb{C}} := \ker(\theta)_{\mathbb{C}} \subset T_{\mathbb{C}}(\mathbb{S}T^{1,0}\Sigma)$ is necessarily trivial and so is $\xi^{1,0}$ trivial. Performing exterior differential on θ^1, we obtain

$$d\theta^1 = \theta^1\wedge\left[\frac{1}{2}(\ln|K|)_{,1}\theta^1 - \frac{1}{2}(\ln|K|)_{,\bar{1}}\theta^{\bar{1}} - \mathbf{i}\mathrm{sgn}(K)\theta\right] = \theta^1\wedge\theta^1_1$$

where we can tell that the torsion τ^1 must vanish here via normalization condition $\tau^1\wedge\theta^{\bar{1}}$.

We further compute $d\theta^1_1$:

$$d\theta^1_1 = \left[\mathrm{sgn}(K) - \frac{(\ln|K|)_{,1}{}^1}{|K|}\right]\theta^1\wedge\theta^{\bar{1}} = \left[\mathrm{sgn}(K) - \frac{\Delta(\ln|K|)}{|K|}\right]\theta^1\wedge\theta^{\bar{1}} = R\theta^1\wedge\theta^{\bar{1}}$$

where $\Delta = h^{\alpha\bar{\beta}}\nabla_{Z_{\bar{\beta}}}\nabla_{Z_\alpha} = h^{1\bar{1}}\nabla_{Z_{\bar{1}}}\nabla_{Z_1}$ is the Laplacian operator with Z_1 the dual frame field of θ^1 and respectively $Z_{\bar{1}}$ for $\theta^{\bar{1}}$. The transgression form is given as

$$\begin{aligned}\tilde{T}C_2(\pi) &= \frac{\mathbf{i}}{8\pi^2}\left[-\frac{1}{2}+\frac{1}{3}\frac{\Delta(\ln|K|)}{K}+\frac{1}{6}\left(\frac{\Delta(\ln|K|)}{K}\right)^2\right]\theta\wedge\theta^1\wedge\theta^{\bar{1}}\\ &= \frac{\mathbf{i}}{8\pi^2}\left[-\frac{1}{2}+\frac{1}{3}\frac{\Delta(\ln|K|)}{K}+\frac{1}{6}\left(\frac{\Delta(\ln|K|)}{K}\right)^2\right]\mathbf{i}K\mathrm{pr}^*\omega^1_1\wedge\mathrm{pr}^*\eta^1\wedge\mathrm{pr}^*\eta^{\bar{1}}.\end{aligned}$$

It remains to compute the integral over $\mathbb{S}T^{1,0}\Sigma$. We should notice that $\mathrm{pr}^*\omega_1^1 = -\mathbf{i}\,\mathrm{sgn}(K)\theta$ integrating along fiber, namely the Reeb orbits $\mathbb{S}^1 \subset \mathbb{C}$, gets $-2\pi\mathbf{i}\,\mathrm{sgn}(K)$. Hence the first term is computed as

$$-\frac{1}{8\pi}\int_\Sigma |K| d\mathrm{Area} = -\frac{|\chi(\Sigma)|}{4}$$

by Gauss-Bonnet theorem where the area form is given by $d\mathrm{Area} = \mathbf{i}\eta^1 \wedge \eta^1 = \frac{\mathbf{i}}{2}\frac{1}{h_{1\bar{1}}} dz \wedge d\bar{z}$. The integrand in second term is $\Delta(\ln|K|)$, by Stoke's theorem, this integral vanishes. The third term is simplified straightforwardly. In the end, we get

$$\mu = -\frac{|\chi(\Sigma)|}{4} + \frac{1}{24}\int_\Sigma \frac{(\Delta(\ln|K|))^2}{|K|} d\mathrm{Area}.$$

From above result, we can see the minimum of the Burns-Epstein is attained only at the constant curvature metrics. By the uniformization theorem, we know the unique negative constant curvature hyperbolic metric does the job.

Burns and Epstein further showed in their later work [31] that for a strictly pseudo-convex domain $N \subset \mathbb{C}^2$ so that $M = \partial N$ is strictly pseudo-convex hypersurface in $\mathbb{C}^2$, the Burns-Epstein invariant is related with Euler characteristic and Chern classes:

$$\int_N c_2(N) - \frac{1}{3}c_1(N)^2 = \chi(N) + \mu(\partial N).$$

Since the region N bounded by such a hypersurface in $\mathbb{C}^2$ would have to be homeomorphic to the $\mathbb{D}^3$, so $\chi(N) \equiv 1$. As a consequence, they got a criterion on CR embeddability of topologically spherical CR manifolds.

Theorem 6.2 ([31] Corollary 5.1)**.** *M is a strictly pseudo-convex CR manifold homeomorphic to $\mathbb{S}^3$. A necessary condition for M to admit a CR embedding into $\mathbb{C}^2$ is: $\mu(M) \geq -1$. If $\mu(M) = -1$ and M embeds in $\mathbb{C}^2$, then M is CR equivalent to $\mathbb{S}^3 = \partial\mathbb{D}^3$.*

Generally, for N a compact strictly pseudo-convex complex 2-manifold with boundary ∂N, there is ([31] Theorem 6.1):

$$\int_N c_2(N) - \frac{1}{3}[c_1(N)^2 - \bar{c}_1(N)^2] = \chi(N) + \mu(\partial N)$$

where $\bar{c}_1$ is a lift of c_1 from $H^2(N;\mathbb{R})$ to $H^2(N,\partial N;\mathbb{R})$ and the integral is independent of the choice of lifting. In particular, the circle bundle $M = \mathbb{S}T^{1,0}\Sigma$ bounds a region $N = \mathbb{D}T^{1,0}\Sigma$ called the unit disk bundle. In this case $\int_N \bar{c}_1(N)^2 = \chi(\Sigma)$. Applying above results we have

$$\int_N c_2(N) - \frac{1}{3}c_1(N)^2 = -\frac{1}{12}\chi(\Sigma) + \frac{1}{24\pi}\int_\Sigma \frac{(\Delta(\ln|K|))^2}{|K|} d\mathrm{Area}.$$

Jih-Hsin Cheng and John M. Lee has generalized Burns-Epstein invariant to arbitrary oriented compact 3-dimensional CR manifolds getting rid of the triviality assumption of holomorphic tangent bundle by slight modification [32]. We still consider a local frame field Z_1 of holomorphic tangent bundle $\xi^{1,0}$ and the admissible 1-form θ^1 dual to it the same as before. Fixing ξ with orientation and using the setting of Tanaka-Webster connection discussed before, they defined the generalized Burns-Epstein functional for a pair of CR structures $(J, \hat{J})$ as

$$\mu(J,\hat{J}) = \frac{1}{8\pi^2}\left[\int_M \left(\frac{1}{6}R^2 + 2|A|^2\right)\theta \wedge d\theta - \int_M \left(\frac{1}{6}\hat{R}^2 + 2|\hat{A}|^2\right)\theta \wedge d\theta + \int_M \frac{2}{3}(\theta_1^1 - \hat{\theta}_1^1) \wedge (d\theta_1^1 + d\hat{\theta}_1^1)\right] \tag{32}$$

($|A|^2 = A_1^{\bar{1}} A_{\bar{1}}^1$) and showed that this is actually independent of choice of θ. Since we have pseudo-Hermitian connection ∇ on the holomorphic tangent bundle $\xi^{1,0}$, by considering it as a complex line bundle whose fiber is $\mathbf{i}$-eigenspace of J, this connection can also be interpreted as Hermitian connection on the complex line bundle. It further gives rise a connection on the principal U(1)-bundle $\mathscr{X}_J := (\xi^{1,0}\backslash\{0\})/\mathbb{R}_+$ associated to the complex line bundle $\xi^{1,0}$.

Lemma 6.3. *Let J and $\hat{J}$ be two oriented CR structures with respect to the same contact structure ξ. Then $\mathscr{X}_J$ and $\mathscr{X}_{\hat{J}}$ are canonically isomorphic as a principal $U(1)$-bundle.*

Proof. Denote $\xi_J^{1,0}$ as the holomorphic tangent bundle associated to J and $\xi_{\hat{J}}^{1,0}$ associated to $\hat{J}$. By restricting the projection $\xi_{\mathbb{C}} = \xi_J^{1,0} \oplus \xi_J^{0,1} \to \xi_J^{1,0}$ to subbundle $\xi_{\hat{J}}^{1,0} \subset \xi_{\mathbb{C}}$, we get a complex linear map $\rho_{\hat{J}J} : \xi_{\hat{J}}^{1,0} \to \xi_J^{1,0}$. To show $\rho_{\hat{J}J}$ induces a U(1)-equivariant isomorphism $\mathscr{X}_{\hat{J}} \to \mathscr{X}_J$, it suffices to show that $\rho_{J\hat{J}} \circ \rho_{\hat{J}J} \equiv \mathrm{id}_{\xi_{\hat{J}}^{1,0}} \bmod \mathbb{R}_+$ where $\mathbb{R}_+$ acts by multiplication. Now we choose local frames $Z_1 \in \xi_J^{1,0}$, $\hat{Z}_1 \in \xi_{\hat{J}}^{1,0}$ and denote θ^1, $\hat{\theta}^1$ as their dual coframes. Then $\rho_{J\hat{J}}(\hat{Z}_1) = \theta^1(\hat{Z}_1) Z_1$ and similar for $\rho_{\hat{J}J}$ so that $(\rho_{J\hat{J}} \circ \rho_{\hat{J}J})(\hat{Z}_1) = \theta^1(\hat{Z}_1)\hat{\theta}^1(Z_1)\hat{Z}_1$. One can write $\hat{Z}_1$ and $\hat{\theta}^1$ in terms of linear combinations $\hat{Z}_1 = uZ_1 + vZ_{\bar{1}}$ and $\hat{\theta}^1 = s\theta^1 + t\theta^{\bar{1}}$ with u, v, s, t complex coefficients so that $\theta^1(\hat{Z}_1)\hat{\theta}^1(Z_1) = us$. The conditions $\hat{\theta}^1(\hat{Z}_1) = 1$, $\hat{\theta}^{\bar{1}}(\hat{Z}_1) = 0$ imply $us + vt = 1$, $v\bar{s} + u\bar{t} = 0$. By assumption, θ is oriented contact form for ξ and $d\theta = \mathbf{i}h_{1\bar{1}}\theta^1 \wedge \theta^{\bar{1}} = \mathbf{i}\hat{g}_{1\bar{1}}\hat{\theta}^1 \wedge \hat{\theta}^{\bar{1}}$ with $h_{1\bar{1}}$ and $\hat{h}_{1\bar{1}}$ positive. Substituting the local frames in $d\theta(\cdot,\cdot)$ and we find $h_{1\bar{1}} = \hat{h}_{1\bar{1}}(|s|^2 - |t|^2) > 0$, therefore $|s|^2 - |t|^2 > 0$ and $us = \frac{|s|^2}{|s|^2 - |t|^2} > 0$. This shows the statement. □

With the help of this lemma, we can think of ∇ and $\hat{\nabla}$ as connections on one fixed principal U(1)-bundle $\mathscr{X}_{\hat{J}}$. It is well known that space of connections is an affine space with tangent space at $\hat{\nabla}$ isomorphic to $\Omega^1(M; \mathbf{i}\mathbb{R})$ where $\mathbf{i}\mathbb{R}$ is the Lie algebra of U(1). In this sense, we can interpret $\theta_1^1 - \hat{\theta}_1^1$ as the translation in the affine space, hence it is a well-defined global purely imaginary-valued 1-form on M. The curvature form of any Hermitian connection on a principal U(1)-bundle over M is globally defined on M. Thus, $d\theta_1^1$ and $d\hat{\theta}_1^1$ are also well-defined global purely imaginary-valued 2-forms. Moreover, the pure imaginary $\mathbf{i}$ gets canceled in the exterior multiplication. In the end, this justifies that the above invariant $\mu(J, \hat{J})$ globally well-defined for any compact CR 3-manifolds.

On a closed 3-manifold M, if the holomorphic tangent bundle is trivial for J, namely $\xi^{1,0}$ admits a global frame field, then the Tanaka-Webster connection θ_1^1 is a global 1-form on M, hence $\hat{\theta}_1^!$ is globally defined as well. By Stoke's theorem:

$$\begin{aligned}
\int_M (\theta_1^1 - \hat{\theta}_1^1) \wedge (d\theta_1^1 + d\hat{\theta}_1^1) &= \int_M (\theta_1^1 \wedge d\theta_1^1 - \hat{\theta}_1^1 \wedge d\hat{\theta}_1^1) - \int_M (\hat{\theta}_1^1 \wedge d\theta_1^1 - \theta_1^1 \wedge d\hat{\theta}_1^1) \\
&= \int_M (\theta_1^1 \wedge d\theta_1^1 - \hat{\theta}_1^1 \wedge d\hat{\theta}_1^1) - \int_M d(\theta_1^1 \wedge \hat{\theta}_1^1) \\
&\overset{\substack{\text{Stokes'} \\ \text{theorem}}}{=} \int_M (\theta_1^1 \wedge d\theta_1^1 - \hat{\theta}_1^1 \wedge d\hat{\theta}_1^1)
\end{aligned}$$

Thus this shows $\mu(J, \hat{J}) = \mu(J) - \mu(\hat{J})$, which is well consistent with the original version (31) defined by Burns and Epstein on closed CR 3-manifolds with trivial holomorphic tangent bundles. However, we should be careful and note that the original version of Burns-Epstein invariant $\mu(J)$ has already been shown to be CR invariant hence it clearly does not depend on the choice of contact form and depends only on the CR structure J with respect to given ξ while here the generalized version $\mu(J, \hat{J})$, we still do not know yet whether it is independent on the choice contact form so far. We will see later that this is indeed the case and it depends only on J and $\hat{J}$.

6.2 Variation of Burns-Epstein invariant

Let $\mathscr{J} \subset \mathrm{End}(\xi)$ be the bundle of oriented CR structures compatible with ξ. For any smooth section $J \in C^\infty(M, \mathscr{J})$, we define a subbundle

$$\mathscr{E}_J := \{E \in \mathrm{End}(\xi) : E \circ J + J \circ E = 0\} \subset \mathrm{End}(\xi)$$

and denote $\Pi_J : \mathrm{End}(\xi) \to \mathscr{E}_J$ the orthogonal projection onto $\mathscr{E}_J$. The local expression for any $E \in \mathrm{End}(\xi)$ with respect to the local frame field $Z_1, Z_{\bar{1}}$ and local coframe field $\theta^1, \theta^{\bar{1}}$ can be written formally as

$$E = E_1^1\theta^1 \otimes Z_1 + E_1^{\bar{1}}\theta^1 \otimes Z_{\bar{1}} + E_{\bar{1}}^1\theta^{\bar{1}} \otimes Z_1 + E_{\bar{1}}^{\bar{1}}\theta^{\bar{1}} \otimes Z_{\bar{1}} \ ,$$

and we have

$$\Pi_J(E) = E_1^{\bar{1}}\theta^1 \otimes Z_{\bar{1}} + E_{\bar{1}}^1\theta^{\bar{1}} \otimes Z_1.$$

In particular, locally

$$J = \mathbf{i}(\theta^1 \otimes Z_1 - \theta^{\bar{1}} \otimes Z_{\bar{1}}).$$

Lemma 6.4. *Let $J \in \mathscr{J}$. There is a smooth embedding $\Phi_J : \mathscr{E}_J \to End(\xi)$ given by $\Phi_J(E) = (1+E_1^{\bar{1}}E_{\bar{1}}^1)^{1/2}J+E$ for all $E \in \mathscr{E}_J$, whose image is $\mathscr{J}$. In particular, $\Pi_J \circ \Phi_J = Id_{\mathscr{E}_J}$ and respectively $\Phi_J \circ \Pi_J = Id_{\mathscr{J}}$.*

Proof. Suppose $E \in \mathscr{J}$ and let $\Pi_J(E) \in \mathscr{E}_J$. Write E in local form, since $E^2(Z_1) = -Z_1$, $E_1^1E_1^1 + E_1^{\bar{1}}E_{\bar{1}}^1 = -1$ holds and hence

$$E_1^1 = \pm\mathbf{i}(1 + E_1^{\bar{1}} + E_1^{\bar{1}}E_{\bar{1}}^1)^{1/2}.$$

The sign is determined by the orientation of $E \in \mathscr{J}$, i.e. $d\theta(X, EX) > 0 \ \forall X \in \xi$. Applying to $Z_1 + Z_{\bar{1}}$, we get

$$d\theta(Z_1+Z_{\bar{1}}, EZ_1+EZ_1) = \mathbf{i}\theta^1(Z_1+Z_{\bar{1}})\theta^{\bar{1}}(EZ_1+EZ_{\bar{1}}) = \mathrm{Im}(E_1^{\bar{1}}+E_{\bar{1}}^{\bar{1}}) = \pm(1+E_1^{\bar{1}}E_{\bar{1}}^1)^{1/2}+\mathrm{Im}\,E_1^{\bar{1}} > 0.$$

However $(1 + E_1^{\bar{1}}E_{\bar{1}}^1)^{1/2} > |\,\mathrm{Im}\,E_1^{\bar{1}}|$, we must choose the positive sign. Therefore we have $E = \Phi_J(\Pi_J(E))$ for all $E \in \mathscr{J}$, which implies $\mathscr{J} \subset \mathrm{im}(\Phi_J)$.

Conversely, for any $E \in \mathscr{E}_J$ we know it can be written locally as $E_1^{\bar{1}}\theta^1 \otimes Z_{\bar{1}} + E_{\bar{1}}^1\theta^{\bar{1}} \otimes Z_1$. Hence

$$\Phi_J(E) = \mathbf{i}(1 + E_1^{\bar{1}}E_{\bar{1}}^1)^{1/2}(\theta^1 \otimes Z_1 - \theta^{\bar{1}} \otimes Z_{\bar{1}}) + E_1^{\bar{1}}\theta^1 \otimes Z_{\bar{1}} + E_{\bar{1}}^1\theta^{\bar{1}} \otimes Z_1 \in \mathscr{J}$$

as we have already verified $\Phi_J(E)^2 = -\mathrm{id}_\xi$ and shown it is positive oriented. So $\Pi_J(\Phi_J(E)) = E$ for all $E \in \mathscr{E}_J$ and $\mathrm{im}(\Phi_J) \subset \mathscr{J}$. Clearly Φ_J is a smooth embedding. □

A deformation of CR structure is a smooth 1-parameter family or smooth path in the space of sections $C^\infty(M, \mathscr{J})$, i.e. $\{J_t \in C^\infty(M, \mathscr{J}) : -\varepsilon < t < \varepsilon\} \subset C^\infty(M, \mathscr{J})$. The tangent space to $C^\infty(M, \mathscr{J})$ as a Fréchet manifold at J is the linear space

$$T_JC^\infty(M, \mathscr{J}) = \{E \in C^\infty(M, \mathrm{End}(\xi)) \,|\, E = \dot{J} \ , \ J_t \in C^\infty(M, \mathscr{J}) \ \forall\, t \in (-\varepsilon, \varepsilon)\}$$

where $\dot{J} = \frac{d}{dt}\big|_{t=0}J_t$. For a given $J \in C^\infty(M, \mathscr{J})$, suppose there is a deformation J_t with $J_0 = J$, by differentiating $J_t^2 = -\mathrm{id}$ and put $\dot{J} = E$ we obtain $E \circ J + J \circ E = 0$, thus $E \in C^\infty(M, \mathscr{E}_J)$. On the other hand, if $E \in C^\infty(M, \mathscr{E}_J)$, we can define a deformation of J by $\Phi_J(tE) = (1+t^2E_1^{\bar{1}}E_{\bar{1}}^1)^{1/2}J+tE$, then clearly $\dot{J} = E$, namely $E \in T_JC^\infty(M, \mathscr{J})$. This means $C^\infty(M, \mathscr{E}_J)$ is exactly the tangent space $T_JC^\infty(M, \mathscr{J})$. Naively, Φ_J plays a role similar to the exponential map in Riemannian geometry, although we have not specified any suitable metric for $C^\infty(M, \mathscr{J})$ here.

Define a L^2-inner product on $C^\infty(M, \mathscr{J})$ via the volume integral

$$(\cdot,\cdot)_J = \int_M \langle \cdot,\cdot\rangle_J \theta\wedge d\theta$$

with Levi form $\langle\cdot,\cdot\rangle_J = d\theta(\cdot, J\cdot)_{\mathbb{C}}$ such that $\langle V, W\rangle_J = h_{1\bar{1}}V^1W^{\bar{1}}$ for $V = V^1Z_1$, $W = W^1Z_1$ in $\xi^{1,0}$. It is a Hermitian inner product on $\xi_{\mathbb{C}} = \xi^{1,0}\oplus\xi^{0,1}$ and further extends to all tensor bundles of $\xi_{\mathbb{C}}$. This L^2-inner product is well-defined since M is compact (closed).

Proposition 6.5. *The first order variation of Burns-Epstein functional with respect to deformation J_t of J and $\hat{J}$ fixed is given by*

$$D\mu(J,\hat{J})|_J = \left.\frac{d}{dt}\right|_{t=0}\mu(J_t,\hat{J}) = -\frac{1}{8\pi^2}(\dot{J}, Q_J)_J.$$

Proof. Fix a unit local frame Z_1 in $\xi^{1,0}$ with dual coframe θ^1. Let $\dot{J} = 2E = 2E_1^{\bar{1}}\theta^1\otimes Z_{\bar{1}} + 2E_{\bar{1}}^1\theta^{\bar{1}}\otimes Z_1$. Denote $Z_{1\,t}$ and θ^1_t the corresponding images of frame and coframe under the isomorphism $\mathscr{X}_J \xrightarrow{\cong} \mathscr{X}_{J_t}$. So $\theta^1(Z_{1\,t})$ is a positive function in t. We can write $\dot{Z}_1 = FZ_1 + G_1^{\bar{1}}Z_{\bar{1}}$ with F real and $G_1^{\bar{1}}$ complex. Since $Z_{1\,t}$ is unit length, for all t it holds that $1 = h_{1\bar{1}\,t} = 2 - \mathbf{i}d\theta(Z_{1\,t}, Z_{\bar{1}\,t})$. Differentiating the equation with respect to t:

$$0 = -2\mathbf{i}[d\theta(\dot{Z}_1, Z_{\bar{1}}) + d\theta(Z_1, \dot{Z}_{\bar{1}})] = -2\mathbf{i}(F+F) \Rightarrow F = 0.$$

Differentiating $J_tZ_{1\,t} = \mathbf{i}Z_{1\,t}$ we get

$$0 = \dot{J}Z_1 + J\dot{Z}_1 - \mathbf{i}\dot{Z}_1 = 2E_1^{\bar{1}}Z_{\bar{1}} - 2\mathbf{i}G_1^{\bar{1}}Z_{\bar{1}} \Rightarrow \dot{Z}_1 = -\mathbf{i}E_1^{\bar{1}}Z_{\bar{1}}.$$

Differentiating $\theta^1_t(Z_{1\,t}) = 1$ and $\theta^1_t(Z_{\bar{1}\,t}) = 0 = \theta^1_t(\mathcal{R})$ yields $\dot{\theta}^1 = -\mathbf{i}E_{\bar{1}}^1\theta^{\bar{1}}$. Inserting this in the equation derived by t-derivative on structure equation for $d\theta$:

$$d\dot{\theta}^1 = \dot{\theta}^1\wedge\theta_1^1 + \theta^1\wedge\dot{\theta}_1^1 + \dot{A}_{\bar{1}}^1\theta\wedge\theta^{\bar{1}} + A_{\bar{1}}^1\theta\wedge\dot{\theta}^{\bar{1}}.$$

Applying lemma 4.2 so that $\theta_1^1 = 0$ at a point in the open neighborhood where the coframe is defined, then putting $\dot{\theta} = -\mathbf{i}E_{\bar{1}}^1\theta 6\bar{1}$ and comparing $\theta\wedge\theta^{\bar{1}}$ term we find

$$\dot{A}_{\bar{1}}^1 = -\mathbf{i}E_{\bar{1},0}^1. \tag{33}$$

Inserting Z_1 in above equation of $d\dot{\theta}^1$, we get

$$\dot{\theta}_1^1 \equiv \mathbf{i}(A_1^{\bar{1}}E_{\bar{1}}^1 + A_{\bar{1}}^1E_1^{\bar{1}})\theta - \mathbf{i}E_{\bar{1},1}^1\theta^{\bar{1}} \bmod \theta^1.$$

$h_{1\bar{1}\,t} = 1$ implies $\theta^1_{1\,t} + \theta^{\bar{1}}_{\bar{1}\,t} = 0$, hence

$$\dot{\theta}_1^1 = \mathbf{i}(A_1^{\bar{1}}E_{\bar{1}}^1 + A_{\bar{1}}^1E_1^{\bar{1}})\theta - \mathbf{i}(E_{1,1}^{\bar{1}}\theta^1 + E_{\bar{1},1}^1\theta^{\bar{1}}). \tag{34}$$

Differentiating the structure equation for $d\theta_1^1$ and we get:

$$\dot{R} = \mathbf{i}(E_{1,\bar{1}}^{\bar{1}}{}^1 - E_{\bar{1},1}^1{}^{\bar{1}}) - (A_1^{\bar{1}}E_{\bar{1}}^1 + A_{\bar{1}}^1E_1^{\bar{1}}). \tag{35}$$

Finally, compute t-derivative for μ and by Stoke's theorem

$$\begin{aligned}
8\pi^2\left.\frac{d}{dt}\right|_{t=0}\mu(J_t,\hat{J}) =& \int_M\left(\frac{1}{3}R\dot{R} + 2\dot{A}_1^{\bar{1}}A_{\bar{1}}^1 + 2\dot{A}_{\bar{1}}^1A_1^{\bar{1}}\right)\theta\wedge d\theta \\
&+ \frac{2}{3}\int_M \dot{\theta}_1^1\wedge(d\theta_1^1 + d\hat{\theta}_1^1) + \frac{2}{3}\int_M(\theta_1^1 - \hat{\theta}_1^1)\wedge d\dot{\theta}_1^1 \\
=& 2\,\mathrm{Re}\int_M\left(\frac{1}{6}R\dot{R} + 2A_1^{\bar{1}}\dot{A}_{\bar{1}}^1\right)\theta\wedge d\theta + \frac{4}{3}\int_M\dot{\theta}_1^1\wedge d\theta_1^1
\end{aligned}$$

Substitute the expression for $\dot{A}^1_{\bar{1}}$, $\dot{\theta}^1_1$ and $\dot{R}$ obtained above, and integrate by parts:

$$\begin{aligned} 8\pi^2 \left.\frac{d}{dt}\right|_{t=0} \mu(J_t, \hat{J}) &= 4\,\mathrm{Re}\int_M \mathbf{i}E^1_{\bar{1}}\left(-\frac{1}{6}R_{,1}{}^{\bar{1}} - \frac{\mathbf{i}}{2}RA^{\bar{1}}_1 + A^{\bar{1}}_{1,0} + \frac{2\mathbf{i}}{3}A^{\bar{1}}_{1,1}{}^1\right)\theta\wedge d\theta \\ &= -(2E, Q_J)_J = -(\dot{J}, Q_J)_J \end{aligned}$$

where $Q_J = \mathbf{i}Q^{\bar{1}}_1\theta^1\otimes Z_{\bar{1}} - \mathbf{i}Q^1_{\bar{1}}\theta^{\bar{1}}\otimes Z_1 = 2\,\mathrm{Re}(\mathbf{i}Q^{\bar{1}}_1\theta^1\otimes Z_{\bar{1}})$ is the Cartan tensor. □

We know from previous section that the necessary and sufficient condition for a CR structure to be spherical is $Q_J = 0$ or $Q^{\bar{1}}_1 = Q^1_{\bar{1}} = 0$. Hence the critical points of the Burns-Epstein invariant are spherical CR structures. Comparing to the Chern-Simons invariant for closed oriented Riemannian 3-manifolds, its critical points of Chern-Simons invariant are conformally flat structures. In fact, similar variational calculus shows that the critical point of Chern-Simons invariant is actually the Cotton tensor, c.f. [16].

So far, we have not yet shown that the generalized Burns-Epstein invariant (32) is independent of the choice of contact forms, or pseudo-Hermitian structure. The very next theorem tells us this is indeed the case.

Theorem 6.6. *$\mu(J, \hat{J})$ depends only on CR structures J and $\hat{J}$, not on the choice of contact form θ.*

Proof. Due to the fact that the space of contact forms is convex, it suffices to show that for any smooth path θ_s in the convex space of contact forms, $\frac{d}{ds}\mu_{\theta_s}(J, \hat{J}) = 0$. Let J_t be a deformation of J and write $\partial_t J_t = 2E_t$. According to the proposition 6.5 above, we have for any independent s and t:

$$\frac{d}{dt}\mu_{\theta_s}(J_t, \hat{J}) = -\frac{1}{8\pi^2}\int_M \langle 2E_t, Q_t\rangle_{J_t}\theta_s\wedge d\theta_s$$

We pick another contact form $\theta' = e^{2f}\theta$, the pseudo-Hermitian metric rescales $h'_{1\bar{1}} = e^{2f}h_{1\bar{1}}$. Recall the structure equation

$$d\theta^1_1 = Rh_{1\bar{1}}\theta^1\wedge\theta^{\bar{1}} + A^{\bar{1}}_{1,\bar{1}}\theta^1\wedge\theta - A^1_{\bar{1},1}\theta^{\bar{1}}\wedge\theta$$

and the expression of $Q^1_{\bar{1}}$ in terms of R and $A^{\bar{1}}_1, A^1_{\bar{1}}$. The invariance of pseudo-Hermitian structure equation implies $R' = e^{-2f}R$, $A'^1_{\bar{1}} = e^{-2f}A^1_{\bar{1}}$ and hence $Q'_J = e^{-4f}Q_J$. Thus in terms of local expression, we find out the integrand is pseudo-conformal invariant:

$$\langle E, Q'_J\rangle_{(J,\theta')}\theta'\wedge d\theta' = \langle E, e^{-4f}Q_J\rangle_{(J,\theta)}e^{4f}\theta\wedge d\theta = \langle E, Q_J\rangle_{(J,\theta)}\theta\wedge d\theta.$$

Without lost of generality, we can put $f = \frac{s}{2}$ and then obtain

$$0 = \frac{\partial}{\partial s}\frac{\partial}{\partial t}\mu_{\theta_s}(J_t, \hat{J}) = \frac{\partial}{\partial t}\frac{\partial}{\partial s}\mu_{\theta_s}(J_t, \hat{J})$$

for any s and t. Using the fact $C^\infty(M, \mathscr{J})$ is connected, $\frac{d}{ds}\mu_{\theta_s}(J, \hat{J})$ depends only on s and not on $J \in C^\infty(M, \mathscr{J})$. Further by substituting $J = \hat{J}$, we get $\mu_{\theta_s}(\hat{J}, \hat{J}) \equiv 0$ for all s and thus $\frac{d}{ds}\mu_{\theta_s}(\hat{J}, \hat{J}) \equiv 0$. But this derivative as just argued is independent on the choice of J implying $\frac{d}{ds}\mu_{\theta_s}(J, \hat{J}) \equiv 0$ for all J and s. In conclusion, $\mu(J, \hat{J})$ does not depend on the choice of θ. □

Viewing the Burns-Epstein invariant as a functional $\mu : C^\infty(M, \mathscr{J}) \to \mathbb{R}$, its gradient (namely Gâteaux derivative) at any $J \in C^\infty(M, \mathscr{J})$ with respect to L^2-norm $\|\cdot\|_J$ is computed as $-\frac{1}{8\pi^2}Q_J$. The gradient flow equation is given by

$$\partial_t J_t = -16\pi^2\,\mathrm{grad}\,\mu(J_t, \hat{J}) = 2Q_{J_t}. \tag{36}$$

The gradient flow is therefore called the Cartan flow. Compute the first variation or gradient along the Cartan flow $\partial_t J_t = 2Q_{J_t}$, we obtain

$$\left.\frac{d}{dt}\right|_{t=0} \mu(J_t, \hat{J}) = -\frac{1}{4\pi^2}\|Q_{J_t}\|^2_{J_t}\Big|_{t=0},$$

hence obviously μ is strictly decreasing in time unless $Q_{J_0} = 0$. To study the second variation of the Burns-Epstein invariant, we need to investigate $\dot{Q}_J = \partial_t Q_{J_t}|_{t=0}$ in more detail.

Definition 6.7. *A differential operator P taking sections of a tensor bundle over M to other sections is said to have weight $\mathrm{w}(P) = m$ if m is the smallest integer such that in any local trivialization the differential operator can be expressed as a polynomial of degree m in vector fields tangent to the contact structure.*

Example. The Reeb vector field for example is an operator on $C^\infty(M, \xi)$ of weight 2, since it can be expressed locally as $[X, Y] \bmod \xi$ for some $X, Y \in \xi$.

Denote $\mathcal{O}_m$ as any operator of weight less than or equal to m. By considering generic deformation $\dot{J} = E$ written in local expression and computing at a point where $\theta^1_1 = 0$ using lemma 4.2 together with (33),(34),(35) and $\dot{h}_{1\bar{1}} = \dot{h}^{1\bar{1}} = 0$, one can find (c.f. [32] (3.2)):

$$\begin{aligned}
\dot{Q}_J &= \partial_t Q_{J_t}|_{t=0} = 2\,\mathrm{Re}[\partial_t(\mathbf{i}Q^{\bar{1}}_{1\,t}\theta^1_t \otimes Z_{\bar{1}\,t})]|_{t=0} \\
&= 2\,\mathrm{Re}[\mathbf{i}\dot{Q}^{\bar{1}}_1\theta^1 \otimes Z_{\bar{1}} + \mathbf{i}Q^{\bar{1}}_1\dot{\theta}^1 \otimes Z_{\bar{1}} + \mathbf{i}Q^{\bar{1}}_1\theta^1 \otimes \dot{Z}_{\bar{1}}] \\
&= 2\,\mathrm{Re}\left[\left(-\frac{1}{6}E^{\bar{1}}_{1,\bar{1}}{}^1{}_1{}^{\bar{1}} + \frac{1}{6}E^{1}_{\bar{1},1}{}^{\bar{1}}{}_1{}^{\bar{1}} + E^{\bar{1}}_{1,00} + \frac{2}{3}\mathbf{i}E^{\bar{1}}_{1,0}{}^1{}_1\right)\theta^1 \otimes Z_{\bar{1}} + \mathcal{O}_3(E)\right]
\end{aligned} \tag{37}$$

This is the image of a weight 4 operator acting on E.

For any $E \in C^\infty(M, \mathrm{End}(\xi))$, a second order differential operator $B_J : C^\infty(M, \mathrm{End}(\xi)) \to C^\infty(M)$ is defined locally by

$$B_J E = E^{\bar{1}}_{1,\bar{1}}{}^1 + E^{1}_{\bar{1},1}{}^{\bar{1}} - \mathbf{i}A^1_{\bar{1}}E^{\bar{1}}_1 + \mathbf{i}A^{\bar{1}}_1 E^1_{\bar{1}}. \tag{38}$$

Its formal adjoint[3] with respect to $\langle\cdot,\cdot\rangle_J$ is the operator $B^*_J : C^\infty(M) \to C^\infty(M, \mathrm{End}(\xi))$ given locally by

$$B^*_J f = (f_{,1}{}^{\bar{1}} + \mathbf{i}A^{\bar{1}}_1 f)\theta^1 \otimes Z_{\bar{1}} + (f_{,\bar{1}}{}^1 - \mathbf{i}A^1_{\bar{1}}f)\theta^{\bar{1}} \otimes Z_1. \tag{39}$$

Proposition 6.8 ([32] Proposition 3.1)**.** *The Cartan tensor satisfies the identity $B_J Q_J = 0$ in general.*

This is proved by using the Bianchi identity (lemma 3.5) for 3-dimensional pseudo-Hermitian manifold

$$R_{,0} = A_{11,}{}^{11} + A_{\bar{1}\bar{1},}{}^{\bar{1}\bar{1}} \tag{40}$$

and the commutation relations for covariant derivatives (lemma 3.6)

$$u_{,1\bar{1}} - u_{,\bar{1}1} = \mathbf{i}h_{1\bar{1}}u_{,0} \tag{41}$$

$$u_{,01} - u_{,10} = A_{11}u_{,}{}^1 \tag{42}$$

$$v_{1,1\bar{1}} - v_{1,\bar{1}1} = \mathbf{i}h_{1\bar{1}}v_{1,0} + Rh_{1\bar{1}}v_1 \tag{43}$$

$$v_{1,01} - v_{1,10} = A_{11}v_{1,}{}^1 - A_{11,}{}^1 v_1 \tag{44}$$

$$v_{1,0\bar{1}} - v_{1,\bar{1}0} = (A^1_{\bar{1}}v_1)_{,1} \tag{45}$$

where u is a real function and and $v_1\theta^1$ is a $(1,0)$-from.

[3]Any differential operator between the smooth sections of two arbitrary Riemannian or Hermitian vector bundles over a oriented Riemannian manifold admits a unique formal adjoint with respect to the corresponding L^2-inner products.

Lemma 6.9. *Let $f \in C^2(M)$, $J \in C^\infty(M, \mathscr{J})$, and X_f be the corresponding contact vector field. Then $L_{X_f} J = 2B_J^* f$.*

Proof. Pick a local frame field $Z_1 \in C^\infty(M, \xi^{1,0})$. Then the contact Hamiltonian vector field corresponding to f is locally written as

$$X_f = \mathbf{i} f_{,}{}^{\bar{1}} Z_{\bar{1}} - \mathbf{i} f_{,}{}^{1} Z_1 - f\mathcal{R}.$$

Computing at a point where $\theta_1^1 = 0$ (applying lemma 3.4):

$$\begin{aligned} L_{X_f}\theta^1 &= i_{X_f} d\theta^1 + d i_{X_f}\theta^1 \\ &= -\mathbf{i} f_{,}{}^{1}{}_{1}\theta^1 + (\mathbf{i} f_{,}{}^{1}{}_{\bar{1}} + A_{\bar{1}}^1 f)\theta^{\bar{1}} + 2(f_{,}{}^{1}{}_{0} + \mathbf{i} f_{,}{}^{1} A_{\bar{1}}^1)\theta \end{aligned}$$

Then dually we obtain

$$L_{X_f} Z_1 = \mathbf{i} f_{,}{}^{1}{}_{1} Z_1 - (\mathbf{i} f_{,}{}^{\bar{1}}{}_{1} - A_1^{\bar{1}} f) Z_{\bar{1}}.$$

Extending J from ξ to TM by setting $J(\mathcal{R}) = 0$, this implies

$$\begin{aligned} L_{X_f} J &= L_{X_f}(\mathbf{i}\theta^1 \otimes Z_1 - \mathbf{i}\theta^{\bar{1}} \otimes Z_{\bar{1}}) \\ &\equiv 2(f_{,}{}^{\bar{1}}{}_{1} + \mathbf{i} A_1^{\bar{1}} f)\theta^1 \otimes Z_{\bar{1}} + 2(f_{,}{}^{1}{}_{\bar{1}} - \mathbf{i} A_{\bar{1}}^1 f)\theta^{\bar{1}} \otimes Z_1 \bmod \theta. \end{aligned}$$

Restricting back to ξ yields the result. □

Remark. It is clear that X_f is an infinitesimal CR diffeomorphism if and only if $B_J^* f = 0$.

Let $(M, \xi, \hat{J})$ be closed strictly pseudo-convex CR 3-manifold and J_t be a deformation of $\hat{J}$. If $J_t = \varphi_t^* \hat{J}$ where φ_t is a smooth 1-parameter family of contactomorphisms, then evidently $\frac{d^2}{dt^2}\big|_{t=0} \mu(J_t, \hat{J}) = 0$: Such deformation has

$$\dot{J}_0 = \lim_{t\to 0} \frac{\varphi_t^* \hat{J} - \hat{J}}{t} = L_{X_f} \hat{J} = 2B_{\hat{J}}^* f$$

for some $f \in C^\infty(M)$, i.e. $\dot{J}_0 \in \mathrm{im}(B_{\hat{J}}^*)$. Then the first t-derivative vanishes identically:

$$\left.\frac{d}{dt}\right|_{t=0} \mu(J_t, \hat{J}) = -(\dot{J}_0, Q_{\hat{J}})_{\hat{J}} = -(2B_{\hat{J}}^* f, Q_{\hat{J}})_{\hat{J}} = -(2f, B_{\hat{J}} Q_{\hat{J}})_{L^2(M)} = 0.$$

The last equality is due to proposition 6.8. On the other hand, the L^2-orthogonal complement to $\mathrm{im}(B_{\hat{J}}^*)$ in $C^\infty(M, \mathscr{E}_{\hat{J}})$ is $\ker(B_{\hat{J}}) = \{E \in C^\infty(M, \mathscr{E}_{\hat{J}}) : B_{\hat{J}} E = 0\}$, namely

$$C^\infty(M, \mathscr{E}_{\hat{J}}) = \mathrm{im}(B_{\hat{J}}^*) \oplus \ker(B_{\hat{J}}).$$

The proof of this decomposition can be found in [33] Proposition 2.4. So we need to compute the second variation for deformation J_t in direction $\dot{J}_0 \in \ker(B_{\hat{J}})$.

Now we carry out the second order variation computation for Burns-Epstein invariant of $\mathbb{S}^3$. The standard complex structure $\hat{J}$ is induced the one from $\mathbb{C}^2$, θ is the standard contact form. We have shown before that the pseudo-Hermitian curvature is constant $\hat{R} = 2$.

Proposition 6.10. *Suppose J_t is a deformation of $\hat{J}$ the standard CR structure on $\mathbb{S}^3$ with $\dot{J}_0 \in \ker(B_{\hat{J}})$. Unless $\dot{J}_0 = 0$, we always have*

$$D^2\mu(J, \hat{J})|_{J=\hat{J}} = \left.\frac{d^2}{dt^2}\right|_{t=0} \mu(J_t, \hat{J}) > 0.$$

Proof. Set $\dot{J}_0 = 2E$ with $E \in \ker(B_{\hat{J}})$ and insert this J_t in the first order variation in proposition 6.5:

$$\frac{d}{dt}\mu(J_t, \hat{J}) = -\frac{1}{8\pi^2}(\dot{J}_t, Q_{J_t})_{J_t}.$$

Differentiate again with respect to t at $t = 0$ and notice $Q_{J_0} = Q_{\hat{J}} = 0$:

$$\left.\frac{d^2}{dt^2}\right|_{t=0} \mu(J_t, \hat{J}) = -\frac{1}{8\pi^2}(2E, \partial_t Q_{J_t}|_{t=0})_{\hat{J}}$$

Due to $\hat{R} = 2$ while $\hat{A} = 0$, then we compute (37) as:

$$\partial_t Q_{J_t}|_{t=0} = 2\mathrm{Re}\left[\left(-\frac{1}{6}E^{\bar{1}}_{1,\bar{1}}{}^{1}{}_{1}{}^{\bar{1}} + \frac{1}{6}E^{1}_{\bar{1},1}{}^{\bar{1}}{}_{1}{}^{\bar{1}} - \frac{\mathbf{i}}{2}\hat{R}E^{\bar{1}}_{1,0} + E^{\bar{1}}_{1,00} + \frac{2}{3}\mathbf{i}E^{\bar{1}}_{1,0}{}^{1}{}_{1}\right)\theta^1 \otimes Z_{\bar{1}}\right].$$

The lower weight terms $\mathcal{O}_3(E)$ vanish since those are all containing $\hat{A}$ and its derivatives which are identically zero. In zero torsion case, $B_{\hat{J}}E = 0$ reduces to $E^{\bar{1}}_{1,\bar{1}}{}^{1} + E^{1}_{\bar{1},1}{}^{\bar{1}} = 0$. Therefore the second order variation becomes

$$\left.\frac{d^2}{dt^2}\right|_{t=0} \mu(J_t, \hat{J}) = \frac{1}{8\pi^2}\int_M 2\mathrm{Re}\left[2E^{1}_{\bar{1}}\left(\frac{1}{3}E^{\bar{1}}_{1,\bar{1}}{}^{1}{}_{1}{}^{\bar{1}} + \frac{\mathbf{i}}{2}\hat{R}E^{\bar{1}}_{1,0} - E^{\bar{1}}_{1,00} - \frac{2}{3}\mathbf{i}E^{\bar{1}}_{1,0}{}^{1}{}_{1}\right)\right]\theta \wedge d\theta.$$

Using the commutation relation (43):

$$\begin{aligned} E^{\bar{1}}_{1,\bar{1}}{}^{1}{}_{1}{}^{\bar{1}} &= (E^{\bar{1}}_{1,\bar{1}}{}^{\bar{1}}{}_{\bar{1}} - \mathbf{i}E^{\bar{1}}_{1,\bar{1}0} - \hat{R}E^{\bar{1}}_{1,\bar{1}})_{,}{}^{\bar{1}} \\ &= E^{\bar{1}}_{1,\bar{1}}{}^{\bar{1}}{}_{\bar{1}}{}^{\bar{1}} - \mathbf{i}E^{\bar{1}}_{1,0}{}^{\bar{1}}{}_{\bar{1}} - \frac{1}{2}\hat{R}E^{\bar{1}}_{1,\bar{1}}{}^{\bar{1}} - \frac{1}{2}\hat{R}E^{\bar{1}}_{1,1}{}^{1} + \frac{\mathbf{i}}{2}\hat{R}E^{\bar{1}}_{1,0} + \hat{R}^2E^{\bar{1}}_{1}, \end{aligned}$$

then integration by parts yields

$$\begin{aligned} \left.\frac{d^2}{dt^2}\right|_{t=0} \mu(J_t, \hat{J}) =& \frac{1}{4\pi^2}2\mathrm{Re}\int_M \left(\frac{1}{6}\hat{R}E_{11,1}E^{11}{}_{,}{}^{1} + \frac{1}{3}E^{1}_{\bar{1},1}E^{\bar{1}}_{1,\bar{1}}{}^{\bar{1}} + \frac{1}{3}\hat{R}^2E^{1}_{\bar{1}}E^{\bar{1}}_{1}\right. \\ &\left. + E^{1}_{\bar{1},0}E^{\bar{1}}_{1,0} - \mathbf{i}E^{1}_{\bar{1},1}{}^{1}E^{\bar{1}}_{1,0} + \frac{2}{3}\mathbf{i}\hat{R}E^{1}_{\bar{1}}E^{\bar{1}}_{1,0} - \frac{1}{6}\hat{R}E^{1}_{\bar{1}}E^{\bar{1}}_{1,\bar{1}}{}^{\bar{1}}\right)\theta \wedge d\theta. \end{aligned}$$

Define four tensor fields as

$$\begin{aligned} T_1 &= E_{11,1}\theta^1 \otimes \theta^1 \otimes \theta^1 + E_{\bar{1}\bar{1},\bar{1}}\theta^{\bar{1}} \otimes \theta^{\bar{1}} \otimes \theta^{\bar{1}} \\ T_2 &= E^{\bar{1}}_{1,\bar{1}}{}^{\bar{1}}\theta^1 \otimes Z_{\bar{1}} + E^{1}_{\bar{1},1}{}^{1}\theta^{\bar{1}} \otimes Z_1 \\ T_3 &= \mathbf{i}E^{\bar{1}}_{1,0}\theta^1 \otimes Z_{\bar{1}} - \mathbf{i}E^{1}_{\bar{1},0}\theta^{\bar{1}} \otimes Z_1 \\ T_4 &= \hat{R}E = \hat{R}(E^{\bar{1}}_{1}\theta^1 \otimes Z_{\bar{1}} + E^{1}_{\bar{1}}\theta^{\bar{1}} \otimes Z_1). \end{aligned}$$

Then the second order varation can be written as

$$\left.\frac{d^2}{dt^2}\right|_{t=0} \mu(J_t, \hat{J}) = \frac{1}{4\pi^2}\left(\frac{1}{6}\hat{R}\|T_1\|^2_{\hat{J}} + \frac{1}{3}\|T_2\|^2_{\hat{J}} + \frac{1}{3}\|T_4\|^2_{\hat{J}} + \|T_3\|^2_{\hat{J}} - (T_3, T_2)_{\hat{J}} + \frac{2}{3}(T_3, T_4)_{\hat{J}} - \frac{1}{6}(T_2, T_4)_{\hat{J}}\right).$$

This is a positive definite quadratic form in (T_1, T_2, T_3, T_4) since $\hat{R} = 2 > 0$. So it is always non-negative, and it is zero only when $E = 0$. In conclusion, $D^2\mu(J, \hat{J})|_{J=\hat{J}} > 0$ on $\mathbb{S}^3$ when restricted to deformation orthogonal to orbits of contactomorphism group. □

In a follow up work [33], Cheng and Lee showed some local slice structures for contactomorphism group in the space of CR structure $C^\infty(M, \mathscr{J})$ involving the Nash-Moser inverse function theorem in the set-up of tame Fréchet manifolds introduced by R. S. Hamilton, c.f. [43]. Then using such results, they proved that the standard CR structure $\hat{J}$ on $\mathbb{S}^3$ is actually a strict local minimum for $\mu(J, \hat{J})$ in the C^∞-topology on $C^\infty(M, \mathscr{J})$.

7 Cartan flow

We have seen that the negative L^2-gradient of Burns-Epstein invariant is the Cartan tensor, and the invariant strictly decreases along the flow until reaching a critical point, i.e. the spherical CR structure. The solution to the flow equation (36) is called the Cartan flow. The Cartan flow can be understood as a sort of geometric flow. The most famous geometric flow on 3-manifolds geometries is the Ricci flow invented by R. S. Hamilton to study the Thurston's geometrization conjecture, and it was G. J. Perelman who filled in the last piece for the proof of the Poincaré conjecture during 2002-2003. Ricci flow can be viewed as gradient flow of Perelman's functional. We will not discuss Ricci flow of course, however, we refer the reader to the book [42] on the subject. The spirit of studying Cartan flow is pretty much in analogy to the Ricci flow. In this section, we make a brief review of the work [32] done by Cheng and Lee on showing the short-time existence of (regularized) Cartan flow, and discuss some CR geometry consequences out of it.

Example. We first study a special solution to the Cartan flow for $\mathbb{S}^3$ with SU(2)-invariant CR structure excluding the standard one (see example (1) in section 4.1). The curvature and torsion are $R = 2\frac{1+s^2}{1-s^2}$, $A^1_{\bar{1}} = -\mathbf{i}\frac{4s}{1-s^2}$, and by (29) we have $Q^1_{\bar{1}} = \frac{\mathbf{i}}{2}RA^1_{\bar{1}} = \frac{4s(1+s^2)}{(1-s^2)^2}$. Set $s = s(t)$, then with $J_{s(t)}$ and $Q_{J_{s(t)}}$ expressed in terms of (co)frames $\theta^1_{s(t)}, \theta^{\bar{1}}_{s(t)}, Z_{1\,s(t)}, Z_{\bar{1}\,s(t)}$, the Cartan flow equation (36) yields:

$$\frac{ds}{dt} = -4s(1+s^2).$$

With initial condition $s(0) = s_0 \in (-1,0)\cup(0,1)$ the solution is given by

$$s(t) = \pm\frac{1}{\sqrt{(1+s_0^{-2})e^{8t}-1}},$$

from its blow-up behavior we know that the SU(2)-invariance preserving Cartan flow exists only for a very short time.

The Cartan flow equation $\partial_t J_t = 2Q_{J_t}$ is a non-linear equation on J_t, and we observe that under transformations of contactomorphism group: For any contactomorphism φ such that $\varphi^*\theta = e^{2f}\theta$ for some $f \in C^\infty(M)$, we have $Q_{\varphi^* J_t} = e^{4f}\varphi^* Q_{J_t}$ due to that Q_{J_t} is pseudo-Hermitian invariant tensor. This will leads to a very large kernel of the principal symbol of the linearized operator DQ_J which means such PDE is highly degenerate. This is quite similar to the story of Ricci flow: The Ricci tensor as a nonlinear partial differential operator of the Riemannian metric has a principal symbol which is non-negative definite and has a non-trivial kernel (i.e. degenerate). This is exactly due to the fact that the Ricci tensor is invariant under diffeomorphism group (c.f. [42] §2.3). For this reason, one finds that the Ricci flow equation is merely weakly parabolic, and can not directly apply maximum principal to show the existence result. In order to make the Ricci flow equation a proper strictly parabolic PDE, DeTurck has implemented a very clever trick to add a term to the Rcci tensor side of the equation breaking the diffeomorphism invariance (c.f. [42] §3). In fact, the diffeomorphism invariance of the Rcci tensor as well as scalar curvature is reflected by certain contracted second Bianchi identity (c.f. [42] §2.2 and §2.3 Page 77). Here in our case, we have seen that before, namely in proposition 6.8, the identity is some sort of contracted Bianchi identity for Cartan tensor. In fact, one can show that the linearization of Q_J at J (recall (37)) satisfies $\mathrm{im}(DQ_J) \subset \ker(B_J)$, and $\ker(B_J)$ is the L^2-complement of the tangent directions to the contactomorphism orbit at J. Although DQ_J is some 4th order (weight 4) operator on $C^\infty(M, \mathscr{E}_J)$, we can nevertheless try to seek for regularized PDE in analogy to regularized Ricci flow equation (or called Ricci-DeTurck flow). Inspired by DeTurck's trick, Cheng and Lee analogously (comparing to the vector field constructed

in [42] §3) define another quadratic operator $F_J : C^\infty(M, \mathrm{End}(\xi)) \to C^\infty(M)$ given locally by

$$F_J E = \mathbf{i} E^1_1 E^{\bar{1}}_{1,\bar{1}}{}^1 - \mathbf{i} E^{\bar{1}}_{\bar{1}} E^1_{1,1}{}^{\bar{1}} + \mathbf{i} E^1_1 E^{\bar{1}}_{1,\bar{1}}{}^{\bar{1}} - \mathbf{i} E^{\bar{1}}_{\bar{1}} E^1_{1,1}{}^1 \tag{46}$$

and picking any $K \in C^\infty(M, \mathscr{J})$ as a fixed background CR structure, they obtain a nonlinear operator $\mathcal{P} : C^\infty(M, \mathscr{J}) \to C^\infty(M, \mathrm{End}(\xi))$ given by

$$\mathcal{P}(J) = -2Q_J + \frac{1}{6} B^*_J F_J K. \tag{47}$$

Observe that $\mathcal{P}(J) \in C^\infty(M, \mathscr{E}_J)$ for any $J \in C^\infty(M, \mathscr{J})$. This $\mathcal{P}$ is a weight 4 quasilinear differential operator. From the proposition 6.8 we see that if $\mathcal{P}(J) = 0$ for a J, then $Q_J = \frac{1}{12} B^*_J F_J K$ and

$$\|Q_J\|^2_J = \frac{1}{12}(Q_J, B^*_J F_J K)_J = \frac{1}{12}(B_J Q_J, F_J K)_{L^2(M)} = 0$$

implies $Q_J = 0$.

All these discussions motivate us to study the regularized Cartan flow equation which is a modification of Cartan flow equation by adding additional fixing term breaking the contactomorphism invariance:

$$\partial_t J_t + \mathcal{P}(J_t) = 0 \tag{48}$$

and $Q_J = 0$ corresponds to a equilibrium (time-independent) solution to such equation.

Proposition 7.1. *Suppose J_t is a C^k-solution with $k \geq 8$ to the regularized Cartan flow equation on $[0, \varepsilon]$. Then there exists a C^{k-4}-solution $\tilde{J}_t$ on $[0, \varepsilon]$ to $\partial \tilde{J}_t = 2Q_{\tilde{J}_t}$.*

Proof. Suppose K is C^k-differentiable. We can define a time-dependent Hamiltonian function $f \in C^{k-2}([0, \varepsilon] \times M)$ by $f_t = \frac{1}{12} F_{J_t} K$ and let $\varphi_t : M \to M$ be the flow generated by contact Hamiltonian vector field X_{f_t}. For a fixed $E \in C^\infty(M, \mathrm{End}(\xi))$, we have

$$\frac{d}{dt} \varphi_t^* E = \varphi_t^* L_{X_{f_t}} E \qquad \forall\, t \in [0, \varepsilon].$$

Set $J^\varphi_t := \varphi_t^* J_t$. By Lemma 6.8

$$\begin{aligned} \partial_t J^\varphi_t &= \varphi_t^* L_{X_{f_t}} J_t + \varphi_t^* \partial_t J_t \\ &= \varphi_t^* \left(2B^*_{J_t} f_t + 2Q_{J_t} - \frac{1}{6} B^*_{J_t} F_{J_t} K \right) = 2\varphi_t^* Q_{J_t}. \end{aligned}$$

Since φ_t is contactomorphism, we can write $\varphi_t^* \theta = e^{2g_t} \theta$ for some function $g \in C^{k-4}([0, \varepsilon] \times M)$. Since $Q_J = Q_{(J,\theta)}$ is a pseudo-Hermitian invariant tensor field,

$$\varphi_t^* Q_{J_t} = Q_{(\varphi_t^* J_t, \varphi_t^* \theta)} = e^{-4g_t} Q_{J^\varphi_t}.$$

Define a function $r \in C^{k-4}([0, \varepsilon] \times M)$ by

$$r(t, x) = \int_0^t e^{-4g_\tau(x)} d\tau$$

and since $[0, \varepsilon] \times M$ is compact by assumption, $\mathrm{im}(r) \subset \mathbb{R}_+$ is also compact. We obtain a space-dependent reparameterization of the time variable by setting $s_x(t) := (r^{-1})|_x \in \{x\} \times [0, \varepsilon] \cong [0, \varepsilon]$

with respect to each point $x \in M$. This is well-defined since the integrand is strictly positive, $r(t,x)$ increases monotonously on $t \in [0,\varepsilon]$ with x fixed. So at each fixed x we have $s_x(t) = t$. Note that comparing at different points, e.g. $x_1 \neq x_2$, $s_{x_1}(t)$ is not necessarily equal to $s_{x_2}(t)$, that is to say, if we fix t and let x vary then $s_x(t) \neq t$. Let $\tilde{J}_t(x) := J^{\varphi}_{s_x(t)}(x)$, then at each point x

$$\partial_t \tilde{J}_t(x) = (\partial_t s_x(t))(\partial_\tau J^{\varphi}_\tau(x))|_{\tau=s_x(t)} = 2Q_{J^{\varphi}_{s_x(t)}}(x) = 2Q_{\tilde{J}_t}(x).$$

Clearly the solution $\tilde{J}_t(x)$ varies C^{k-4}-smoothly on x. □

Lemma 7.2 ([32] Lemma 3.2)**.** *Let J_t be any deformation of J with $\dot{J} = 2E$. Then*

$$\partial_t B^*_{J_t} F_{J_t} K|_{t=0} = Re[(2E^{\bar{1}}_{1,\bar{1}}{}^1{}_1{}^{\bar{1}} + 2E^{1}_{\bar{1},1}{}^{\bar{1}}{}_1{}^{\bar{1}})\theta^1 \otimes Z_{\bar{1}} + \mathcal{O}_3(E)].$$

The proof involves merely cumbersome computations, which will be omitted here. The details can be found in [32] Lemma 3.2.

Now for each $\alpha \in \mathbb{C}$, we define a linear differential operator $\mathcal{L}_\alpha$ acting on tensor field. Let $\mathscr{T}$ be any complex tensor bundle of the tensor power of $\xi_{\mathbb{C}}$ and $\xi^*_{\mathbb{C}}$. First we define an differential operator $\mathfrak{D} : C^\infty(M, \mathscr{T}) \to C^\infty(M, \mathscr{T} \otimes \xi^*)$ of weight 1 by

$$\mathfrak{D}T = \nabla T|_\xi = \nabla_{Z_1} T \otimes \theta^1 + \nabla_{Z_{\bar{1}}} T \otimes \theta^{\bar{1}} \tag{49}$$

and its formal adjoint $\mathfrak{D}^* : C^\infty(M, \mathscr{T} \otimes \xi^*) \to C^\infty(M, \mathscr{T})$. Then, we define the quadratic operator $\mathcal{L}_\alpha : C^\infty(M, \mathscr{T}) \to C^\infty(M, \mathscr{T})$ of weight 2 by

$$\mathcal{L}_\alpha = \mathfrak{D}^*\mathfrak{D} + \mathbf{i}\alpha\nabla_{\mathcal{R}}. \tag{50}$$

This is called the *generalized Folland-Stein operator*. Its adjoint is $\mathcal{L}^*_\alpha = \mathcal{L}_{\bar{\alpha}}$.

For $\mathscr{T} = \mathscr{E}_J$, we have:

$$\mathcal{L}_\alpha E = 2\,\mathrm{Re}[(-E^{\bar{1}}_{1,1}{}^1 - E^{\bar{1}}_{1,\bar{1}}{}^{\bar{1}} + \mathbf{i}\alpha E^{\bar{1}}_{1,0})\theta^1 \otimes Z_{\bar{1}}]. \tag{51}$$

Lemma 7.3. *The linearizion of $\mathcal{P}$ at J is $D\mathcal{P}_J = \frac{1}{12}\mathcal{L}^*_\alpha\mathcal{L}_\alpha + \mathcal{O}_3$ where $\alpha = 4 + \mathbf{i}\sqrt{3}$.*

Proof. From previous lemma 7.2 and the computation of $\dot{Q}_J$, we have

$$D\mathcal{P}_J(2E) = 2\,\mathrm{Re}\left[\left(\frac{2}{3}E^{\bar{1}}_{1,\bar{1}}{}^1{}_1{}^{\bar{1}} - 2E^{\bar{1}}_{1,00} - \frac{4}{3}\mathbf{i}E^{\bar{1}}_{1,0}{}^1{}_1\right)\theta^1 \otimes Z_{\bar{1}} + \mathcal{O}_3(E)\right].$$

On the other hand, using the commutation relation for pseudo-Hermitian covariant derivatives (41) and compute directly

$$\mathcal{L}^*_\alpha\mathcal{L}_\alpha(2E) = 2\mathrm{Re}\left[\left(8E^{\bar{1}}_{1,\bar{1}}{}^1{}_1{}^{\bar{1}} + (4\,\mathrm{Re}\,\alpha - 2|\alpha|^2 - 2)E^{\bar{1}}_{1,00} + (16 - 8\,\mathrm{Re}\,\alpha)\mathbf{i}E^{\bar{1}}_{1,0}{}^1{}_1\right)\theta^1 \otimes Z_{\bar{1}} + \mathcal{O}_3(E)\right].$$

Substituting $\alpha = 4 + \mathbf{i}\sqrt{3}$, we obtain the result. □

For a fixed CR structure $\hat{J}$, we can choose an arbitrary smooth section $K \in C^\infty(M, \mathscr{E}_J)$, e.g. setting $K = \hat{J}$. We define a map $\widehat{\mathcal{P}} : C^\infty(M, \mathscr{E}_{\hat{J}}) \to C^\infty(M, \mathscr{E}_{\hat{J}})$ as

$$\widehat{\mathcal{P}} := \Pi_{\hat{J}} \circ \mathcal{P} \circ \Phi_{\hat{J}}. \tag{52}$$

Note that the restriction of projection $\Pi_{\hat{J}} : \mathrm{End}(\xi) \to \mathscr{E}_{\hat{J}}$ induces an isomorphism $\mathscr{E}_J \xrightarrow{\cong} \mathscr{E}_{\hat{J}}$. The linearization of this operator is $P_t = \Pi_{\hat{J}} \circ D\mathcal{P}_{J_t} \circ D\Phi_{\hat{J}}$, this is a weight 4 operator for each t. Since $\Phi_{\hat{J}}(0) = \hat{J}$, the initial value problem of the regularized Cartan flow equation now becomes: For some $\tau > 0$, $u \in C^\infty([0,\tau] \times M, \mathscr{E}_{\hat{J}})$ solving the equations

$$\partial_t u_t + \widehat{\mathcal{P}}(u_t) = 0 \tag{53}$$

$$u_0 = 0 \tag{54}$$

Now we wan to extend the notion of weight of a time independent differential operators to operators involving ∂_t as well as spatial derivatives. If P is a linear differential operator on section of any tensor bundle $\mathscr{T}$ (we only concern $\mathscr{T} = \mathscr{E}_{\hat{J}}$) over $[0,\tau] \times M$, we define the weight of P by setting $w(\partial_t) = 4$ and extending the definition to polynomials in ∂_t and spatial parts in the obvious way.

Following the general principle of PDE theory, one would first prove the existence and uniqueness result on some weaker Banach/Hilbert space than the Fréchet space C^∞, and then counting on regularity theory on that Banach/Hilbert space to verify the existence and uniqueness of classical solution. The function space we need here is called the Folland-Stein spaces $S^2_k(M, \mathscr{E}_{\hat{J}})$ built from space of L^2-sections $L^2(M, \mathscr{E}_{\hat{J}})$ with derivative operator (49). Roughly speaking, these are Sobolev-type spaces with Hilbert inner product $(\cdot,\cdot)_{S^2_k}$. Under such Hilbert space setting, the operator (50) is understood as a second order subelliptic operator, and thus $D\mathcal{P}$ is a 4th order subelliptic operator. We leave the details for subelliptic operators and Folland-Stein space theory to appendix B. Since the solution is time-dependent section u_t, one further define some Hilbert spaces from the Folland-Stein spaces: For any real number $\tau > 0$ and positive integer k, we define a Hilbert space of sections depending on one parameter $t \in [0,\tau]$ by

$$S^2_{k,\tau}(M, \mathscr{E}_{\hat{J}}) := L^2([0,\tau], S^2_k(M, \mathscr{E}_{\hat{J}}))$$

with norm

$$\|u\|^2_{S^2_{k,\tau}} := \int_0^\tau \|u_t\|^2_{S^2_k} dt.$$

For any $k > 0$, we define a Hilbert space by

$$E_{k,\tau} := \{u \in S^2_{0,\tau} \,|\, Pu \in S^2_{0,\tau} \text{ whenever } w(P) \le k\}$$

which can also be written as

$$E_{k,\tau} = \{u \in S^2_{k,\tau} \,|\, \partial_t^j u \in S^2_{k-4j,\tau} \text{ whenever } 0 \le 4j \le k\}.$$

Therefore it has a norm

$$\|u\|^2_{E_{k,\tau}} := \sum_{j=0}^{[k/4]} \int_0^\tau \|\partial_t^j u_t\|^2_{S^2_{k-4j,\tau}} dt.$$

For $k \ge 4$, we also define subspaces of $E_{k,\tau}$:

$$A_{k,\tau} := \{u \in E_{k,\tau} \,|\, u_0 = 0\}$$
$$B_{k,\tau} := \{u \in E_{k,\tau} \,|\, u_\tau = 0\}$$

with subspace norms. Of course, above construction generally works for any tensor bundle $\mathscr{T}$. The non-linear operator $\widehat{\mathcal{P}}$ extends to a C^1-map from $A_{k,\tau} \to E_{k-4,\tau}$ for all $k \ge 14$ ([32] Lemma 5.4).

These spaces are the basic set-up for the initial value problem of regularized Cartan flow equation: Any $u \in S^2_{0,\tau}$ with $\partial_t u \in S^2_{0,\tau}$ is an absolutely continuous function $[0,\tau] \to S^2_0$, hence the boundary

conditions $u_0 = 0$ and $u_\tau = 0$ make sense. The linear operator P_t on sections depending smoothly on $t \in [0, \tau]$ of weight 4 for each t, then $\partial_t + P_t$ maps $A_{k+4,\tau}$ continuously to $E_{k,\tau}$ for any $k \geq 0$.

Based on the Folland-Stein spaces theory in appendix B.2, we proves several technical propositions below, which lead to the short time existence result for regularized Cartan flow equation in the end.

Proposition 7.4 ([32] Corollary 4.3)**.** *$\exists C > 0$ such that for all $u \in S^2_{k+2}$*

$$\|u\|_{S^2_{k+2}} \leq C\|(\mathcal{L}_0 + 1)u\|_{S^2_k}.$$

Proof. By [32] Proposition 4.2 and the property (e) in appendix B.2, we have

$$\|u\|_{S^2_{k+2}} \leq C_k(\|\mathcal{L}_0 u\|_{S^2_k} + \|u\|_{L^2}) \leq C_k\|(\mathcal{L}_0 + 1)u\|_{S^2_k} + (C_k + C_\varepsilon)\|u\|_{L^2} + \varepsilon\|u\|_{S^2_{k+2}}$$

for any $\varepsilon > 0$. Then by definition of $\mathcal{L}_0$, we estimate:

$$\|u\|^2_{L^2} = (u, u)_J \leq (u, u)_J + (\mathfrak{D}u, \mathfrak{D}u)_J = (u, (\mathcal{L}_0 + 1)u)_J \leq \|u\|_{L^2}\|(\mathcal{L}_0 + 1)u\|_{L^2}.$$

Thus $\|u\|_{L^2} \leq \|(\mathcal{L}_0 + 1)\|_{L^2} \leq \|(\mathcal{L}_0 + 1)u\|_{S^2_k}$ and the result follows. □

Proposition 7.5 ([32] Proposition 4.4)**.** *Suppose $\alpha \in \mathbb{C}$ is admissible, and let $P = \mathcal{L}^*_\alpha \mathcal{L}_\alpha + S$ where S is differential operator with $\mathrm{w}(S) \leq 3$. For any integer $k \geq 0$, there exist positive constant C, λ such that*

$$\mathrm{Re}((\mathcal{L}_0 + 1)^{k+1}u, (\mathcal{L}_0 + 1)^k(P + \lambda)u)_J \geq C\|u\|_{S^2_{2k+4}} \qquad \forall\, u \in S^2_{2k+4}.$$

Proof. For $u \in C^\infty(M, \mathscr{E}_J)$, due to self-adjointness of $\mathcal{L}_0 + 1$, we have

$$\begin{aligned}((\mathcal{L}_0 + 1)^{k+1}u, (\mathcal{L}_0 + 1)^k Pu)_J &= (\mathcal{L}_\alpha(\mathcal{L}_0 + 1)^{2k+2}u, \mathcal{L}_\alpha u)_J + (u, (\mathcal{L}_0 + 1)^{2k+2}Su)_J \\ &= ((\mathcal{L}_0 + 1)^{2k+2}\mathcal{L}_\alpha u, \mathcal{L}_\alpha u)_J + (u, Xu)\end{aligned}$$

where X is some operator with $w(X) \leq 4k+7$ and we can write it as $X = \sum_j V_j W_j$, $w(V_j) \leq 2k+3$, $w(W_j) \leq 2k+4$. Here we have observed that $[\mathcal{L}_0 + 1, \mathcal{L}_\alpha] = \mathbf{i}\alpha[\mathcal{L}_0, \mathcal{R}]$ which can be shown to be weight 2 operator by using (41)-(42). By proposition 7.4 above and property (a) in appendix B.2, we find estimate:

$$\begin{aligned}\mathrm{Re}((\mathcal{L}_0 + 1)^{k+2}u, (\mathcal{L}_0 + 1)^k Pu)_J &= \|(\mathcal{L}_0 + 1)^{k+1}\mathcal{L}_\alpha u\|^2_{L^2} + \mathrm{Re}\sum_j (V_j^* u, W_j u)_J \\ &\geq C\|\mathcal{L}_\alpha u\|^2_{S^2_{2k+2}} - C\|u\|_{S^2_{2k+3}}\|u\|_{S^2_{2k+4}}.\end{aligned}$$

By property (e) in appendix B.2, for any $\varepsilon > 0$ we have

$$\|u\|_{S^2_{2k+3}}\|u\|_{S^2_{2k+4}} \leq \varepsilon\|u\|_{S^2_{2k+4}} + C_\varepsilon\|u\|_{S^2_{2k+2}}.$$

Applying again [32] Proposition 4.2 and using $\|u\|_{L^2} \leq \|(\mathcal{L}_0 + 1)u\|_{L^2}$ iteratively, we obtain the desired inequality in the statement by setting λ to some proper positive constant.Since both side of the desired inequality are continuous in S^2_{2k+4} and C^∞ is dense in S^2_{2k+4}, the same estimate holds for all $u \in S^2_{2k+4}$. □

For any $u, v \in C^\infty([0, \tau] \times M)$, we perform the time integration below:

$$\int_0^\tau [((\mathcal{L}_0+1)^k \partial_t u_t, (\mathcal{L}_0+1)^{k+2}v_t)_J + ((\mathcal{L}_0+1)^{k+2}u_t, (\mathcal{L}_0+1)^k \partial_t v_t)_J]\, dt = ((\mathcal{L}_0+1)^{k+1}u_t, (\mathcal{L}_0+1)^{k+1}v_t)_J\Big|_{t=0}^{t=\tau}.$$

The right hand side has a unique continuous extension to all $u, v \in E_{2k+4,\tau}$ and this equation still holds. Basically, the left hand side is continuous in $u, v \in E_{2k+4,\tau}$ and C^∞ is dense in $E_{2k+4,\tau}$, we can simply use the same formula to define the right hand side.

Proposition 7.6 ([32] Theorm 4.6)**.** *Let P_t be a time-dependent linear differential operator on sections of $\mathscr{E}_J$ over M involving only spatial derivatives of weight ≤ 4 and depending smoothly on $t \in [0,1]$, with $P_0 = \mathcal{L}_\alpha^* \mathcal{L}_\alpha + S$ where α is admissible and S is an operator of weight ≤ 3. For any integer $k \geq 0$, there exist $\tau \in (0,1]$ such that $\partial_t + P_t$ is a bounded isomorphism from $A_{2k+4,\tau}$ to $E_{2k,\tau}$.*

Proof. Since P_t is smooth in t, by picking sufficient small $\tau \in (0,1]$ we can ensure the coefficients of $P_t - P_0$ are as small as we want in $C^{2k}(M)$ for each $t \in [0,\tau]$: For a small τ, for some $\varepsilon > 0$ we have

$$\|P_t u - P_0 u\|_{S^2_{2k}} \leq \varepsilon \|u\|_{S^2_{2k+4}} \qquad \forall\, u \in S^2_{2k+4}\,,\ t \in [0,\tau]. \tag{55}$$

Putting λ as in proposition 7.5, we define a Hermitian bilinear form:

$$\mathcal{A} : B_{2k+4,\tau} \times S^2_{2k+4,\tau} \to \mathbb{C}$$
$$(v,u) \mapsto \int_0^\tau [((\mathcal{L}_0+1)^{k+2} v_t, (\mathcal{L}_0+1)^k (P_t+\lambda) u_t)_J - ((\mathcal{L}_0+1)^k \partial_t v_t, (\mathcal{L}_0+1)^{k+2} u_t)_J]\, dt.$$

Claim 1: For any $v \in B_{2k+4,\tau}$ the linear functional $u \mapsto \overline{\mathcal{A}(v,u)}$ is bounded in $S^2_{2k+4,\tau}$.

Proof of claim 1: This is clear by previous estimates.

Claim 2: $\exists\, C > 0$, $C\|v\|^2_{S^2_{2k+4,\tau}} \leq |\mathcal{A}(v,v)|\ \forall\, v \in B_{2k+4,\tau}$.

Proof of claim 2: By proposition 7.5 and (55), by choosing τ so small that $C_1 > \varepsilon C_2$ for two positive constant $C_1, C_2 > 0$, for every $t \in [0,\tau]$ there is:

$$\begin{aligned}\mathrm{Re}((\mathcal{L}_0+1)^{k+2} v_t, (\mathcal{L}_0+1)^k (P_t+\lambda) v_t)_J &= \mathrm{Re}((\mathcal{L}_0+1)^{k+2} v_t, (\mathcal{L}_0+1)^k (P_0+\lambda) v_t)_J \\ &\quad + \mathrm{Re}((\mathcal{L}_0+1)^{k+2} v_t, (\mathcal{L}_0+1)^k (P_t - P_0) v_t)_J \\ &\geq C_1 \|v_t\|^2_{S^2_{2k+4}} - \varepsilon C_2 \|v_t\|^2_{S^2_{2k+4}}.\end{aligned}$$

Regarding the second term in the bilinear form, for any $v \in B_{2k+4,\tau}$ by the time integration introduced above we find:

$$\mathrm{Re} \int_0^\tau ((\mathcal{L}_0+1)^k \partial_t v_t, (\mathcal{L}_0+1)^{k+2} v_t)_J\, dt = -\frac{1}{2} \|(\mathcal{L}_0+1)^{k+1} v_0\|^2_{L^2}$$

and thus

$$\mathrm{Re}\, \mathcal{A}(v,v) = \mathrm{Re} \int_0^\tau ((\mathcal{L}_0+1)^{k+1} v_t, (\mathcal{L}_0+1)^k (P_t+\lambda) v_t)_J\, dt + \frac{1}{2} \|(\mathcal{L}_0+1)^{k+1} v_0\|^2_{L^2} \geq C \int_0^\tau \|v_t\|^2_{S^2_{2k+4}}\, dt.$$

This finishes the proof of the claim.

The claim 1 and claim 2 meet the conditions of a generalization of Lax-Milgram lemma.[4] Then for any linear functional $F : S^2_{2k+4} \to \mathbb{C}$, there is $\tilde{u} \in S^2_{2k+4}$ with $\|\tilde{u}\|_{S^2_{2k+4,\tau}} \leq C\|F\|_{\mathrm{Op.}}$ such that $\mathcal{A}(v,\tilde{u}) = F(v)\ \forall\, v \in B_{2k+4,\tau}$.

[4] This generalization is due to J.L. Lions ([44] Lemma 41.2): Let $\mathcal{H}$ be a Hilbert space and $\mathcal{V}$ is a linear subspace, $\mathcal{A} : \mathcal{V} \times \mathcal{H} \to \mathbb{C}$ is a sesquilinear functional satisfying the properties:
(1) For each fixed $v \in \mathcal{V}$, $u \mapsto \mathcal{A}(v,u)$ is a continuous linear functional on $\mathcal{H}$.
(2) $\exists\, C > 0$ such that for every $v \in \mathcal{V}$: $C\|v\|^2_{\mathcal{H}} \leq |\mathcal{A}(v,v)|$.
Then there is a bounded linear map G of anti-dual $\bar{\mathcal{H}}^*$ into $\mathcal{H}$ with norm $\leq C$ such that for every continuous anti-linear functional F on $\mathcal{H}$ we have $\mathcal{A}(GF,v) = F(v)\ \forall\, v \in \mathcal{V}$.

Now pick a $f \in E_{2k,\tau}$, we take $F : S^2_{2k+4,\tau} \to \mathbb{C}$ as the linear functional

$$F(v) = \int_0^\tau ((\mathcal{L}_0+1)^{k+2} v_t, (\mathcal{L}_0+1)^k (e^{-\lambda t} f_t))_J \, dt.$$

There exists $\tilde{u} \in S^2_{2k+4,\tau}$ such that

$$\begin{aligned}\int_0^\tau [((\mathcal{L}_0+1)^{k+2} v_t, (\mathcal{L}_0+1)^k (P_t+\lambda)\tilde{u}_t)_J - ((\mathcal{L}_0+1)^k \partial_t v_t, (\mathcal{L}_0+1)^{k+2}\tilde{u}_t)_J] \, dt \\ = \int_0^\tau ((\mathcal{L}_0+1)^{k+2} v_t, (\mathcal{L}_0+1)^k (e^{-\lambda t} f_t))_J \, dt \qquad \forall\, v \in B_{2k+4,\tau}.\end{aligned}$$

In particular, for $v \in C_0^\infty((0,\tau)\times M)$, this $\tilde{u}$ is a weak solution (i.e. a solution in distribution sense) to the equation

$$(\mathcal{L}_0+1)^{k+2}[(\partial_t + P_t + \lambda)\tilde{u}_t - e^{-\lambda t} f_t] = 0.$$

Since $\mathcal{L}_0 + 1$ is hypoelliptic, this implies that $(\partial_t + P_t + \lambda)\tilde{u}_t - e^{-\lambda t} f_t \in C^\infty(M)$ for each t (see appendix B.1), and also $(\mathcal{L}_0+1)|_{S^2_2}$ is injective by (a),(b) in appendix B.2, we actually have

$$(\partial_t + P_t + \lambda)\tilde{u}_t = e^{-\lambda t} f_t.$$

Therefore $u = e^{\lambda t}\tilde{u}_t$ solves the equation

$$(\partial_t + P_t) u_t = f_t. \tag{56}$$

The operator norm of F is then bounded by $C\|e^{-\lambda t} f\|_{2k,\tau} \le C\|f\|_{2k,\tau}$, and thus we get the estimate $\|u\|_{2k+4,\tau} \le e^\lambda C \|f\|_{2k,\tau}$. Furthermore $\partial_t u_t = f_t - P_t u_t \in S^2_{2k,\tau}$. Suppose $\partial_t^j u \in S^2_{2k-4j+4,\tau}$, differentiating (56), we obtain $\partial_t^{j+1} u_t = \partial_t^j f_t - \partial_t^j (P_t u_t) \in S^2_{2k-4j,\tau}$ for $0 \le 4j \le 2k$, so $u \in E_{2k+4,\tau}$. We can conclude that for any $v \in B_{2k+4}$:

$$0 = \mathcal{A}(v,\tilde{u}) - \int_0^\tau ((\mathcal{L}_0+1)^{k+2} v_t, (\mathcal{L}_0+1)^k (e^{-\lambda t} f_t))_J \, dt = ((\mathcal{L}_0+1)^{k+1} v_0, (\mathcal{L}_0+1)^{k+1}\tilde{u}_t)_J.$$

For any $w \in C^\infty(M)$, we can find $v \in B_{2k+4,\tau}$ such that $v_0 = w$, this shows that $(\mathcal{L}_0+1)^{2k+2}\tilde{u}_0 = 0$ as distribution, hence $u_0 = \tilde{u}_0 = 0$. Thus $u \in A_{2k+4,\tau}$.

Finally, for uniqueness of solution to (56) with initial condition $u_0 = 0$, suppose $u \in A_{2k+4,\tau}$ solves $(\partial_t + P_t)u_t = 0$, setting $\tilde{u}_t = e^{-\lambda t} u_t$, the proposition 7.5 implies $\exists\, C > 0$:

$$\begin{aligned} C\int_0^\tau \|\tilde{u}_t\|^2_{S^2_4} \, dt &\le \operatorname{Re}\int_0^\tau ((\mathcal{L}_0+1)^2\tilde{u}_t, (P_t+\lambda)\tilde{u}_t)_J \, dt \\ &= -\operatorname{Re}\int_0^\tau ((\mathcal{L}_0+1)^2\tilde{u}_t, \partial_t\tilde{u}_t)_J \, dt = -\frac{1}{2}\|(\mathcal{L}_0+1)\tilde{u}_\tau\|^2_{L^2} \le 0.\end{aligned}$$

Therefore $u = \tilde{u} = 0$. □

The strategy to prove the short-time existence is firstly construct an infinite-order formal power series in t solving (53)-(54) recursively, that is: For any $N \in \mathbb{N}$, a section $\tilde{u}_t$ depending on $t \in [0,1]$ smoothly solving

$$\begin{aligned}\partial_t \tilde{u}_t + \widehat{\mathcal{P}}(\tilde{u}_t) &= O(t^N) \\ \tilde{u}_0 &= 0.\end{aligned}$$

Denote P_t be the linearization of $\widehat{\mathcal{P}} : A_{k,\tau} \to E_{k-4,\tau}$ $(k \geq 14)$ at $\tilde{u}_t$, i.e. $P_t = \Pi_{\hat{J}} \circ D\mathcal{P}_{\tilde{J}_t} \circ (D\Phi_{\hat{J}})_{\tilde{u}_t}$ where $\tilde{J}_t = \Phi_{\hat{J}}(\tilde{u}_t)$. At $t = 0$, $P_0 = \frac{1}{12}\mathcal{L}_\alpha^* \mathcal{L}_\alpha + S$ where $\mathcal{L}_\alpha$ is now defined for $\hat{J}$ and $\mathrm{w}(S) \leq 3$, so P_t satisfies the condition in proposition 7.6. This implies that $\partial_t + P_t$ is a bounded isomorphism from $A_{2k+4,\tau}$ to $E_{2k,\tau}$ for sufficiently small τ, and the inverse function theorem[5] implies that $\partial_t + \widehat{\mathcal{P}}$ has a C^1-inverse on some neighborhood of $\tilde{u}_t \in A_{k,\tau}$ $(k \geq 14)$. The section $\tilde{f}_t = (\partial_t + \widehat{\mathcal{P}})\tilde{u}_t$ vanishes to infinite order at $t = 0$, so for any $\varepsilon \in [0, \tau]$ the section $f_t^\varepsilon := \begin{cases} 0 & , t \in [0, \varepsilon] \\ \tilde{f}_{t-\varepsilon} & , t \in [\varepsilon, \tau] \end{cases}$ is smooth in t. Then $\forall\, \delta > 0$, $\exists\, \varepsilon > 0$ such that $\|f^\varepsilon - \tilde{f}\|_{E_{k-4,\tau}} \leq \delta$ $(k \geq 14)$. Note that the subscript of variable t is dropped since under the norm $\|\cdot\|_{E_{k-4,\tau}}$ it is integrated out. So there exits $u_t \in A_{k,\tau}$ solving $(\partial_t + \widehat{\mathcal{P}})u_t = f_t^\varepsilon$ by the C^1-inverse of the differential operator, and on $t \in [0, \varepsilon]$ we find that u_t is the desired solution to (53)-(54). Thus this procedure demonstrates that for any $k \geq 14$ and a small $\varepsilon > 0$, there exists a solution $u \in A_{k,\varepsilon}$ to (53)-(54) ([32] Theorem 5.5). This is the central result in [32]. For any m, by choosing large enough k we can find a C^m solution u_t (due to property (c) in appendix B). Then in short time interval $[0, \varepsilon]$ we have $J_t = \Phi_{\hat{J}}(u_t)$ with $J_0 = \hat{J}$ solving the regularized Cartan flow equation in original form with corresponding short-time existence. We summarize the result as the following theorem.

Theorem 7.7 ([32] Theorem B)**.** *Let $\hat{J}$ be any smooth oriented CR structure on (M, ξ). For any integer $m \in \mathbb{N}$ (finite), there exists $\varepsilon > 0$ such that the initial value problem of regularized Cartan flow equation (48) with $J_0 = \hat{J}$ has a solution $J \in C^m$ on time interval $[0, \varepsilon]$.*

Remark. The short-time existence here is not a C^∞-smooth result (since $m = \infty$ requires $k = \infty$).

Whether there exists a unique smooth (C^∞ in all variables) solution to the regularized Cartan flow equation is still unknown. Existence and uniqueness of such classical solution is expected to rely on some estimates and regularity results in terms of Folland-Stein spaces for linear subelliptic equations whose coefficients are only finitely differentiable.

There is however a weaker uniqueness result shown by Jih-Hsin Cheng:

Theorem 7.8 ([34] Theorem 5.2)**.** *For large enough m (e.g. $m \geq 14$), the C^m-solution of the initial value problem of regularized Cartan flow equation (48) is unique in a small interval $[0, \varepsilon']$ with $\varepsilon' < \varepsilon$.*

A natural step moving forward is to attack the long-time existence problem of the Cartan flow. However, long-time existence type problems are very difficult and one should not expect the solutions always exist in general. For instance, one interesting question is asking under which conditions the Cartan flow equation $\partial_t J_t = 2Q_J$ can have long-time solution (i.e. Cartan flow exists on the entire $[0, \infty]$) so that the initial CR structure J_0 under the flow converging to the spherical CR structure $\hat{J}$ as $t \to \infty$ (note that ξ is fixed up to isotopy). We do not even know the answer for $(\mathbb{S}^3, \xi_{\mathrm{std}})$ yet. In general, one possible way inspired by the results of Ricci flow is to start with the initial CR structure on a closed CR 3-manifold admitting strictly positive pseudo-Hermitian curvature and vanishing pseudo-Hermitian torsion. At best, we might hope for prolonging the short-time Cartan flow to its maximal existence interval and ending up with curvature explosion in maximum finite time, thereafter one may attempt to analogously construct certain normalized Cartan flow by adding to the Cartan curvature some averaged curvature term so that this normalized Cartan flow might exist on entire time interval $[0, \infty]$. This is beyond scope since there are fundamental analysis difficulties here:

[5]Inverse function theorem for Banach spaces: Suppose $\mathcal{B}_1$ and $\mathcal{B}_2$ are Banach spaces. Let U be an open neighborhood of the origin in $\mathcal{B}_1$ and $F : U \to \mathcal{B}_2$ a continuously differentiable map, and assume that the Fréchet derivative $DF_0 : \mathcal{B}_1 \to \mathcal{B}_2$ at 0 is a bounded linear isomorphism of $\mathcal{B}_1$ onto $\mathcal{B}_2$. Then there exists an open neighborhood V of $F(0) \in \mathcal{B}_2$ and a continuously differentiable map $G : V \to \mathcal{B}_1$ such that $F \circ G = \mathrm{Id}_{\mathcal{B}_2}$ and $G \circ F = \mathrm{Id}_{\mathcal{B}_1}$. Moreover, $G(y)$ is the only sufficiently small solution x of the equation $F(x) = y$ where $x \in \mathcal{B}_1, y \in \mathcal{B}_2$.

Unlike what Hamilton has done in his famous paper showing long-time existence for the Ricci flow on 3-manifolds initially with strictly positive Ricci curvature, there seems to be no suitable maximum principle available for the 4th order subelliptic operators here. Hence even provided with well-posted questions, we would still need to develop some L^2-estimates for the solutions of regularized Cartan flow equation.

In the light of the short-time existence result, Jih-Hsin Cheng has studied the behavior of holomorphic fillability of CR structure under regularized Cartan flow.

Theorem 7.9 ([35] Corollary C)**.** *Suppose J_0 is holomorphic fillable in the convex side with respect to θ with $A_{J_0,\theta} = 0$ and $R_{J_0,\theta} < 0$. Solution to the regularized Cartan flow equation (48) with $K = J_0$ exists on $[0, \varepsilon)$, then $A_{J_t,\theta}$ stays vanishing on $[0, \varepsilon)$, and J_t stays holomorphic fillable in the convex side with respect to θ on $[0, \varepsilon'')$ where $\varepsilon'' \leq \varepsilon$.*

Remark. The holomorphically fillability in the convex side with respect to θ means the Reeb direction $\mathcal{R}$ for θ under the action of complex structure of the holomorphic filling points inwards. The parameter ε is as in theorem 7.7, and ε'' is chosen sufficiently small such that $R_{J_t,\theta} < 0 \ \forall\, t \in [0, \varepsilon'')$.

In [36], Cheng proposed that if one can show generic long-time existence for Cartan flow converging to the standard CR structure on $(\mathbb{S}^3, \xi_{\text{std}})$ starting with any initial CR structure, then it might provide a new geometric analysis proof for the Smale conjecture which says that $\text{Diff}(\mathbb{S}^3)$ is homotopy equivalent to O(4) (In 1983, Hatcher gave a combinatorial proof). He has mentioned that by viewing the Burns-Epstein invariant as some Morse function on the space of CR structures, one may attempt to study the topology of this infinite dimensional space from Morse theory point of view. Potentially, many aspects of the Ricci flow on 3-manifolds could be parallelly investigated on the Cartan flow. Obviously, these expectations are beyond reachable comparing to what we can achieve so far. Nevertheless, it might still be a considerable project for the future study.

Appendices

A Chern-Weil theory and transgression

A.1 Chern-Weil theory

There are many materials on Chern-Weil theory, we choose [2] and [3] in our favor.

Suppose a vector bundle is equipped with connection, the curvature measure the local deviation of the vector bundle from flatness. For a flat vector bundle over a simply connected base manifold, it immediately yields the triviality. This suggest there is a relationship between characteristics class and curvature. The Chern-Weil method is a way to justify this relationship and represent these classes in de Rham cohomolgy.

We shall first recall symmetric algebras and polynomial algebras: Let V be a finite dimensional $\mathbb{K}$-vector space ($\mathbb{K}=\mathbb{R},\mathbb{C}$). For $k\geq 1$, S^kV^* denote the vector space of symmetric $\mathbb{K}$-linear functions $P:V\times\cdots\times V\to\mathbb{K}$ which is identified with the map $P:V\otimes\cdots\otimes V\to\mathbb{K}$ which is invariant under the action from permutation group, namely

$$P(v_{\sigma(1)}\otimes\cdots\otimes v_{\sigma(k)})=P(v_1\otimes\cdots\otimes v_k)\ \ \forall\,\sigma\in\mathfrak{S}_k.$$

There is a natural composition product $\circ:S^kV^*\otimes S^lV^*\to S^{k+l}V^*$ defined by

$$(P\circ Q)(v_1\otimes\cdots\otimes v_{k+l})=\frac{1}{(k+l)!}\sum_{\sigma\in\mathfrak{S}_{k+l}}P(v_{\sigma(1)}\otimes\cdots\otimes v_{\sigma(k)})Q(v_{\sigma(k+1)}\otimes\cdots\otimes v_{\sigma(k+l)}).$$

This gives rise a graded algebra, the symmetric algebra $S(V^*)=\bigoplus_{k=0}^{\infty}S^kV^*$. There is a basic fact of the symmetric algebra: There is a canonical isomorphism $S^kV^*\cong\mathrm{gr}^k\,\mathbb{K}[x_1,\cdots,x_n]$ given by

$$\widetilde{P}(x_1,\ldots,x_n)=P(v\otimes\cdots\otimes v)$$

with $v=\sum_{i=1}^n x_i\mathbf{e}_i$ and $V=\mathrm{span}_{\mathbb{K}}\{\mathbf{e}_i:i=1,\ldots,n\}$. So $S(V^*)\cong\mathbb{K}[x_1,\ldots,x_n]$.

The adjoint representation of Lie group G on its Lie algebra $\mathfrak{g}$ induces an action of G on $S^k(\mathfrak{g}^*)$ for every k:

$$(g\cdot P)(X_1\otimes\cdots\otimes X_k)=P(\mathrm{Ad}(g^{-1})X_1\otimes\cdots\otimes\mathrm{Ad}(g^{-1})X_k)$$

where $X_1,\ldots,X_k\in\mathfrak{g}$ and $g\in G$.

Definition. *$P\in S^k\mathfrak{g}^*$ is called invariant if $g\cdot P=P\ \forall\,g\in G$. The set of invariant elements in $S(\mathfrak{g}^*)$ is called an invariant polynomial ring denoted $S(\mathfrak{g}^*)^G=\bigoplus_{k=0}^{\infty}S^k(\mathfrak{g}^*)^G$, which is a subring of $S(\mathfrak{g}^*)$.*

Now we consider a principal G-bundle $\mathscr{P}\xrightarrow{\pi}M$ with connection ω and Ω is the corresponding curvature form. For $k\geq 1$ we have $\Omega^k=\Omega\wedge\cdots\wedge\Omega\in\Omega^{2k}(\mathscr{P};\mathfrak{g}^{\otimes k})$ and $P\in S^k(\mathfrak{g}^*)^G$ defines a linear map $P:\mathfrak{g}\otimes\cdots\otimes\mathfrak{g}\to\mathbb{K}$ giving a $2k$-form $P(\Omega^k)\in\Omega^{2k}(\mathscr{P})$. Since Ω is horizontal Ad-equivariant and P is invariant, then $P(\Omega^k)$ is a basic $2k$-form. There is a unique primitive $2k$-form $P(F_\omega^k)$ on M which pulls back to $P(\Omega^k)$ by $\pi^*:\Omega^{2k}(M)\to\Omega^{2k}(\mathscr{P})$ where $F_\omega\in\Omega^2(M;\mathrm{Ad}\,P)$ is the unique curvature 2-form on M corresponding to the curvature 2-form $\Omega\in\Omega^2(\mathscr{P};\mathfrak{g})$ on $\mathscr{P}$.

Proposition. *In above setting, we have:*

i) $P(F_\omega^k) \in \Omega^{2k}(M)$ *is closed.*

ii) For $P \in S^k(\mathfrak{g}^*)^G, Q \in S^l(\mathfrak{g}^*)^G$, $(P \circ Q)(F_\omega^{k+l}) = P(F_\omega^k) \wedge Q(F_\omega^l)$

iiii) For a bundle map $\bar{f} : \mathscr{P}' \to \mathscr{P}$ *(*$f : M' \to M$ *is the underlying map between base manifolds), then for a connection* ω *in* E, $\omega' = \bar{f}^*\omega$ *is the induced connection on* $\mathscr{P}'$*, then* $P(F_{\omega'}^k) = f^*P(F_\omega^k)$.

Proof. Since $\pi^* : \Omega^{2k}(M) \to \Omega^{2k}(\mathscr{P})$ is injective, it suffices to show the statements in $\Omega^{2k}(\mathscr{P})$.

i) Using the symmetry of P we get:

$$dP(\Omega^k) = kP(d\Omega \wedge \Omega^{k-1}) \overset{\text{Bianchi identity}}{=} kP([\Omega, \omega] \wedge \Omega^{k-1})$$

The form above is horizontal, and $[\Omega, \omega]$ vanishes on horizontal vectors implies that $dP(\Omega^k) = 0 \in \Omega^{2k+1}(\mathscr{P})$.

ii) Let T_σ be permutation action on $\mathfrak{g}^{\otimes(k+l)}$ given by $T_\sigma(X_1 \otimes \cdots \otimes X_{k+l}) = X_{\sigma(1)} \otimes \cdots \otimes X_{\sigma(k+l)}$. $\Omega^{k+l} = T_\sigma \circ \Omega^{k+l} = T_\sigma \circ \Omega^{k+l} = T_\sigma \circ (\Omega^k \wedge \Omega^l)$ since Ω is degree 2. Hence

$$(P \circ Q)(\Omega^{k+l}) = \frac{1}{(k+l)!} \sum_{\sigma \in \mathfrak{S}_{k+l}} (P \otimes Q) \circ T_\sigma \circ (\Omega^{k+l}) = \frac{1}{(k+l)!} \sum_{\sigma \in \mathfrak{S}_{k+l}} P(\Omega^k) \wedge Q(\Omega^l) = P(\Omega^k) \wedge Q(\Omega^l)$$

iii) Since $\Omega' = \bar{f}^*\Omega$ in $\Omega^2(\mathscr{P})$, clearly $P(\Omega'^k) = \bar{f}^*P(\Omega^k)$. □

Definition. $\mathfrak{W}(\mathscr{P}, \cdot) : S^k(\mathfrak{g}^*)^G \to H_{dR}^{2k}(M)$ *given by* $\mathfrak{W}(\mathscr{P}, P) = [P(F_\omega^k)]$ *is called the Weil homomorphism.*

Theorem. *The Weil homomorphism satisfies the following properties:*

i) $\mathfrak{W}(\mathscr{P}, P) \in H_{dR}^{2k}(M)$ *does not depend on connection* ω *(hence also not on A), and depends only on the isomorphism class of* $\mathscr{P}$.

ii) $\mathfrak{W}(\mathscr{P}, \cdot) : (S(\mathfrak{g}^*)^G, \circ) \to (H_{dR}^\bullet(M), \wedge)$ *is a ring homorphism.*

iii) For a bundle map $\bar{f} : \mathscr{P}' \to \mathscr{P}$, $f : M' \to M$*, it follows* $\mathfrak{W}(\mathscr{P}', P) = f^*\mathfrak{W}(\mathscr{P}, P)$.

Proof. We only show i): Let ω_0 and ω_1 be two connections in $\mathscr{P}$. Consider the product bundle $(\mathscr{P} \times \mathbb{R}, \pi \times \mathrm{id}, M \times \mathbb{R})$ and we can think of ω_0, ω_1 as connections on $\mathscr{P} \times \mathbb{R}$. Then there is a connection $\tilde{\omega} \in \Omega^1(\mathscr{P} \times \mathbb{R}; \mathfrak{g})$ defined as $\tilde{\omega}_{(u,t)} = (1-t)\omega_{0u} + t\omega_{1u}$, $(u, t) \in \mathscr{P} \times \mathbb{R}$ (we have used the fact that convex combination of connections is again a connection). Let $i_{0,1} : M \to M \times \mathbb{R}$ be the inclusions given by $i_0(u) = (u, 0)$ and $i_1(u) = (u, 1)$. Clearly $i_s^*\tilde{\omega} = \omega_s$ $(s = 0, 1)$, hence $i_s^*\tilde{\Omega} = \Omega_s$ $(s = 0, 1)$ and due to Poincaré lemma, there is a homotopy operator $h : \Omega^k(\mathscr{P} \times \mathbb{R}) \to \Omega^{k-1}(\mathscr{P})$ such that

$$dh(\tilde{\omega}) + h(d\tilde{\omega}) = i_1^*\tilde{\omega} - i_0^*\tilde{\omega},$$

and we obtain

$$dh(P(\tilde{\Omega}^k)) = i_1^*P(\tilde{\Omega}^k) - i_0^*P(\tilde{\Omega}^k) = P(\Omega_1^k) - P(\Omega_0^k)$$

due to $dP(\tilde{\Omega}^k) = 0$ by previous proposition. Thus $P(\Omega_1^k)$ and $P(\Omega_0^k)$ differing by an exact form represent the same cohomology class. The reset of statements follow from previous proposition. □

Definition. *A characteristic class* c *with (*$\mathbb{R}$ *coefficients) for a principal* G*-bundle is a cohomology class* $c(\mathscr{P}) \in H_{dR}^\bullet(M)$ *such that for a bundle map* $\bar{f} : \mathscr{P}' \to \mathscr{P}$, $f : M' \to M$ *it follows* $c(\mathscr{P}') = f^*c(\mathscr{P})$.

Thus from the theorem we know, fixing an invariant polynomial $P \in S^k(\mathfrak{g}^*)^G$, the Weil homomorphism $\mathfrak{W}(\cdot, P) : \mathscr{P} \mapsto \mathfrak{W}(\mathscr{P}, P)$ defines a characteristic class.

Remark.

• The set of characteristic classes (with $\mathbb{R}$ coefficients) is a ring denoted as $H_G^\bullet(M;\mathbb{R})$ (equivariant cohomology). H. Cartan proved that if G is a compact Lie group, then $\mathfrak{W} : S(\mathfrak{g}^*)^G \to H_G^\bullet(M;\mathbb{R})$ is an isomorphism.

• The coefficient can also be extended to $\mathbb{C}$, i.e. $H_{\mathrm{dR}}^\bullet(M;\mathbb{C})$.

Example. For matrix Lie groups and Lie algebras, the algebra of invariant polynomial functions can be generated from determinant. The relevant statements and proofs are referred to [3] Chapter XII §2.

(1) $G = \mathrm{GL}(n,\mathbb{R})$, $\mathfrak{g} = \mathfrak{gl}_n(\mathbb{R})$, $\det(\lambda I - \frac{1}{2\pi}X) = \sum_{k=0}^n P_{k/2}(X,\dots,X)\lambda^{n-k}$, $X \in \mathfrak{gl}_n(\mathbb{R})$. The Ad-invariant polynomial $P_{k/2}$ is called the $k/2$-th Pontrjagin polynomial. For $\mathrm{GL}(n,\mathbb{R})$-bundle $\mathscr{P}$, the Pontrjagin class is defined as $p_{k/2}(\mathscr{P}) = \mathfrak{W}(\mathscr{P}, P_{k/2}) \in H_{\mathrm{dR}}^{2k}(M)$. For a rank-$n$ real vector bundle $\mathscr{V}$, its Pontrjagin class is given by its frame bundle $p_{k/2}(\mathscr{V}) = p_{k/2}(\mathrm{Fr}(\mathscr{V}))$.

(2) $G = \mathrm{O}(n) \subset \mathrm{GL}(n,\mathbb{R})$, $\mathfrak{g} = \mathfrak{o}(n)$ consists of skew-symmetric matrices. Hence by transposing, $\det(\lambda I - \frac{1}{2\pi}X) = \det(\lambda I + \frac{1}{2\pi}X)$, $X \in \mathfrak{o}(n)$. It follows that the restriction of $P_{k/2}$ on $\mathfrak{o}(n)$ vanishes for odd k, thus only $P_l \in S^{2l}(\mathfrak{o}(n)^*)^{\mathrm{O}(n)}$ for $l = 0,\dots,[n/2]$ are considered. Since every vector bundle can equip a Riemannian metric making its frame bundle reduce from $\mathrm{GL}(n,\mathbb{R})$-bundle to $\mathrm{O}(n)$-bundle. Therefore for every $\mathrm{GL}(n,\mathbb{R})$-bundle $\mathscr{P}$, we actually have $p_{k/2}(\mathscr{P}) = 0$ for k odd.

(3) $G = \mathrm{SO}(2m) \subset \mathrm{O}(2m)$, $\mathfrak{g} = \mathfrak{so}(2m) = \mathfrak{o}(2m)$. The Pontrjagin polynomials $P_l \in S^{2l}(\mathfrak{so}(2m)^*)^{\mathrm{SO}(2m)}$, $l = 0, 1, \dots, m$ are invariant polynomials. Another type of invariant polynomials are called Pfaffian polynomials:

$$\mathrm{Pf}(X,\dots,X) = \frac{1}{2^{2m}\pi^m m!}\sum_{\sigma\in\mathfrak{S}_{2m}} \mathrm{sgn}(\sigma) a_{\sigma(1)\sigma(2)}\cdots a_{\sigma(2m-1)\sigma(2m)}$$

where $X = (a_{ij})$ and $a_{ij} = -a_{ji}$. Pf is invariant for $\mathrm{SO}(2m)$ not for $\mathrm{O}(2m)$ (there would be a minus sign). For $\mathrm{SO}(2m)$-bundle $\mathscr{P}$, $e(\mathscr{P}) = \mathfrak{W}(\mathscr{P}, \mathrm{Pf}) \in H_{\mathrm{dR}}^{2m}(M)$ is called the Euler class. If $\mathscr{P}$ is the reduction to $\mathrm{SO}(2m)$-bundle of the frame bundle of an oriented vector bundle V over M, then $e(\mathscr{V}) = e(\mathscr{P})$ is the Euler class for V. For a compact oriented Riemannian manifold of dimension $2m$, the Euler characteristic can be given by

$$\chi(M) = \langle e(TM), [M]\rangle = \int_M \mathrm{Pf}(F_\omega^m)$$

where ω is the connection for reduction to $\mathrm{SO}(2m)$-bundle of the orthogonal frame bundle of TM determined by orientation, descending to TM the connection A is usually chosen to be the Levi-Civita connection. This is the higher dimensional version of Gauss-Bonnet theorem.

(4) $G = \mathrm{GL}(n,\mathbb{C})$, $\mathfrak{g} = \mathfrak{gl}_n(\mathbb{C})$. $\det(\lambda I - \frac{1}{2\pi\mathbf{i}}X) = \sum_{k=0}^n C_k(X,\dots,X)\lambda^{n-k}$, $X \in \mathfrak{gl}_n(\mathbb{C})$. The invariant polynomials C_k are called Chern polynomials. For a $\mathrm{GL}(n,\mathbb{C})$-bundle $\mathscr{P}$ the characteristic class $c_k(\mathscr{P}) = \mathfrak{W}(\mathscr{P}, C_k) \in H_{\mathrm{dR}}^{2k}(M;\mathbb{C})$ is called Chern class. For a rank-n complex vector bundle $\mathscr{V}$, its Chern class is defined as the Chern class of its complex frame bundle, i.e. $c_k(\mathscr{V}) = c_k(\mathrm{Fr}(\mathscr{V}))$.

Writing them out explicitly:

$$\begin{aligned}
C_0(\Omega) &= 1 \\
C_1(\Omega) &= \frac{\mathbf{i}}{2\pi}\operatorname{tr}(\Omega) \\
C_2(\Omega) &= \frac{\mathbf{i}}{8\pi}[\operatorname{tr}(\Omega\wedge\Omega) - \operatorname{tr}(\Omega)\wedge\operatorname{tr}(\Omega)] \\
&\vdots \\
C_k(\Omega) &= \left(\frac{\mathbf{i}}{2\pi}\right)^k \frac{1}{k!}\sum_{\substack{i_1,\dots,i_k\\ j_1,\dots,j_k}} \delta^{j_1}_{i_1}\cdots\delta^{j_k}_{i_k}\Omega^{i_1}_{j_1}\wedge\cdots\wedge\Omega^{i_k}_{j_k}.
\end{aligned}$$

Note that the restriction of C_k to $\mathfrak{gl}_n(\mathbb{R})$ yields $\mathbf{i}^k C_k(X,\dots,X) = P_{k/2}(X,\dots,X)$, hence if we extend a $\mathrm{GL}(n,\mathbb{R})$-bundle $\mathscr{P}$ to a $\mathrm{GL}(n,\mathbb{C})$-bundle $\mathscr{P}_{\mathbb{C}} = \mathscr{P}\otimes\mathbb{C}$ then for $k = 2l$ we have $p_l(\mathscr{P}) = (-1)^l c_{2l}(\mathscr{P}_{\mathbb{C}}) \in H^{4l}_{\mathrm{dR}}(M;\mathbb{C})$. Furthermore, every complex vector bundle can equip a Hermitian metric, therefore the frame bundle as a $\mathrm{GL}(n,\mathbb{C})$-bundle can be reduced to a $\mathrm{U}(n)$-bundle. It follows that every $\mathrm{GL}(n,\mathbb{C})$-bundle admits a reduction to $\mathrm{U}(n)$-bundle, on which C_k are real-valued (by Hermitian conjugation). This implies the Chern classes of $\mathrm{U}(n)$-bundle are real cohomology classes and consequently all Chern classes for a $\mathrm{GL}(n,\mathbb{C})$-bundle are real cohomology classes.

A.2 Transgression

Consider generally a fiber bundle $F \to E \xrightarrow{\pi} M$. In [5], the transgression or transgressive element is defined to be an element in the Leray-Serre spectral sequence in the setting of singular cohomology in real coefficient: $\omega \in H^q(F) \hookrightarrow E_2^{0,q} = H^0(\mathcal{U}, \mathscr{H}^q)$ lives to E_{q+1}, i.e. $d_2\omega = d_3\omega = \cdots = d_q\omega = 0$. Here $\mathcal{U}$ is a good cover of M such that $\pi^{-1}(U)$ is also a cover of E. $H^\bullet(\mathcal{U},\mathscr{H}^q)$ is the Čech cohomology valued in the locally constant presheaf $\mathscr{H}^q$ defined by $\mathscr{H}^q(U) = H^q(\pi^{-1}(U))$ for $U\in\mathcal{U}$.

Proposition ([5] Proposition 18.13)**.** *Let $\pi : E \to M$ be a fiber bundle with smooth fiber F. An element $\omega \in H^q(F)$ is transgressive if and only if it is the restriction of a global form ψ on E such that $d\psi = \pi^*\tau$ for some form τ on M.*

Remark. Since π^* is injective and $\pi^* d\tau = d^2\psi = 0$, this leads to $d\tau = 0$ hence τ represents a cohomology class on M.

Now we come back to the principal bundle. Consider a G-bundle $\mathscr{P} \xrightarrow{\pi} M$ and pick a connection 1-form ω. The form $P(\Omega^k) = \pi^* P(F_\omega^k)$ is closed for any $P \in S^k(\mathfrak{g}^*)^G$ and determines an element in $H^{2k}(M;\mathbb{R})$ which is independent of ω. While this may be an non-trivial element in the cohomology of M, it always pulls back along $\mathrm{pr} : \pi^*\mathscr{P} \to \mathscr{P}$ to a trivial cohomology class in the cohomology of $\mathscr{P}$ because this pullback bundle is a trivial G-bundle over $\mathscr{P}$, and thus all of its characteristic classes vanish. Namely, $\mathrm{pr}^* P(\Omega^k) = P(\Omega^k)$ is exact when considered as a form on $\mathscr{P}$. So this fits into the proposition above. Hence one expects to find a primitive of $P(\Omega^k)$ on $\mathscr{P}$ and calls it the transgression.

Proposition ([15] Proposition 3.2 and 3.6)**.** *Set $\varphi_t = t\Omega + (t^2-t)\frac{1}{2}[\omega,\omega]$ and*

$$TP(\omega) = k\int_0^1 P(\omega\wedge\varphi_t^{k-1})dt.$$

Then $dTP(\omega) = P(\Omega^k) \in \Omega^{2k}(\mathscr{P})$. The transgression form is unique up to exact term.

Chern and Simons have derived the transgression form explicitly (c.f. [15] (3.5)):

$$TP(\omega) = \sum_{i=0}^{k-1} (-1)^i \frac{k!(k-1)!}{2^i (k+i)!(k-1-i)!} P(\omega \wedge [\omega,\omega]^i \wedge \Omega^{k-i-1}).$$

Proposition ([15] Proposition 3.8)**.** *Let $\{\omega_t\}_{t\in[0,1]}$ be a smooth 1-parameter family of connection 1-forms on $\mathscr{P}$. Set $\omega = \omega_t|_{t=0}$ and $\dot{\omega} = \frac{d}{dt}\omega_t\big|_{t=0}$. Then*

$$\frac{d}{dt} TP(\omega_t)\bigg|_{t=0} = kP(\dot{\omega} \wedge \Omega^{k-1}) + \textit{exact terms.}$$

Theorem ([15] Theorem 3.9)**.** *Let $\mathscr{P} \xrightarrow{\pi} M$ be a principal bundle with connection ω and* $\dim M = n$.

- *If $2k-1 = n$, then $TP(\omega)$ is closed and $[TP(\omega)] \in H^n(\mathscr{P};\mathbb{R})$ depends on ω.*
- *If $2k-1 > n$, then $TP(\omega)$ is closed and $[TP(\omega)] \in H^{2k-1}(\mathscr{P};\mathbb{R})$ is independent of ω.*

If for some connection ω and curvature Ω, the primary class $P(\Omega^k) = 0$ as a form, then $TP(\omega)$ is a closed form, so we get a cohomology class on $\mathscr{P}$. Although this class is still on the principal bundle, but we can pull it down to M along some global section (if exists) to obtain a well-defined cohomology class on M. However, under structure group G action on the section (i.e. changing frames if we think of principal bundle as some frame bundle), such a representative may end up with an additional closed term. When M is compact, after integrating such form over M, it may turn out that the value will be some real number modulo integer (integral periods) which is contributed from the extra closed term. So in general we would get a cohomology class lies in $H^{2k-1}(M;\mathbb{R}/\mathbb{Z})$ from the transgression form in $\Omega^{2k-1}(\mathscr{P})$, c.f. [15] Proposition 3.15 and Theorem 3.16. In such sense, this is called a *secondary characteristic class*. The secondary characteristic class clearly depends on connection unlike the usual characteristic class, so it actually reflects some geometric information rather than topology. This is the general philosophy underlying the Chern-Simons conformal invariant as well as Chern's proof for the generalized Gauss-Bonnet theorem (which is called the Gauss-Bonnet-Chern theorem nowadays) by constructing the secondary characteristic class associated to Pfaffian polynomial on even-dimensional compact oriented Riemannian manifold.

B Subelliptic operators and Folland-Stein spaces

B.1 Subelliptic operators

Definition. *Let M be a smooth manifold, a formally self-adjoint second order differential operator $P : C^\infty(M) \to C^\infty(M)$ is said to be subelliptic of order $\epsilon \in (0,1]$ at $x \in M$ if there is an open neighborhood $U \subseteq M$ of x such that*

$$\|u\|^2_{H^\epsilon} \leq C(|(Pu,u)_{L^2}| + \|u\|^2_{L^2}) \qquad \forall u \in C^\infty_0(U)$$

for some positive constant C. P is (globally) subelliptic of order ϵ if it is so at any point $x \in M$.

Recall $H^\epsilon := W^{2,\epsilon}$ is the Sobolev space, whose norm can be defined via Fourier transformation as $\|u\|^2_{H^\epsilon} = \int (1+|p|^2)^\epsilon |\hat{u}(p)|^2 dp$.

Definition. *Let L be a differential operator with its formal adjoint P^*. If $\varphi \in C^\infty_0(M)'$ is a distribution on M, then we can define another distribution $P\varphi(u) = \varphi(P^*u)$ for any $u \in C^\infty_0(M)$. L is said to be hypoelliptic if $P\varphi \in C^\infty(M)$ implies $\varphi \in C^\infty(M)$.*

It was shown by J.J. Kohn and L. Nirenberg that any subelliptic operator P of order ϵ is hypoelliptic and satisfies the following estimate (c.f. [18] page 116):

$$\|u\|^2_{H^{s+2\epsilon}} \leq C_s(\|Pu\|^2_{H^s} + \|u\|^2_{L^2}) \ , \ s \geq 0 \ , \ \forall u \in C^\infty_0(U).$$

An important lemma by E. V. Radkevic states:

Lemma ([18] Lemma 2.1)**.** *Let $K \subseteq M$ be a compact set and let $\{Z_1, \ldots, Z_N\}$ be a system of complex vector fields on M such that*

i) $span_{\mathbb{C}}\{Z_1, \ldots, Z_N\}$ is closed under complex conjugation,

ii) $\{Z_{1,x}, \ldots, Z_{N,x}\} \cup \{[Z_i, Z_j]_x : 1 \leq i,j \leq N\}$ span $T_xM \otimes \mathbb{C}$ for any $x \in K$.

Then there is a constant $C > 0$ such that

$$\|u\|^2_{H^{1/2}} \leq C\left(\sum_{j=1}^N \|Z_j u\|_{L^2} + \|u\|_{L^2}\right) \qquad \forall\, u \in C^\infty_0(K).$$

We want to study the most fundamental subelliptic operator on strictly pseudo-convex CR manifolds. First we take a look at the so-called Folland-Stein operator introduced by G. B. Folland and E. M. Stein acting on real-valued functions on the Heisenberg group $\mathbf{H}_n$ (c.f. [41] (5.5)).

Definition. *Let $Z_{\bar{\mu}}$ be the Lewy operators on the Heisenberg group $\mathbf{H}_n$ with respect to the canonical contact form θ, i.e. $L_\theta(Z_\mu, Z_{\bar{\mu}'}) = \delta_{\mu\mu'}$, and the Reeb vector field $\mathcal{R} = \frac{\partial}{\partial t}$. The second order differential operators given by*

$$\mathcal{L}_\alpha = -\frac{1}{2}\sum_{\mu=1}^n (Z_\mu Z_{\bar{\mu}} + Z_{\bar{\mu}} Z_\mu) + \mathbf{i}\alpha\mathcal{R}, \quad \alpha \in \mathbb{C},$$

are called the Folland-Stein operators.

One can prove that with respect to the L^2-inner product for complex-valued functions on $\mathbb{H}_n$, the adjoint of Folland-Stein operators are $\mathcal{L}^*_\alpha = \mathcal{L}_{\bar{\alpha}}$ (c.f. [18] Lemma 3.7). This is then used to show that the Folland-Stein operators are locally invertible:

Proposition ([18] Proposition 3.2). *For any $f \in C_0^\infty(\mathbf{H}_n)$ and any admissible value of $\alpha \in \mathbb{C}$, we have*

$$\mathcal{L}_\alpha K_\alpha f = K_\alpha \mathcal{L}_\alpha f = f.$$

In above statement, the operator $K_\alpha : C_0^\infty(\mathbf{H}_n) \to C^\infty(\mathbf{H}_n)$ is defined to be the convolution $f * (\frac{\varphi_\alpha}{c_\alpha})$ where $\varphi_\alpha = (|z|^2 - \mathbf{i}t)^{-(n+\alpha)/2}(|z|^2 + \mathbf{i}t)^{-(n-\alpha)/2}$ and $c_\alpha := \frac{2^{2(1-n)}\pi^{n+1}}{\Gamma(\frac{n+\alpha}{2})\Gamma(\frac{n-\alpha}{2})}$, and $\alpha \in \mathbb{C}$ is said to be admissible when $c_\alpha \neq 0$ (this means $\alpha \notin \{\pm n, \pm(n+2), \pm(n+4), \dots\}$). In fact, $\mathcal{L}_\alpha \varphi_\alpha = c_\alpha \delta$ in distribution sense where δ is the Dirac distribution (c.f. [18] Theorem 3.9).

Corollary ([18] Corollary 3.1). *Let $\alpha \in \mathbb{C}$ be admissible. Then*

i) For any $f \in \mathcal{E}'(\mathbf{H}_n)$ we have $\mathcal{L}_\alpha K_\alpha f = K_\alpha \mathcal{L}_\alpha f = f$.

ii) $\mathcal{L}_\alpha$ is locally sovable, i.e. $\forall\, g \in \mathcal{E}'(\mathbf{H}_n)$, $\exists\, f \in \mathcal{D}'(\mathbf{H}_n)$ such that $\mathcal{L}_\alpha f = g$.

iii) The equation $\mathcal{L}_\alpha f = 0$ has no non-trivial solution in $\mathcal{E}'(\mathbf{H}_n)$.

Theorem ([18] Theorem 2.2). *The differential operator $\mathcal{L}_0$ on $\mathbf{H}_n$ which is written explicitly as*

$$\mathcal{L}_0 = \sum_{\mu=1}^{n} \left[-\frac{\partial^2}{\partial z_\mu \partial \bar{z}_\mu} - |z_\mu|^2 \frac{\partial^2}{\partial t^2} + \mathbf{i}\frac{\partial}{\partial t}\left(z_\mu \frac{\partial}{\partial z_\mu} - \bar{z}_\mu \frac{\partial}{\partial \bar{z}_\mu} \right) \right]$$

is left-invariant and is subelliptic of order $1/2$.

On every $2n+1$ dimensional strictly pseudo-convex CR manifold, around any point there exists a local open neighborhood contactomorphic to the Heisenberg group $\mathbf{H}_n$ with its canonical contact structure, and the contact form on that neighborhood is equal to the pullback of canonical contact form on $\mathbf{H}_n$. Such coordinate charts are called the Folland-Stein normal coordinates at that point. Under the normal coordinates, the vector fields on the CR manifold can be locally written as the linear combinations of the Lewy operators on $\mathbf{H}_n$. This is easy consequence of the contact Darboux's theorem. We refer to [18] §3.2 for more details. Therefore, the Heisenberg group is not just the flat space in CR geometry, but also the model for the analysis studying on the CR manifolds. This motivates to define Folland-Stein operators in terms of Tanak-Webster covariant derivatives with respect to the *pseudo-Hermitian frame* on strictly pseudo-convex CR manifolds (the frame defined via the Levi form as $L_\theta(Z_\alpha, Z_{\bar{\beta}}) = \delta_{\alpha\bar{\beta}}$).

Actually, $\mathcal{L}_0 = \frac{1}{2}\Delta_b$ where Δ_b is the sub-Laplacian operator on $\mathbf{H}_n$. The sub-Laplacian Δ_b on general strictly pseudo-convex CR manifolds are defined similarly by the pseudo-Hermitian frames. Hence consequently, Δ_b is hypoelliptic by the result of L. Hörmander. The sub-Laplacian operator is the most fundamental differential operator on strictly pseudo-convex CR manifolds. For example, it is heavily involved in the study of CR Yamabe problem. Subelliptic theory on strictly pseudo-convex CR manifolds is believed to play the prominent role as the elliptic theory on Riemannian manifolds.

To establish some Sobolev-type estimate for Folland-Stein operators with $\alpha \neq 0$ on $\mathbf{H}_n$ and generalize to arbitrary strictly pseudo-convex CR manifolds, we need to work in different setting rather than classical Sobolev spaces. This is when the Folland-Stein space theory comes in.

B.2 Folland-Stein spaces

G.B. Folland and E.M. Stein has defined some weighted Sobolev-type spaces called Folland-Stein spaces to carry out some estimates on the Folland-Stein operators on the $(2n+1)$-dimensional real Heisenberg group $\mathbf{H}_n$ [41]. Via the Folland-Stein normal coordinates, these notions can be defined on any strictly pseudo-convex CR manifolds (M, θ) of $2n+1$ dimension.

Definition. *Let U be a relatively compact open subset of a normal coordinate neighborhood in M. Put $X_j = \operatorname{Re} Z_j$, $X_{j+n} = \operatorname{Im} Z_j$ where $\{Z_j\}_{1\le j\le n}$ is a pseudo-Hermitian frame. Put $X^\mu = X_{\mu_1}\cdots X_{\mu_k}$ where $\mu = (\mu_1,\ldots,\mu_k)$ with $1 \le \mu_j \le 2n$ is a multi-index of length $\ell(\mu) = k \in \mathbb{N}_0$. Consider the norms defined by*

$$\|f\|_{S^p_k(U)} := \sup_{\ell(\mu)\le k} \|X^\mu f\|_{L^p(U)}$$

where

$$\|f\|_{L^p(U)} = \left(\int_U |f|^p\, \theta\wedge(d\theta)^n\right)^{\frac{1}{p}}.$$

The Folland-Stein space $S^p_k(U)$ is the completion of $C^\infty_0(U)$ under the norm $\|\cdot\|_{S^p_k(U)}$.

On a compact strictly pseudo-convex CR manifold (M,θ), let $\{U_i\}_{i\in I}$ be a finite open cover and $\{\rho_i\}_{i\in I}$ be a smooth partition of unity subordinate to the cover. We define the Folland-Stein spaces on entire manifold as

$$S^p_k(M) = \{f \in L^1(M)\,|\,\rho_i f \in S^p_k(U)\ \forall\, i\in I\}.$$

There is relation between Folland-Stein space and classical Sobolev spaces.

Theorem ([41] Theorem 19.1)**.** *On a open subset U of strictly pseudo-convex CR manifold (M,θ), $S^p_k(U) \subset W^{p,k/2}_{loc}(U)$ for $k\in\mathbb{N}_0$ and $1<p<\infty$.*

Using singular integral operator theory adapted to strictly pseudo-convex CR manifolds (actually the proof of local invertibility of Folland-Stein operators is also based on such technique), Folland and Stein are able to prove some Sobolev-type embedding theorems:

Theorem ([18] Theorem 3.13 and 3.15)**.** *On compact strictly pseudo-convex CR manifold (M,θ) of $2n+1$ dimension, we have:*

- *If $\frac{1}{q} = \frac{1}{p} - \frac{k}{2(n+1)}$ and $1<p<q<\infty$, then $S^p_k(M)\subset L^q(M)$.*
- *If $1<p<q<\infty$ and $\frac{1}{q} > \frac{1}{p} - \frac{1}{2(n+1)}$, then the unit ball in $S^p_1(M)$ is compact in $L^q(M)$.*

On Heisenberg group $\mathbf{H}_n$, they have developed the regularity theory for the Folland-Stein operators. Firstly, there is a L^p-estimate:

Theorem ([41] Theorem 9.4)**.** *For any admissible α,$1<p<\infty$, and $k\in\mathbb{N}_0$, we have*

$$\|f\|_{S^p_{k+2}} \le C_{p,k}(\|\mathcal{L}_\alpha f\|_{S^p_k} + \|f\|_{L^p}) \qquad \forall\, f\in C^\infty_0.$$

By applying this estimate, they obtain the L^p-regularity theorem:

Theorem ([41] Theorem 9.5)**.** *Suppose α is admissible and $f,g\in\mathcal{D}'$ satisfying $\mathcal{L}_\alpha f = g$ on $U\subset\mathbf{H}_n$.*

- *If $g\in S^p_{k,loc}(U)$ with $1<p<\infty$, then $f\in S^p_{k+2,loc}(U)$.*
- *If $g\in L^p_{loc}(U)$ and $\frac{1}{q} = \frac{1}{p} - \frac{1}{n+1} > 0$, then $f\in L^p_{loc}(U)$ when $p>1$ and $f\in L^{q-\epsilon}_{loc}(U)$ for any $\epsilon>0$ when $p=1$.*

Via normal coordinates, one can expect to have completely similar result for the Folland-Stein type operators constructed in terms of pseudo-Hermitian frames on any strictly pseudo-convex CR manifold.

All the constructions and the results above should hold similarly for sections of tensor bundle $\mathscr{T}$ built from the tensor power of $\xi_{\mathbb{C}}$ and $\xi_{\mathbb{C}}^*$ on any strictly pseudo-convex CR manifold (M,θ) of $2n+1$ dimension. This is because we can work on any local trivializing neighborhood U of the bundle $\mathscr{T}$, then the function spaces would simply be $C_0^\infty(U,\mathbb{R}^N), L^p(U,\mathbb{R}^N), S_k^p(U,\mathbb{R}^N)$ where N is the complex rank of $\mathscr{T}$. Following [32], we define the Folland-Stein spaces of sections of $\mathscr{T}$ on a compact strictly pseudo-convex 3-manifold (M,θ) by

$$S_k^2(M,\mathscr{T}) := \{u \in L^2(M,\mathscr{T}) \,|\, Pu \in L^2(M,\mathscr{T}) \text{ whenever } \mathrm{w}(P) \leq k\}\ ,\ k \geq 0,$$

where we choose the L^2-norm as $\|\cdot\|_J$ associated to the oriented CR structure (J,θ,ξ). This gives a norm on S_k^2:

$$\|u\|_{S_k^2}^2 = \sum_{0\leq j\leq k} \|\mathfrak{D}^j u\|_J^2$$

where the weight 1 operator $\mathfrak{D} : C^\infty(M,\mathscr{T}) \to C^\infty(M,\mathscr{T}\otimes\xi^*)$ is defined as (49). Recall on some trivializing neighborhood we can always find a pseudo-Hermitian frame such that the Christoffel symbol vanishes (lemma 3.4). Hence the spaces S_k^2 are well-defined via the norm completion in the previous way, and it is a Hilbert space under the inner product. The following are some important facts about S_k^2:

(a) Any weight m operator maps $S_{k+m}^2 \to S_k^2$ continuously for all $k \geq 0$.

(b) $S_k^2 \subset L^q$ provided $q \geq 2$ and $k \geq 2 - \frac{4}{q}$.

(c) $S_k^2 \subset C^0$ provided $k > 2$. More generally, $S_k^2 \subset C^m$ provided $k > 2m+2$.

(d) The inclusion $S_{k+1}^2 \subset S_k^2$ is compact for $k \geq 0$.

(e) For any $k_1 < k_2 < k_3$ and any $\varepsilon > 0$, there is a constant C_ε such that

$$\|u\|_{S_{k_2}^2} \leq \varepsilon \|u\|_{S_{k_3}^2} + C_\varepsilon \|u\|_{S_{k_1}^2} \qquad \forall u \in S_{k_3}^2.$$

Recall the definition of $\mathcal{L}_\alpha$ in section 7, i.e. (50), and consider $\mathscr{T} = \mathscr{E}_J$. So this operator of our interest is simply regarded as covariant derivative generalization of the original Folland-Stein operators to the tensor bundle. The following regularity theorem is essentially based on the results by Folland-Stein for $\mathcal{L}_\alpha$ on $\mathbf{H}_n$ stated before.

Theorem ([32] Proposition 4.2)**.** *Assume $\alpha \in \mathbb{C}$ is not an odd integer (i.e. admissible), $u \in L^2$ and $\mathcal{L}_\alpha u \in S_k^2$ in the distribution sense. Then $u \in S_{k+2}^2$ and*

$$\|u\|_{S_{k+2}^2} \leq C_k(\|\mathcal{L}_\alpha u\|_{S_k^2} + \|u\|_{L^2})$$

for some positive constant C_k.

Comparing to the hypoelliptic characterization of usual subelliptic operator, we say that $\mathcal{L}_\alpha$ is a second order subelliptic differential operator of weight 2 with respect to the Folland-Stein spaces S_k^2.

References

[1] Michael Spivak, *A Comprehensive Introduction to Differential Geometry, Vol. 2*, 3rd Edition, Publish or Perish, 1999.

[2] Johan L. Dupont, *Curvature and Characteristic Classes*, Lecture Notes in Mathematics, Springer-Verlag Berlin Heidelberg, 1978.

[3] S. Kobayashi, K. Nomizu, *Foundations of Differential Geometry* (Wiley Classics Library) Volume 2, ISBN 0-470-49648-7.

[4] Helga Baum, *Eichfeldtheorie: Eine Einführung in die Differentialgeometrie auf Faserbündeln*, 2. Auflage, Springer-Verlag Berlin-Heidelberg, 2014.

[5] Raoul Bott, Loring W. Tu, *Differential Forms in Algebraic Topology*, Graduate Texts in Mathematics 82, Springer (1982).

[6] Thomas Andrew Ivey, J. M. Landsberg, *Cartan for Beginners: Differential Geometry Via Moving Frames and Exterior Differential Systems*, Graduate Studies in Mathematics, vol. 61, American Mathematical Society, 2003.

[7] Shoshichi Kobayashi, *Transformation Groups in Differential Geometry*, Springer-Verlag Berlin Heidelberg 1995.

[8] Victor Guillemin, *The integrability problem for G-structures*, Trans. Amer. Math. Soc. 116 (1965), 544–560.

[9] R.W. Sharpe, *Differential Geometry - Cartan's Generalization of Klein's Erlangen Program*, Graduate Texts in Mathematics (166), Springer-Verlag New York, 1997.

[10] Ana Cannas da Silva, *Lectures on symplectic geometry*, Lecture Notes in Mathematics, 1764, Springer-Verlag, Berlin, 2008.

[11] B. Aebischer, M. Borer, M. Kälin, C. Leuenberger, H. M. Reimann, *Symplectic Geometry*, An Introduction based on the Seminar in Bern, 1992. Progress in Mathematics, Birkhäuser.

[12] Hansjörg Geiges, *An Introduction to Contact Topology*, Cambridge Studies in Advanced Mathematics, Band 109. Cambridge University Press, 2008.

[13] Yakov Eliashberg, *Symplectic geometry of plurisubharmonic functions*. In: Hurtubise J., Lalonde F., Sabidussi G. (eds) Gauge Theory and Symplectic Geometry. NATO ASI Series (Series C: Mathematical and Physical Sciences), vol 488. Springer, Dordrecht. 1997.

[14] Kai Cieliebak and Yakov Eliashberg, *Stein to Weinstein and Back: Symplectic Geometry of Affine Complex Manifolds*, American Mathematical Society, 2012.

[15] Shiing-shen Chern, James Simons, *Characteristic forms and geometric invariants*, The Annals of Mathematics, Second Series, Vol. 99, No. 1 (1974).

[16] Sergiu Moroianu, *The Cotton tensor and Chern-Simons invariants in dimension 3: an introduction*, Buletinul Academiei de Ştiinţe a Republicii Moldova. Matematica, Number 2(78), 2015, Pages 3–20, ISSN 1024–7696.

[17] Howard J. Jacobowitz, *An Introduction to CR Structures*, Mathematical Surveys and Monographs Volume: 32; 1990.

[18] Sorin Dragomir, Giuseppe Tomassini, *Differential Geometry and Analysis on CR Manifolds*, Birkhäuser 2006.

[19] E. Cartan, *Sur la géométrie pseudo-conforme des hypersurfaces de l' espace de deux variables complexe*, I, Ann. Mat. 11 (1932) 17-90; II, Ann. Sc. Norm. Sup. Pisa 1 (1932) 333-354.

[20] S. Webster, *Pseudo-hermitian structures on a real hypersurface*, J. Differ. Geom. 13, 25-41 (1978).

[21] S. S. Chern, J. Moser, *Real hypersurfaces in complex manifolds*, Acta Math. 133, 219-271 (1974).

[22] Howard J. Jacobowitz, *Chains in CR geometry*, J. Differential Geom. Volume 21, Number 2 (1985), 163-194.

[23] J. M. Lee, *Pseudo-Einstein structures on CR manifolds*, Am. J. Math. 110 (1988), 157-178.

[24] Noboru Tanaka, *On non-degenerate real hypersurfaces, graded Lie algebras and Cartan connections*, Japan. J. Math. Vol. 2, No. 1, (1976).

[25] William M. Goldman, *Conformally Flat Manifolds with Nilpotent Holonomy and the Uniformization Problem for 3-Manifolds*, Transactions of the American Mathematical Society Vol. 278, No. 2 (Aug., 1983), pp. 573-583.

[26] Le Anbo, *Cartan connections for CR manifolds*, Manuscripta Math. 122 (2007), no. 2, 245-264.

[27] Elisha Falbel and Nikolay Gusevskii, Spherical CR-Manifolds of Dimension 3, Bol. Soc. Bras. Mat., Vol. 25, N. 1, 21-56. 1994, Sociedade Brasileira de Mathemática.

[28] D. Burns, and S. Shnider, *Spherical hypersurfaces in complex manifolds*, Invent. Math., 33, 223-246 (1976).

[29] William M. Goldman, *Locally homogeneous geometric manifolds*, Proceedings of International Congress of Mathematics, Hyderabad, India, 2010.

[30] D.M. Burns, Jr. and C.L. Epstein, *A global invariant for three dimensional CR-manifolds*, Invent. Math. 92 (1988) 333-348.

[31] D. Burns and C. L. Epstein, *Characteristic numbers of bounded domains*, Acta Math. Volume 164 (1990), 29-71.

[32] J. H. Cheng and J. M. Lee, *The Burns-Epstein invariant and deformation of CR structures*, Duke Math. J. 60 (1990), 221-254.

[33] J. H. Cheng and J. M. Lee, *A local slice theorem for 3-dimensional CR structures*, Amer. J. Math., 117 (1995), 1249-1298.

[34] Jih-Hsin Cheng, *Deformation of fillable CR structures*, math.DG/0202051 (2002).

[35] Jih-Hsin Cheng, *Flows and a tangency condition for embeddable CR structures in dimension 3*, Methods and Applications of Analysis. Vol. 21, No. 3, pp. 337–356, September 2014.

[36] Jih-Hsin Cheng, *The Geometry of Cauchy-Riemann Manifolds in Dimention Three*, Proceedings of the Second Asian Mathematical Conference 1995.

[37] K. Hirachi, Q-prime curvature on CR manifolds, Differential Geom. Appl. 33 (2014) no. suppl., 213–245.

[38] Khoi The Vu, *On the Burns-Epstein invariants of spherical CR 3-manifolds*, Ann. Inst. Fourier (Grenoble) 61 (2011), no. 2, 775-797.

[39] Howard Jacobowitz, *Locally CR spherical three manifolds*, arXiv: 1303.5286 (2013).

[40] Stefan P. Ivanov, Dimiter N. Vassilev, *Extremals for the Sobolev Inequality and the Quaternionic Contact Yamabe Problem*, 1st Edition, World Scientific (2011).

[41] G.B. Folland, E. M. Stein, *Estimates for the $\bar{\partial}_b$ complex and analysis on Heisenberg group*, Communications on Pure and Applied Mathematics 27, 429–522 (1974).

[42] Bennett Chow, Dan Knopf, *The Ricci Flow: An Introduction*, Mathematical Surveys and Monographs, Band 110. American Mathematical Society, 2004.

[43] R. S. Hamilton, *The inverse function theorem of Nash and Moser*, Bull. Amer. Math. Soc. 7 (1982).

[44] F. Treves, *Basic Linear Partial Differential Equations*, Academic Press, New York (1975).